# City charges for competing delivery carriers: exact best responses under probability-box travel times and leader pricing without equilibrium search

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**Abstract**

A city that charges delivery carriers faces a bilevel problem with three stacked difficulties. Each carrier solves an NP-hard routing problem under probability-box travel times with unknown dependence; the carriers play a Cournot-type congestion game; and the usual search over charges solves one equilibrium per grid point, about two thousand routing solves per city. We give three algorithms. Budget-resource labelling prices routes exactly, and branch-and-price makes each carrier's integer plan exact. It proves optimality at about 25 to 40 customers per carrier with far fewer labels than scenario labelling; probability boxes buy a guaranteed upper bound and a lower overtime risk at a higher expected cost than sample averages. Dual implementation pricing makes a social target an equilibrium by best-response cuts. Aggregate-response pricing sets charges from observed zone totals and contracts at a rate independent of instance size. On nine metropolitan areas built from OpenStreetMap and gridded population, with ten seeds in three cities, dual implementation pricing reached 1.07–1.23 times the best grid point's gain with 5.7–8.3% of its calls, and 1.00–1.02 of the social optimum's gain (grid: 0.815–0.938); aggregate-response pricing reached 0.933–0.995. Pure equilibria exist under exposure-based congestion, and deciding their existence is strongly NP-complete under vehicle-count congestion; the belief-width result builds on earlier work on approximate equilibria. Levels of the price of anarchy, service and gains depend on the criterion the carriers use: with sample-average carriers service roughly doubles. Orderings and policy directions are robust.

*Keywords:* bilevel optimization, vehicle routing, imprecise probabilities, congestion games, branch-and-price, road pricing

## 1. Introduction

Several delivery firms serve the same neighbourhoods, sell to overlapping customers and slow one another down on the same streets. City authorities respond with instruments that act on all of them: congestion and low-emission zones, kerbside rules, and charges or subsidies per delivery [1, 2, 3, 4]. Choosing such an instrument means predicting how the carriers react, and the reaction is itself the outcome of routing decisions that are hard to compute.

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We treat the city as the leader of a bilevel problem. Three difficulties are stacked.

(1) *Each follower solves an NP-hard routing problem under imprecise probabilities.* A carrier has a few weeks of travel-time observations. These support bounds on the distribution function of each arc time (a probability box) and say nothing about the dependence between arcs. The robust cost of a route is its worst-case expected loss over all joint laws compatible with the boxes, and the carrier's best response is a vehicle routing problem with this cost.

(2) *The followers play a congestion game.* The price a carrier obtains falls with the total deliveries in a zone (Cournot competition) and travel times rise with vehicle density. If congestion depends on vehicle-hours (exposure), the game has an exact potential and pure equilibria exist. If it depends on vehicle counts, pure equilibria may fail to exist and deciding whether one exists is strongly NP-complete (Section 5).

(3) *The leader's problem is a bilevel program whose lower level is the game above.* The usual approach evaluates a grid of charges and computes one equilibrium per grid point. With 63 grid points this took 1 921–2 419 routing-oracle calls per city and seed in our experiments, each call an NP-hard problem.

This paper gives algorithms for each layer and tests them on nine metropolitan areas.

*Contributions, in order, with their novelty level.*

(i) *The problem and the model* (Section 3): a bilevel model with a public leader above carriers who choose routing plans, interact through zone aggregates of deliveries and vehicle-hours, and plan with probability-box travel times. The two congestion regimes, exposure and vehicle count, differ in their existence properties (Section 5).

(ii) *Budget-resource labelling (BRL)* (Section 4.1): exact pricing of routes under probability boxes with unknown dependence. The per-arc budget of the stop-loss decomposition [5, 6] becomes a decision resource of the pricing problem, one per kink of the loss function, and labels are pruned by a dominance rule restricted to a computable completion window, which we prove exact. With branch-and-price each carrier's integer plan is exact: all nine metropolitan instances are solved to proven optimality, with label sets 1.8–2.9 times smaller than without the window, at about 25 to 40 customers per carrier.

(iii) *Leader pricing without equilibrium search* (Sections 4.4 and 4.5). *Dual implementation pricing (DIP)* computes a socially good target profile and then finds charges under which it is an equilibrium, by cutting planes whose separation step is one best response per carrier. *Aggregate-response pricing (ARP)* lets the leader see only the zone aggregates of the market after each posted charge; its iteration contracts at a rate $\varrho = \max_z \kappa_z/(b_z+\kappa_z)$ that does not depend on the number of zones, carriers, lots, routing costs or probability boxes, and a marginal-tariff variant needs no iteration. The methods build on the Pigou–Cournot tax, toll polyhedra, inverse optimisation, trial-and-error tolls and marginal-contribution tariffs [7, 8, 9, 10, 11, 12, 13]; new here are the potential–welfare identity of the lumpy Cournot–congestion game with loss bounds, best responses of the carriers as the separation step, the instance-free rate, and the shift of the implementing charges under Γ-maximin beliefs.

(iv) *Theory* (Section 5): a credal absorption theorem. Above the explicit belief width $\sigma_{\max} \min\{N-1,\ d+1,\ 1+\sqrt{d \ln(2d)/2}\}$, where $\sigma_{\max}$ is the largest range of a plan's contribution in units of the belief weights, a mixed equilibrium rounds to an exact pure imprecise equilibrium. The result builds on the approximate-equilibrium theory of Liu et al. [14]; on our city instances ($N \leq 6$, $d \geq 84$) the term that applies is $(N-1)\bar{\rho}$, which grows with $N$ and is attained at $N = 2$. Pure equilibria exist under exposure-based congestion, and deciding whether one exists is strongly NP-complete under vehicle-count congestion [15, 16]; $\Gamma$-maximin beliefs reduce to a surcharge game. The discrete or algebraic cores of several results, including parts of the DIP, tariff, ARP and BRL statements, are checked in Lean 4 (Section 5.9).

(v) *Evidence* (Sections 6 and 7): nine metropolitan areas built from OpenStreetMap, gridded population (GHSL, epoch 2025) and web-sourced operating parameters with quality grades; an exact integer oracle; ten random draws (seeds) of the city slice for New Delhi, Buenos Aires and London and nine for the other cities, with bootstrap confidence intervals; results on algorithm performance, a public-instance benchmark and scale limits of BRL, an out-of-sample comparison of the p-box criterion with sample-average, Wasserstein and CVaR criteria, equilibria, charges, the price of anarchy and imprecision, reported against both the grid and the social optimum; and a robustness study of the game results under other carrier criteria. The instance pipeline is released as a benchmark generator for multi-carrier routing games.

*Scope.* The instances are stylised slices of nine cities with proxy demand, so the findings concern effects and orderings, not forecasts for a particular city. The levels of the price of anarchy, served share and leader gain describe carriers who plan against the guaranteed upper bound of the probability-box criterion; the orderings and the policy directions carry over to carriers who plan with sample averages or Wasserstein balls, the levels do not (Section 7.6). The leader's optimum is defined with optimistic equilibrium selection. The relation of each method to prior work is stated with the method and in Section 2.

*Organisation.* Section 2 places the work in the literature, Section 3 defines the model, Section 4 gives BRL, the equilibrium engine, DIP and ARP, Section 5 the equilibrium theory, Section 6 the data, and Section 7 the computational study. Sections 8 and 9 give implications and limits. Proofs are in the appendices.

## 2. Related work

*Competing carriers and leaders above carriers.* Carriers competing in sequential auctions for requests were studied by simulation by Figliozzi et al. [17, 18]. Stylised games treat shared collection points and parcel lockers [19, 20], and ride-hailing and crowdsourcing models give equilibrium evidence on competition for workers and customers, with queues instead of routes [21, 22, 23, 24, 25]. Cerulli et al. [26] place a platform above carriers who solve profitable tour problems, but the carriers do not interact. Catalano and Migliore [27] cast terminal design as a Stackelberg game. Curb management supplies pricing leaders [4, 28], and policy work documents the restrictions cities use [1, 2, 3]. Collaborative routing [29, 30, 31] gives savings that bound what cooperation could recover, not a price of anarchy.

*Routing under uncertainty.* Robust and distributionally robust routing keeps the combinatorial structure when the ambiguity set is suitable [32, 33, 34, 35, 36]. Closest to BRL, Ghosal et al. [37] (see also 38 for related distributionally robust bounds) show that a capacitated routing problem with ambiguous demand marginals and unknown dependence reduces, through a comonotone worst case, to a scenario problem solved by branch-and-cut. The comonotone worst case and the stop-loss decomposition of comonotone sums are due to Meilijson and Nádas [5] [see also 6, 39, 40], and worst-case marginals of a probability box under monotone losses are standard [41]. Branch-price-and-cut under mean-covariance travel times [42] and path dominance by second-order stochastic dominance for independent arc times [43] address other dependence models. None of these adds competition among carriers.

*Games, aggregation and ambiguity.* Potential and congestion games guarantee pure equilibria [44, 45], and selfish-routing theory bounds their inefficiency [46, 47, 48]. A carrier chooses tours, so its strategy set is finite and nonconvex. Liu et al. [14] prove, with a Shapley–Folkman argument in the tradition of Starr [49], that nonconvex aggregative games have approximate pure equilibria whose error does not grow with the number of players. Approximate pure equilibria in large games go back to Kalai [50] and Azrieli and Shmaya [51], and finding them can be PPAD-complete [52]. Equilibrium under ambiguous beliefs has been studied as equilibrium in beliefs [53, 54], as pure equilibrium in beliefs with ambiguous payoffs [55], and as robust-optimisation equilibrium [56, 57]; Crespi et al. [57] show that robust equilibria are approximate equilibria of the nominal game, the converse direction of our absorption result. Lanzetti et al. [58] and Feik et al. [59] protect against the worst case in a Wasserstein ball centred on the realised play; they obtain existence for convex or mixed games, whereas for finite plan sets the worst-case criterion has no pure equilibrium in general (our remark on $\Gamma$-maximin). Deciding pure-equilibrium existence is NP-complete in polymatrix and integer programming games [15, 60] and is hard for graphical games [16]; integer programming games are surveyed by Carvalho et al. [61].

*Tolls, implementation and adaptive pricing.* Corrective taxes for market power and externalities go back to Buchanan [7] and Barnett [62]. Toll polyhedra that implement a target flow are classical for nonatomic routing [8, 63], cost-balancing tolls exist for atomic games [64], and Bhaskar et al. [65] compute implementing tolls from an equilibrium oracle. The cutting-plane loop of DIP is an inverse-optimisation cutting plane [9, 66, 67], and best-response cuts are used for equilibria of integer programming games [68, 69, 70]. Implementing a target by incentives is studied by Monderer and Tennenholtz [71], and taxes in atomic congestion games by Caragiannis et al. [72]. Marginal-contribution tariffs make welfare the potential of the game [11, 73]. Trial-and-error implementation of marginal-cost pricing from observed flows is due to Yang et al. [10], with extensions by Han and Yang [74] and Ye et al. [75]; evolutionary implementation of pricing to Sandholm [12], and adaptive incentive design with learning agents, for atomic Cournot games with continuous strategies in the conference paper and for networked aggregative games and nonatomic routing in the journal paper, to Maheshwari et al. [13, 76]. The closest leader-side method for combinatorial games updates tolls by zeroth-order steps around an equilibrium solver, for nonatomic populations, with a freight tour example [77]; like a grid, it needs an equilibrium solve per outer step. Our novelty relative to these is stated in Sections 4.4 and 4.5. Our searches were web and index searches. Beauchêne [55]

and Liu–Ma–Ma (2026) were read only at abstract or metadata level, other neighbours at snippet level, and a Scopus and Web of Science search has not been done.

## 3. Model

Notation: $[n] = \{1, \dots, n\}$; for $v \in \mathbb{R}^d$ and $w \in \mathbb{R}^d_{>0}$, $\|v\|_{w,\infty} = \max_k |v_k|/w_k$ and $\|v\|_M^2 = \langle v, Mv\rangle$.

### *3.1. Carriers, plans and congestion*

The city is divided into zones $z \in [Z]$. There are $N$ carriers. Carrier $i$ has a finite plan set $\mathcal{P}_i$; a plan is a set of depot-to-depot routes over the carrier's customer zones. A plan $p$ determines deliveries $q_{iz}(p)$, vehicles $n_{iz}(p)$ and vehicle-hours $\tau_{iz}(p)$ in each zone, and a routing cost $C_i(p)$ (Section 3.2). Write $Q_z = \sum_i q_{iz}$ and $T_z = \sum_i \tau_{iz}$. Inverse demand in zone $z$ is $\alpha_z - b_z Q_z$ with $b_z > 0$, and a vehicle-hour costs $\gamma(\xi_z + c_z X_z)$, where $\gamma > 0$ is the hourly cost, $\xi_z$ the free-flow travel-time factor with mean $\mu_z$, $c_z > 0$ the congestion slope and $X_z$ the congestion variable. The city charges $s_z$ per delivery and $\theta_z$ per unit of the congestion variable. Carrier $i$'s profit is

$$u_i = \sum_z \left[q_{iz}(\alpha_z - b_z Q_z) - \gamma \tau_{iz}(\xi_z + c_z X_z) - s_z q_{iz} - \theta_z x_{iz}\right] - C_i(p). \tag{1}$$

Two congestion regimes are distinguished.

- **D1 (exposure-based):** $x_{iz} = \tau_{iz}$ and $X_z = \sum_i \tau_{iz}$.
- **D2 (vehicle-count):** $x_{iz} = n_{iz}$ and $X_z = \sum_i n_{iz}$.

Under D1 the carriers' interaction depends on the same quantity that determines the carrier's own congestion cost. Under D2 the externality depends on a different quantity (counts) than the carrier's own exposure (hours), which breaks the potential structure (Section 5).

For the theory it is convenient to write the interaction abstractly. Let $d = 2Z$ with coordinates $(z, Q)$ and $(z, X)$.

**Assumption A1** (Finite plans)**.** Plan $p \in \mathcal{P}_i$ has a contribution $a_i(p) \in \mathbb{R}^d_{\geq 0}$ to the aggregate, an exposure $\varphi_i(p) \in \mathbb{R}^d_{\geq 0}$ to the aggregate, a traffic exposure $t_i(p) \in \mathbb{R}^Z_{\geq 0}$ and a private profit $\pi_i(p)$. At a pure profile $\sigma$, $A(\sigma) = \sum_i a_i(\sigma_i)$ and $A_{-i} = A - a_i(\sigma_i)$.

**Assumption A2** (Affine coupling)**.** There are a diagonal $M \geq 0$ and $\gamma > 0$ such that, for a rival aggregate $m \in \mathbb{R}^d_{\geq 0}$ and traffic factor $\xi \in \mathbb{R}^Z_{\geq 0}$, carrier $i$ obtains $u_i(p; m, \xi) = \pi_i(p) - \langle \varphi_i(p), Mm\rangle - \gamma\langle t_i(p), \xi\rangle$.

Equation (1) has this form with $a_{i,(z,Q)} = \varphi_{i,(z,Q)} = q_{iz}$, $M_{(z,Q)} = b_z$, $a_{i,(z,X)} = x_{iz}$, $\varphi_{i,(z,X)} = \tau_{iz}$, $M_{(z,X)} = \gamma c_z$ and $t_{iz} = \tau_{iz}$. The game is *of type D1* if $\varphi_i = a_i$ for all carriers and plans, and *of type D2* otherwise. Social welfare, with charges treated as transfers, is

$$W(\sigma) = \sum_z \left[\alpha_z Q_z - \tfrac{b_z}{2} Q_z^2 - \gamma \mu_z T_z - \gamma c_z T_z X_z\right] - \sum_i C_i(\sigma_i), \tag{2}$$

that is, consumer surplus minus the congestion cost that carriers impose on each other, minus routing costs; in the experiments the traffic term is part of the routing cost and the social cost is $\mathrm{SC} = -W$. The affine coupling excludes power-law congestion such as BPR curves, which we linearise (Section 6).

*3.2. The routing layer*

For every arc $a$ the carrier knows a probability box: distribution functions $\underline{F}_a \le \overline{F}_a$ on a bounded support $[0, B_a]$, for example a Dvoretzky–Kiefer–Wolfowitz band around the empirical distribution of $n$ observed days, $\underline{F}_a(x) = \max\{F_n(x) - \varepsilon, 0\}$ with $\varepsilon = \sqrt{\ln(2/\alpha)/(2n)}$ [78, 79]. The dependence between arcs is unknown, so the joint credal set is the Fréchet class $\mathcal{M}$ of all laws whose $a$-th marginal lies between $\underline{F}_a$ and $\overline{F}_a$. A route of duration $T_r = \sum_{a \in r} X_a$ incurs a convex piecewise-linear loss

$$\phi(T) = c_0 + \gamma T + \sum_{j=1}^{H} \rho_j \, (T - \Theta_j)^+ , \qquad \gamma \ge 0, \ \rho_j > 0, \ 0 \le \Theta_1 < \cdots < \Theta_H. \tag{3}$$

In the experiments $H = 1$: a fixed cost, an hourly cost and an overtime penalty after the shift length. The robust route cost is the upper prevision $\overline{\mathbb{E}}[\phi(T_r)] = \sup_{P \in \mathcal{M}} \mathbb{E}_P \phi(T_r)$, and $C_i(p) = \sum_{r \in p} \overline{\mathbb{E}}[\phi(T_r)]$. A carrier's best response to prices on its deliveries and hours is therefore a prize-collecting routing problem in which every route is priced by $\overline{\mathbb{E}}[\phi]$. We call this problem the *carrier oracle*:

**Assumption A3** (Oracle)**.** For each $i$ there is an oracle that, for price vectors $\lambda^a, \lambda^\varphi \in \mathbb{R}^d$, $\nu \in \mathbb{R}^Z$ and a diagonal $Q \succeq 0$, returns a plan in $\arg\max_{p \in \mathcal{P}_i} \{\pi_i(p) + \frac{1}{2}\langle a_i(p), Q a_i(p)\rangle - \langle \lambda^a, a_i(p)\rangle - \langle \lambda^\varphi, \varphi_i(p)\rangle - \langle \nu, t_i(p)\rangle\}$.

A best response at the belief $(m, \xi)$ is the call $(0, Mm, \gamma\xi, 0)$. Section 4.1 gives an exact implementation of the oracle. We count *oracle calls* as the cost unit of every method below.

*3.3. The leader's problem*

The city chooses charges $u$ from an admissible set $\mathcal{U}$. For a charge $u$ let $\mathrm{NE}(u)$ be the set of pure Nash equilibria of the carriers' game. With optimistic selection the leader's problem is

$$\min_{u \in \mathcal{U}, \ \sigma} \ \mathrm{SC}(\sigma) \qquad \text{s.t.} \qquad \sigma \in \mathrm{NE}(u). \tag{4}$$

In the D1 instance used for computation (Section 4.4), each carrier serves its customer zones in all-or-nothing lots, and the charges reach the game through one number per zone. The three methods compared in Section 7 differ in what the leader may use: the *grid* and DIP use the carriers' routing oracle, ARP uses only the zone aggregates the market produces after each posted charge.

## 4. Algorithms

The algorithms form a stack. BRL (Section 4.1) and branch-and-price (Section 4.2) implement the carrier oracle exactly. An equilibrium engine (Section 4.3) calls the oracle to compute a pure equilibrium for a given charge. DIP (Section 4.4) and ARP (Section 4.5) choose the charges; the grid baseline calls the engine once per grid point.

*4.1. Budget-resource labelling (BRL)*

Let $\overline{Q}_a(u) = \inf\{x : \underline{F}_a(x) \geq u\}$ be the upper quantile of arc $a$, $\bar{x}_a = \int_0^1 \overline{Q}_a(u)\,du$ and $\overline{\Pi}_a(d) = \int_0^1 \big(\overline{Q}_a(u) - d\big)^+ du$, the stop-loss transform of the lower distribution function. $\overline{\Pi}_a$ is convex and nonincreasing with slopes in $[-1, 0]$.

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**Theorem 1** (Upper prevision and budget decomposition)**.** *For every route $r$,*

$$\overline{\mathbb{E}}[\phi(T_r)] = \int_0^1 \phi\Big(\sum_{a\in r} \overline{Q}_a(u)\Big)du = c_0 + \sum_{a\in r} \gamma \bar{x}_a + \sum_{j=1}^{H} \rho_j \min\Big\{\sum_{a\in r} \overline{\Pi}_a(d_a^j) : \sum_{a\in r} d_a^j \leq \Theta_j\Big\}.$$

*The supremum is attained by the comonotone coupling of the lower CDFs; each minimum is attained, at an integer vector when quantiles and thresholds are integers.*

The first equality and the stop-loss identity for comonotone sums with fixed marginals are due to Meilijson and Nádas [5] (see also 39, 40), and the choice of the lower CDF as worst marginal is standard [41]. We re-prove the statement in Appendix B because the algorithm relies on the inequality-budget form and on integrality. With a common quantile grid of $L$ levels (weights $\omega_l$), the route profile $S_{r,l} = \sum_{a\in r} q_{a,l}$ gives $\overline{\mathbb{E}}[\phi(T_r)] = c_0 + f_r(T_{\max})$ for $H = 1$, where $f_r(D) = \gamma \sum_l \omega_l S_{r,l} + \rho \sum_l \omega_l \,(S_{r,l} - D)^+$.

**Theorem 2** (Budget-resource reformulation)**.** *Let $y$ be dual prices and define the arc costs*

$$c_a(d^1, \ldots, d^H) = \gamma \bar{x}_a + \sum_j \rho_j \overline{\Pi}_a(d^j).$$

*The dependence-robust pricing problem*

$$\min_{r\in\mathcal{R}} \Big\{\overline{\mathbb{E}}[\phi(T_r)] - \sum_{i\in r} y_i\Big\}$$

*equals an elementary shortest path problem in which every arc chooses a consumption $d_a^j$ of each of $H$ budget resources, paid through the separable convex cost $c_a$ and subject to the budget constraints $\sum_{a\in r} d_a^j \leq \Theta_j$, $j \leq H$. The number of resources is $H$; it does not depend on $L$, the sample size or the number of arcs.*

A scenario view of the same problem, which is what the comonotone reduction of Ghosal et al. [37] gives, needs the $L$ level sums $(S_{p,l})_l$ as resources and compares labels pointwise. BRL instead compares labels through their cost functions, but only where a completion can still reach. For a partial path (label) $p$ ending at $v$ with level profile $S_p$, collected duals $y(p)$, visited set $\mathcal{V}(p)$ and load $\ell(p)$, concatenation with a completion $q$ is an inf-convolution, $f_{p\oplus q}(T_{\max}) = \min_D\{f_p(D) + f_q(T_{\max} - D)\}$ (Lemma 46). Let $C_{\min}(v)$ be a lower bound on the duration of any completion from $v$ and define the completion window $\mathcal{W}(p) = [\underline{D}(p), \overline{D}(p)]$ with $\overline{D}(p) = \min\{S_{p,L}, T_{\max} - C_{\min}(v)\}$ and $\underline{D}(p) = \min\{S_{p,1}, \overline{D}(p)\}$.

**Algorithm 1** BRL pricing for the dependence-robust routing problem ($H = 1$)

**Require:** graph, demands, capacity; level weights $\omega$; arc level profiles $q_{a,\cdot}$; $\gamma, \rho, T_{\max}, c_0$; duals $y$

$C_{\min}(v) \leftarrow$ shortest path from $v$ to the depot with arc lengths $q_{a,1}$
buckets $\mathcal{B}_v \leftarrow \emptyset$; queue $\leftarrow$ {depot label}
**while** queue nonempty **do**
    pop $p$; record the closed route $p \oplus (v, 0)$ with reduced cost $c_0 + f_{p\oplus(v,0)}(T_{\max}) - y(p)$
    **for** each customer $j \notin \mathcal{V}(p)$ that fits the capacity **do**
        $p' \leftarrow$ extension of $p$ to $j$ ($S_{p'} = S_p + q_{(v,j),\cdot}$, $y(p') = y(p) + y_j$)
        **if** some $o \in \mathcal{B}_j$ functionally dominates $p'$ **then** discard $p'$
        **else** delete the labels of $\mathcal{B}_j$ dominated by $p'$; insert and push $p'$
**return** recorded routes sorted by reduced cost

**Definition 3** (Functional dominance)**.** Label $p_1$ functionally dominates $p_2$, both ending at $v$, if $\mathcal{V}(p_1) \subseteq \mathcal{V}(p_2)$, $\ell(p_1) \le \ell(p_2)$ and $(f_{p_1}(D) - y(p_1)) - (f_{p_2}(D) - y(p_2)) \le 0$ for all $D \in \mathcal{W}(p_2)$.

The difference is piecewise linear with breakpoints at the level sums, so the test needs at most $2L + 2$ evaluations.

**Theorem 4** (Validity and exactness)**.** *Functional dominance implies that every completion of $p_2$ is a completion of $p_1$ with no larger reduced cost. The labelling algorithm that discards dominated labels (Algorithm 1) returns the minimum reduced cost exactly, with a route attaining it.*

The window is what makes the rule exact and strong at the same time. The slope of $f_p$ lies in $[-\rho, 0]$ and that of $D \mapsto f_q(T_{\max} - D)$ in $[0, \rho]$, so an optimal split never lies above $\overline{D}(p)$, and below $S_{p,1}$ the difference of two label functions is nondecreasing (Lemma 47). Quantile-vector dominance and sector dominance are special cases of functional dominance, and neither converse holds (Proposition 48). Under independent arc times concatenation is a convolution of laws rather than an inf-convolution and the method does not apply. The natural shortcut of fixing a single risk level for all routes gives a lower bound but is not exact (Proposition 49).

On the city instances the pricing runs over ng-routes [80]: the visited set is replaced by the ng-memory, and Theorem 4 holds verbatim, with the weaker ng-relaxation bound. Column generation [81, 82] with BRL pricing returns the exact LP bound of the dependence-robust routing problem when no negative reduced cost remains; an integer program over the generated columns (price-and-branch) gives a feasible plan but is not an exact integer method. Section 4.2 makes the integer step exact.

*Relation to prior work.* The upper-prevision formula and the stop-loss decomposition are due to Meilijson–Nádas and Tchen; Theorem 1 is re-proved in Appendix B because the algorithm uses its inequality-budget form and its integrality. The comonotone reduction for ambiguous marginals appears in routing in Ghosal et al. [37], which solves the demand-uncertain problem by branch-and-cut on scenarios. BRL contributes the per-arc budget as a decision resource of the pricing problem (Theorem 2) and the completion-window functional dominance with

its exactness proof (Theorem 4); on the metro instances the window keeps label sets 1.8–2.9 times smaller than the same comparison without it (Section 7.1). Resource-extension-function dominance [83, 84] is the closest labelling technique; a direct comparison is left for future work.

*4.2. Branch-and-price with BRL pricing*

Price-and-branch, an integer program over the columns of the root relaxation, can miss the integer optimum. We therefore embed the BRL pricer in branch-and-price. The master is a set-partitioning model for the routing benchmark, or a prize-collecting set-packing model for the carrier oracle, with one column pool shared by all nodes. Each node solves its LP by column generation with the exact BRL pricer, restricted to the node's allowed arcs with $C_{\min}$ recomputed on them; functional dominance remains valid because the restrictions are local.

- *Branching* is on the aggregated arc flow $f_{ij}$: $f_{ij} = 0$ deletes the arc; $f_{ij} = 1$ deletes the other arcs leaving $i$ and entering $j$. Integral arc flows imply an integral plan.
- *Fleet size.* When every customer must be served, a row for the minimum number of routes is added at the root and a fractional route count is branched on first.
- *ng-routes* [80]. The visited set is replaced by the ng-memory, and Theorem 4 holds verbatim with the weaker relaxation. If a node solution contains a non-elementary column, the ng-sets along its cycle are enlarged; reported plans are elementary.
- *Bounds.* The node LP value and the Lagrangian bound during column generation are valid lower bounds, marked invalid if a pricing call hits the label limit. Upper bounds come from integral nodes and from an integer program over the generated elementary columns. Nodes are chosen by best bound; the method stops at a relative gap of $10^{-6}$.

Since ng-routes relax elementary routes, a zero gap proves optimality for the elementary problem. This is standard branch-and-price machinery; its purpose here is to remove the inexact integer step, not to add a branching rule. Column generation [81, 82] with exact pricing returns the exact LP bound, which also gives every integer plan a valid optimality gap.

*4.3. The equilibrium engine*

For a given charge, a pure equilibrium of the D1 game is computed in two phases. In the first phase, zone-market clearing (ZMC) queries every carrier's oracle at a common zone-price vector for eight rounds, using the sign of the clearing residual to update the prices (Appendix C). A restricted master over the collected plans and derandomised rounding give a start profile. In the second phase, sequential best responses (one oracle call each) run until no carrier improves; since D1 has an exact potential (Proposition 26) this terminates at a pure Nash equilibrium. Each carrier's column pool is kept across calls, because route costs do not depend on prices. In the city experiments a carrier chooses node-disjoint routes from its depot over 22 customer zones; each served zone adds its whole lot (all-or-nothing) together with prize, own Cournot and congestion terms, charges and prices as node costs, and each route costs its BRL upper expectation.

*Remark* 5 (ZMC and its negative result). ZMC has a clearing identity that bounds the potential gap and each carrier's regret by the clearing error, and a price-bracketing rate that does not depend on $d$ (Appendix C). On synthetic games with non-separable plan sets, it gave the smallest potential gap in 17 of 20 instances at an equal call budget (Table F.24). On the nine city games it did not beat plain best response: it used 36–65 calls, against 8–17 for best response, and was strictly better than all other methods in 0 cities (Section 7.3). We use it as the starting phase of the engine and make no claim for it.

### *4.4. Dual implementation pricing (DIP)*

A direct approach searches a grid of charges and solves one equilibrium per grid point; with 63 points this costs between 1 921 and 2 419 oracle calls per city and seed in the three cities where the exact grid is complete (Section 7.5.1). DIP replaces the search by an implementation step. It computes the profile the city wants and then computes charges under which that profile is an equilibrium, using the carriers' own oracle to separate candidate charges. No equilibrium is solved for the leader's decision.

*Setting.* We use the D1 instance of Section 3 in cost form. Carrier $i$ serves a subset of its customer zones with all-or-nothing lots $d_{iz} > 0$, so $q_{iz} \in \{0, d_{iz}\}$ and $q_{iz}^2 = d_{iz} q_{iz}$; vehicle-hours are $\tau_{iz} = r\, q_{iz}$ for one ratio $r > 0$ (assumption (S3)); $R_i(p) \geq 0$ is the robust routing cost of plan $p$; $g_z = \gamma c_z$. The city posts $s_z$ per delivery and $t_z$ per vehicle-hour. Carrier $i$'s cost and the social cost (charges are transfers) are

$$J_i(p; \sigma_{-i}) = R_i(p) + \sum_z \big[ q_{iz}(-\alpha_z + b_z(q_{iz} + Q_{-i,z}) + s_z) + \tau_{iz}(g_z(\tau_{iz} + X_{-i,z}) + t_z) \big],$$

$$\mathrm{SC}(\sigma) = \sum_i R_i(\sigma_i) + \sum_z \big[ -\alpha_z Q_z + \tfrac{b_z}{2} Q_z^2 + g_z X_z^2 \big].$$

Write $\kappa_z = g_z r^2$ (congestion slope in delivery units), $m_z = b_z + \kappa_z$ (the slope the carriers face) and $\bar{d}_z$ for the mean lot in zone $z$.

**Proposition 6** (One instrument per zone)**.** *(a) For every plan, $\sum_z (s_z q_{iz} + t_z \tau_{iz}) = \langle u, q_i \rangle$ with $u_z = s_z + r t_z$, so the game depends on the charges only through $u \in \mathbb{R}^Z$. (b) Every $u$ is realised by a uniform per-delivery charge $s = \min_z u_z$ over zones with lots and non-negative zone congestion charges $t_z = (u_z - s)/r$. (c) The grid of Section 7.5 is the two-parameter family $u = s\mathbf{1} + \vartheta\, r\, g \odot X^{\mathrm{NE}}$.*

Under $u$ the game has the exact potential $\Phi_u(\sigma) = \sum_i R_i(\sigma_i) + \sum_z [-\alpha_z Q_z + \frac{m_z}{2}(Q_z^2 + \sum_i q_{iz}^2) + u_z Q_z]$ (Proposition 26).

**Lemma 7** (Potential–welfare identity)**.** *Fix $Q^\circ \in \mathbb{R}^Z$, let $u_z^{\tan} = \kappa_z Q_z^\circ - \frac{m_z}{2} \bar{d}_z$ and $v = u - u^{\tan}$. For every pure profile $\sigma$,*

$$\Phi_u(\sigma) = \mathrm{SC}(\sigma) - \sum_z \tfrac{\kappa_z}{2}(Q_z - Q_z^\circ)^2 + L(\sigma) + \langle v, Q \rangle + \sum_z \tfrac{\kappa_z}{2}(Q_z^\circ)^2, \qquad L(\sigma) = \sum_{i,z} \tfrac{m_z}{2}(d_{iz} - \bar{d}_z) q_{iz}.$$

**Algorithm 2** DIP: dual implementation pricing

**Target.** Column generation on the convexified social cost (the carriers' oracle at social prices), rounding and coordinate polish $\Rightarrow$ pure profile $\sigma^*$ with aggregate $Q^*$
**Reference.** $u^{\mathrm{ref}} \leftarrow \kappa \odot Q^* - \frac{m}{2} \odot \bar{d}$ (tangent charge); cuts $\mathcal{C} \leftarrow \emptyset$; $u \leftarrow u^{\mathrm{ref}}$
**repeat**
    for each $i$: $b_i \leftarrow$ best response to $\sigma^*_{-i}$ under $u \triangleright N$ routing calls, no equilibrium solve
    for each $i$ with $J^u_i(b_i) < J^u_i(\sigma^*_i)$: add $\langle q_i(b_i) - q_i(\sigma^*_i), u\rangle \geq J^0_i(\sigma^*_i) - J^0_i(b_i) + \varepsilon$ to $\mathcal{C}$
    **if** no cut was added: **return** $u$ $\quad\triangleright\ \sigma^*$ is an exact pure equilibrium under $u$
    $u \leftarrow \arg\min\{\|u - u^{\mathrm{ref}}\|_1 : u \in \mathcal{U} \text{ satisfies } \mathcal{C}\}$ (LP); **if** infeasible: **break**
**until**
**Fallback** (only if $P(\sigma^*) = \emptyset$): best-response dynamics from $\sigma^*$ under the least-violating charge and under $u^{\mathrm{ref}}$; keep the better equilibrium
**Instruments.** $s = \min_z u_z$, $t_z = (u_z - s)/r$ (Proposition 6)

The *tangent charge* $u^{\tan}$ is the classical corrective tax of a Cournot market with an externality: the marginal external damage at the target aggregate minus a market-power subsidy [7, 62]. Two details are specific to all-or-nothing lots: the subsidy is half a lot, $\frac{m_z}{2}\bar{d}_z$ per delivery, and it includes the carriers' own congestion. With carrier-specific charges $U_{iz} = \kappa_z Q^*_{-i,z} - \frac{b_z}{2} d_{iz}$ every carrier's cost differences equal the social-cost differences, so every social optimum is an equilibrium; this is the classical marginal-contribution charge [11, 73] (Proposition 56 in Appendix D). A city cannot charge per carrier, and the next results measure what flat zone charges lose. In the convexified (mixed) game let $x^*$ minimise the convexified social cost $\overline{\mathrm{SC}}$, with aggregate $\bar{Q}^*$.

**Theorem 8** (Loss under asymmetric lots)**.** *Let $u = \kappa \odot \bar{Q}^* - \frac{m}{2} \odot \bar{d}$. Every convexified equilibrium $x^u$ satisfies $0 \leq \overline{\mathrm{SC}}(x^u) - \overline{\mathrm{SC}}(x^*) \leq (1+\varrho)E$, where $\varrho = \max_z \kappa_z/(b_z + \kappa_z) < 1$ and $E = \sum_{i,z} \frac{m_z}{2}|d_{iz} - \bar{d}_z|\, d_{iz}$ measures how unequal the lots in a zone are.*

**Theorem 9** (Exact implementation with symmetric lots)**.** *If all lots in a zone are equal, the tangent charge makes $x^*$ a convexified equilibrium, and every convexified equilibrium under it has aggregate $\bar{Q}^*$ and social cost $\overline{\mathrm{SC}}(x^*)$.*

*Sketch.* The identity holds for the convexified potential and social cost with the same algebra. Because $\overline{\mathrm{SC}}$ is linear plus a quadratic in $\bar{Q}$, its expansion around $x^*$ is exact; comparing it with the identity at a minimiser of the potential bounds the aggregate deviation by the lot-asymmetry term $L$, and $|L| \leq E$. With equal lots $L \equiv 0$.

*The pure game.* For a pure target $\sigma^*$ and an admissible box $\mathcal{U}$ of charges let

$$P(\sigma^*) = \big\{u \in \mathcal{U} :\ \langle q_i(p) - q_i(\sigma^*_i), u\rangle \geq J^0_i(\sigma^*_i; \sigma^*_{-i}) - J^0_i(p; \sigma^*_{-i})\ \ \forall i,\ \forall p \in \mathcal{P}_i\big\},$$

with $J^0_i$ the cost at zero charges. Since $J^u_i = J^0_i + \langle u, q_i\rangle$, $P(\sigma^*)$ is exactly the set of charges under which $\sigma^*$ is a pure Nash equilibrium: a polyhedron, possibly empty. Toll polyhedra of this kind are classical for nonatomic routing [8, 63] and inverse optimisation [9].

**Theorem 10** (Separation and finite termination)**.** *(a) $u \in P(\sigma^*)$ if and only if no carrier has a strictly improving best response to $\sigma^*_{-i}$ under $u$; an improving best response gives that carrier's*

*most violated constraint. (b) With $\varepsilon = 0$, Algorithm 2 stops after at most $1 + \sum_i (|\mathcal{P}_i| - 1)$ rounds, either with $u \in P(\sigma^*)$ or with an infeasible LP, which happens only if $P(\sigma^*) = \emptyset$. Each round costs $N$ oracle calls and no equilibrium solve.*

**Proposition 11** (Single-level form)**.** *With optimistic selection, the leader's value is $V^\star = \min\{\mathrm{SC}(\sigma) : \sigma \text{ pure},\ P(\sigma) \neq \emptyset\} \geq \min_\sigma \mathrm{SC}(\sigma)$. If Algorithm 2 ends in its main loop, the leader's loss is at most the suboptimality of the target, $\mathrm{SC}(\sigma^*) - \min_\sigma \mathrm{SC}(\sigma)$; in particular DIP is at least as good as any grid over any subfamily of charges when its target is a social optimum.*

**Theorem 12** (Loss of the fallback)**.** *For any charge $u$ with $v = u - u^{\tan}$ and $Q^\circ = Q^*$, the equilibrium $\sigma^u$ reached from $\sigma^*$ by improving moves satisfies $\mathrm{SC}(\sigma^u) - \mathrm{SC}(\sigma^*) \leq \sum_z \frac{\kappa_z}{2}(Q^u_z - Q^*_z)^2 + L(\sigma^*) - L(\sigma^u) + \langle v, Q^* - Q^u \rangle$.*

**Corollary 13** (Oracle calls)**.** *DIP uses the calls of the social target, $N$ calls per cutting-plane round, and, only if $P(\sigma^*)$ is empty, the calls of the fallback dynamics. A grid of $G$ points with an equilibrium solve of $C_{\mathrm{eq}}$ calls per point uses $GC_{\mathrm{eq}}$.*

**Theorem 14** (Imprecision shifts the charge)**.** *Under $\Gamma$-maximin credal boxes of width $\delta$ (Lemma 25), the charges that implement $\sigma^*$ are $P(\sigma^*) - \hat{s}(\delta)$ with $\hat{s}_z(\delta) = \delta(b_z w_{(z,Q)} + r\, g_z w_{(z,X)})$. One run of DIP therefore gives $u(\delta) = u(0) - \hat{s}(\delta)$ for every width at no further oracle cost, and the leader attains the same social cost at every width as long as $u(\delta)$ is admissible. For E-admissibility, maximality and interval dominance, $u(0)$ works at every width.*

**Proposition 15** (When one instrument suffices)**.** *A uniform per-delivery charge implements the convexified optimum under equal lots if the tangent charge is the same in every active zone. With the calibration of Section 6 ($\alpha_z = 1.5\,$fee, $b_z = \alpha_z/(2Q^{\mathrm{full}}_z)$) and $n_z$ carriers with equal lots in zone $z$, the congestion-free tangent charge is $u_z = -0.375\,\mathrm{fee}/n_z$: a subsidy of 37.5% of the fee in single-carrier zones and 19% in two-carrier zones. The congestion charge contributes only spatial differentiation.*

Proofs are in Appendix D. Theorem 10(a) is what makes the method cheap: the separation oracle is one best response per carrier at the fixed target, not an equilibrium solve. Selection remains optimistic: the target is an exact equilibrium under the returned charges, but other equilibria may coexist.

*Relation to prior work.* DIP combines the Pigou–Cournot corrective tax [7, 62], marginal-contribution utilities [11, 73], toll polyhedra for implementing a target [8, 63, 64, 65], and an inverse-optimisation cutting plane [9, 66, 67] with best-response cuts as used for integer programming games [68]; implementation of a target by incentives and taxes in atomic congestion games is studied by Monderer and Tennenholtz [71] and Caragiannis et al. [72]. The separation step of Theorem 10 is one best response per carrier at the fixed target, so no equilibrium is solved, whereas the closest leader-side method [77] solves one per outer step. New here are the exact potential–welfare identity of the lumpy Cournot–congestion game with loss bounds in the lot-asymmetry number $E$, the translation of the implementing charges by the $\Gamma$-maximin surcharge, and the competitor-count rule.

**Algorithm 3** ARP flat rule

**Require:** slopes $b, g, r$; damping $\theta$

1: $u \leftarrow 0$; observe $(Q, n)$ of the unpriced equilibrium
2: **repeat**
3: update the lot estimates $\bar{d}_z$ from $(Q, n)$ and the previous observations
4: $u_z \leftarrow (1-\theta)u_z + \theta[\kappa_z Q_z - \frac{m_z}{2}\bar{d}_z]$ for all $z$
5: post $u$; the carriers best-respond from their current plans until nobody moves; observe $(Q, n)$
6: **until** $u = \kappa Q - \frac{m}{2}\bar{d}$ (fixed point) or the posted charge and the market state repeat (cycle)

*4.5. Aggregate-response pricing (ARP)*

DIP and the grid need the carriers' routing oracle, which a real city does not have. ARP asks how much of the leader's gain is reachable from what the city can observe. The city knows its own demand and congestion slopes $(b, g, r)$ and hence $\kappa_z = g_z r^2$ and $m_z = b_z + \kappa_z$. It posts a per-delivery-equivalent charge $u \in \mathbb{R}^Z$ (realised, as in Proposition 6, by a uniform per-delivery charge plus non-negative zone congestion charges), the carriers settle in a pure equilibrium by their own best responses, and the city observes the deliveries $Q_z$ and the number $n_z$ of active carriers in each zone, nothing else.

*The flat rule.* With an estimate $\bar{d}_z$ of the mean lot, the city updates

$$u_z^{t+1} = (1-\theta_z)u_z^t + \theta_z\big[\kappa_z Q_z^t - \tfrac{m_z}{2}\bar{d}_z\big], \qquad \theta_z \in (0, 1]. \tag{5}$$

With $\theta = 1$ this is DIP's tangent charge evaluated at the observed aggregate instead of a computed target. The lot estimate is $Q_z/n_z$; when $n_z$ changes by one between observations, the jump of $Q_z$ is the lot of the carrier that entered or left (Lemma 59); for a zone never served nothing is learned and the city-wide mean observed lot is used.

*The marginal tariff (ARP-M).* The flat rule charges each delivery the externality at the observed aggregate. The exact externality of carrier $i$'s lot on the others is $\kappa_z d_{iz} Q_{-i,z}$, and its market-power wedge is $\frac{b_z}{2}d_{iz}^2$; both are functions of the carrier's own deliveries and of the zone total, so an anonymous nonlinear tariff can bill them exactly, in the limit of re-tangenting at every transaction.

**Theorem 16** (Marginal tariff; the mechanism is classical)**.** *Let every carrier pay, in every zone, $T_z(q_{iz}, Q_z) = \kappa_z q_{iz}(Q_z - q_{iz}) - \frac{b_z}{2}q_{iz}^2$. Then (a) $J_i^T(p; \sigma_{-i}) = \mathrm{SC}(p, \sigma_{-i}) - \mathrm{SC}(\emptyset_i, \sigma_{-i})$ for all $i$, $p$ and $\sigma_{-i}$, so* SC *is an exact potential of the tariff game; (b) the pure equilibria are exactly the profiles that no single carrier can improve socially, every social optimum is a pure equilibrium, and best-response dynamics from any profile never increase* SC *and stop at such an equilibrium; (c) given the rivals, the tariff is linear in the carrier's own service indicators, so a best response is one oracle call with the zone prices shifted by $\kappa_z Q_{-i,z} - \frac{b_z}{2}d_{iz}$.*

The mechanism is the marginal-contribution tariff of Wolpert and Tumer [11], Sandholm [12] and Marden and Wierman [73]; what is stated here is its lumpy form in the city game and the consequence that DIP's implementation step is unnecessary under it. The tariff needs billing on own deliveries and the zone total. It leaves equilibrium selection open: the carriers'

dynamics stop at a unilateral social optimum, which may be a local optimum of SC. In the experiments we run it with the carriers' own best responses from the status quo (tariff+BR) and with the potential-minimising solver of the engine on the social-cost potential (tariff+SO).

*Theory of the flat rule.* Convexify the plan sets: let $\mathcal{X} = \prod_i \Delta(\mathcal{P}_i)$, $\bar{q}_i(x_i) = \mathbb{E}_{x_i} q_i$, $\bar{Q}(x) = \sum_i \bar{q}_i$ and $\mathcal{Q} = \bar{Q}(\mathcal{X})$. With a reference lot vector $\bar{d}$ define the carriers' aggregate value function $G(Q) = \min\{\sum_i \mathbb{E}_{x_i}[R_i + \sum_z \frac{m_z}{2}(d_{iz} - \bar{d}_z)q_{iz}] : \bar{Q}(x) = Q\}$ and $F(Q) = G(Q) + \sum_z[-\alpha_z Q_z + \frac{m_z}{2}Q_z^2 + \frac{m_z}{2}\bar{d}_z Q_z] + \iota_{\mathcal{Q}}(Q)$, where $R_i$ is the routing cost. Let $L_{\bar{d}}(\sigma) = \sum_{i,z} \frac{m_z}{2}(d_{iz} - \bar{d}_z)q_{iz}$ be the lot-asymmetry term of Section 4.4 with reference lots $\bar{d}$, $E_{\bar{d}} = \sum_{i,z} \frac{m_z}{2}|d_{iz} - \bar{d}_z|d_{iz}$, and $g(Q) = \sum_z[\frac{\kappa_z}{2}Q_z^2 - \frac{m_z}{2}\bar{d}_z Q_z]$. All statements below are proved in Appendix E; $\|v\|_w^2 = \sum_z w_z v_z^2$.

**Theorem 17** (The response is a proximal map)**.** *$F$ is 1-strongly convex in $\|\cdot\|_m$. For every charge $u$, all convexified equilibria share one aggregate $Q(u) = \arg\min_Q\{F(Q) + \langle u, Q\rangle\} = \nabla F^*(-u)$, and $\|Q(u_1) - Q(u_2)\|_m \le \|u_1 - u_2\|_{m^{-1}}$. The leader's problem $\min \overline{\mathrm{SC}} + L_{\bar{d}}$ is the Fenchel dual $\min_u D(u)$, $D(u) = F^*(-u) + \sum_z (u_z + \frac{m_z}{2}\bar{d}_z)^2/(2\kappa_z)$, whose gradient $\kappa^{-1} \odot (u + \frac{m}{2}\bar{d}) - Q(u)$ is observable from the aggregate alone. Its unique minimiser $u^*$ is the unique fixed point of $u \mapsto \kappa \odot Q(u) - \frac{m}{2} \odot \bar{d}$. With symmetric lots within zones the fixed point implements a convexified social optimum; in general it is within $E_{\bar{d}}$ of one.*

**Theorem 18** (Instance-free contraction)**.** *Let $\varrho = \max_z \kappa_z/m_z < 1$ over zones with a lot. In the convexified game, (a) the plain rule ($\theta = 1$) satisfies $\|u^{t+1} - u^*\|_{\kappa^{-1}} \le \varrho\|u^t - u^*\|_{\kappa^{-1}}$; (b) the damped rule with $\theta = 2/(2+\varrho)$ contracts at ratio $\varrho/(2+\varrho) < 1/3$. The ratios depend on the instance only through $\varrho$, not on $Z$, $N$, the number of customers, the lots, the routing costs or the probability boxes.*

The rate comes from the Cournot slope $b_z$: without the demand side ($b = 0$), $\varrho = 1$ and there is no contraction, which is why the nonatomic literature uses averaging [10]. The flat rule is gradient descent on $D$ in the $\kappa^{-1}$ geometry; this identification, and the resulting rate in a lumpy NP-hard routing game, is the new part.

**Theorem 19** (Pure game and identification)**.** *(a) If $\sigma$ is a pure equilibrium under $u = \kappa \odot Q(\sigma) - \frac{m}{2} \odot \bar{d} + v$, then for every carrier $i$ and plan $p$, with $\sigma' = (p, \sigma_{-i})$ and $\Delta Q = Q(\sigma') - Q(\sigma)$, $\mathrm{SC}(\sigma') \ge \mathrm{SC}(\sigma) + [L_{\bar{d}}(\sigma) - L_{\bar{d}}(\sigma')] + \sum_z \frac{\kappa_z}{2}\Delta Q_z^2 - \langle v, \Delta Q\rangle$: at an exact fixed point with symmetric lots, no carrier can lower the social cost unilaterally. (b) If the observed aggregate $Q^t$ differs from the convexified response by $e^t$, then $\|u^{t+1} - u^*\|_{\kappa^{-1}} \le \varrho\|u^t - u^*\|_{\kappa^{-1}} + \|e^t\|_\kappa$, hence $\limsup \|u^t - u^*\|_{\kappa^{-1}} \le \sup \|e^t\|_\kappa/(1-\varrho)$; if the carriers settle in potential minimisers, $\|e^t\|_\kappa \le \sqrt{2\varrho\eta_t}$ with $\eta_t$ the rounding gap of the potential. (c) The convexified response map determines $F$ up to a constant, so the convexified optimum is all that an observation-only scheme can target; the pure social optimum is not identified (Example 58).*

**Proposition 20** (Imprecise carriers)**.** *Under $\Gamma$-maximin beliefs of width $\delta$ the response to $u$ is the precise response to $u + \hat{s}(\delta)$ (Lemma 25), so the flat rule needs no knowledge of $\delta$ and contracts at the same rates; it only moves the fixed point, and the loss in $\overline{\mathrm{SC}} + L_{\bar{d}}$ against the precise fixed point is at most $\frac{1}{2}\|\hat{s}(\delta)\|_{m^{-1}}^2$. Under E-admissibility of width $\delta$, a pure target is implemented exactly once the posted charge is within the surcharge band $|u_z - u_z^*| \le \hat{s}_z(\delta)$ of an implementing charge (with a side condition on rival volumes, Appendix E), after at most $\lceil \log(\cdot)/\log(1/\varrho) \rceil$ rounds.*

*Relation to prior work.* ARP follows trial-and-error marginal-cost pricing [10, 74, 75], evolutionary implementation of pricing [12] and adaptive incentive design [13, 76], whose fixed points are socially optimal, together with marginal-contribution tariffs. New here are the identification of the equilibrium response as a proximal map with instance-free rates in the lumpy routing game (Theorems 17 and 18), the fixed-point and identification results for the pure game (Theorem 19; part (b) is conditional on potential minimisation), and the absorption of the leader's error by the carriers' credal width (Proposition 20).

## 5. Equilibrium theory

This section states the results on pure and imprecise equilibria; complete proofs are in Appendix A. Results labelled *classical* are known and cited.

### *5.1. Imprecise beliefs*

**Assumption A4** (Traffic box)**.** $\Xi = \prod_z [\underline{\mu}_z, \overline{\mu}_z] \subset \mathbb{R}^Z_{\geq 0}$ with widths $\varpi_z = \overline{\mu}_z - \underline{\mu}_z$; a nominal mean $\hat{\mu} \in \Xi$ is fixed and the true mean $\mu$ lies in $\Xi$.

**Assumption A5** (Credal sets, clipped and consistent)**.** Fix weights $w \in \mathbb{R}^d_{>0}$ and a width $\delta \geq 0$. For $c \in \mathbb{R}^d_{\geq 0}$ let $K^\delta(c) = \{m \in \mathbb{R}^d_{\geq 0} : |m_k - c_k| \leq \delta w_k \ \forall k\}$. At a pure profile $\sigma$, carrier $i$'s credal set $\mathcal{M}_i^\delta(\sigma)$ is the set of Borel probability laws $P$ on $\mathbb{R}^d_{\geq 0} \times \mathbb{R}^Z_{\geq 0}$ with finite first moments such that $\mathbb{E}_P[m] \in K^\delta(A_{-i}(\sigma))$ and $\mathbb{E}_P[\xi] \in \Xi$.

The box is centred at the realised rival aggregate, which is the consistency requirement of equilibrium in beliefs [53], and it is clipped at zero, so every belief is a feasible aggregate. The width $\delta$ measures how much a carrier is unsure about its rivals' activity, in units $w_k$.

**Definition 21** (Decision criteria; 85)**.** For a finite option set $\mathcal{P}$ and a set $\mathcal{M}$ of laws, with $\underline{\mathbb{E}} = \inf_{P \in \mathcal{M}} \mathbb{E}_P$ and $\overline{\mathbb{E}} = \sup_{P \in \mathcal{M}} \mathbb{E}_P$, an option $p$ is *E-admissible* if it maximises $\mathbb{E}_P[u(\cdot)]$ for some $P \in \mathcal{M}$; *maximal* if no $p'$ has $\underline{\mathbb{E}}[u(p') - u(p)] > 0$; *interval-undominated* (ID) if $\overline{\mathbb{E}}[u(p)] \geq \max_{p'} \underline{\mathbb{E}}[u(p')]$; and $\Gamma$-*maximin* if $\underline{\mathbb{E}}[u(p)] = \max_{p'} \underline{\mathbb{E}}[u(p')]$.

**Definition 22** (Imprecise credal Nash equilibrium)**.** For $C \in \{E, M, ID, \Gamma\}$ and $\delta \geq 0$, $\mathrm{ICNE}_C(\delta)$ is the set of pure profiles $\sigma$ in which every $\sigma_i$ is $C$-admissible for $(\mathcal{P}_i, \mathcal{M}_i^\delta(\sigma))$ with payoff $u_i$. The existence threshold is $\delta_0^C = \inf\{\delta \geq 0 : \mathrm{ICNE}_C(\delta) \neq \emptyset\}$.

### *5.2. Reduction to mean boxes and the criteria*

Write $\bar{u}_i(p; m, \xi)$ for the payoff of Assumption A2 evaluated at the means $(m, \xi)$ and $K_i^\delta(\sigma) = K^\delta(A_{-i}(\sigma)) \times \Xi$.

**Lemma 23** (Reduction of the credal set to its mean box)**.** *Under Assumptions A1–A5: (a) for every $P \in \mathcal{M}_i^\delta(\sigma)$ and every $p$, $\mathbb{E}_P[u_i(p)] = \bar{u}_i(p; \mathbb{E}_P m, \mathbb{E}_P \xi)$; (b) $P \mapsto (\mathbb{E}_P m, \mathbb{E}_P \xi)$ maps $\mathcal{M}_i^\delta(\sigma)$ onto $K_i^\delta(\sigma)$; (c) hence each criterion of Definition 21 coincides with the same criterion evaluated over the compact box $K_i^\delta(\sigma)$, and infima and suprema are attained.*

**Lemma 24** (Nesting and monotonicity)**.** *For every $\delta \geq 0$,*

$$\mathrm{ICNE}_E(\delta) \subseteq \mathrm{ICNE}_M(\delta) \subseteq \mathrm{ICNE}_{ID}(\delta) \quad \textit{and} \quad \mathrm{ICNE}_\Gamma(\delta) \subseteq \mathrm{ICNE}_M(\delta).$$

*For $C \in \{E, M, ID\}$ and $\delta \leq \delta'$, $\mathrm{ICNE}_C(\delta) \subseteq \mathrm{ICNE}_C(\delta')$. For $C = \Gamma$ the monotonicity fails (Example 40 in Appendix A).*

The inclusions between criteria are due to Troffaes [85]. Monotonicity in $\delta$ says that more imprecision can only enlarge the set of E-admissible, maximal or interval-undominated equilibria; $\Gamma$-maximin behaves differently because the worst case moves with $\delta$.

**Lemma 25** ($\Gamma$-maximin surcharge)**.** *If $\varphi_i(p), t_i(p) \geq 0$, the lower expectation over $K^\delta(c) \times \Xi$ is attained, for every plan simultaneously, at the upper corner $(c + \delta w, \overline{\mu})$ and is linear in mixtures. Consequently* $\mathrm{ICNE}_\Gamma(\delta)$ *is the set of pure Nash equilibria of the precise game $G^\Gamma_\delta$ with private profits*

$$\pi_i^\Gamma(p) = \pi_i(p) - \delta\langle \varphi_i(p), Mw\rangle - \gamma\langle t_i(p), \overline{\mu}\rangle,$$

*the same coupling, and traffic set to zero.*

The surcharge $\delta\langle \varphi_i(p), Mw\rangle$ is what a cautious carrier adds to the cost of activity in zone $k$ because it fears up to $\delta w_k$ more rival activity than it observes. Section 7 uses it to compute $\Gamma$-maximin equilibria of the city games.

*5.3. Existence under exposure-based congestion*

**Proposition 26** (Existence under D1 at every width)**.** *Under D1, for every $\hat{\xi} \in \Xi$,*

$$\Psi_{\hat{\xi}}(\sigma) = \textstyle\sum_i \left[\pi_i(\sigma_i) + \frac{1}{2}\|a_i(\sigma_i)\|_M^2 - \gamma\langle t_i(\sigma_i), \hat{\xi}\rangle\right] - \frac{1}{2}\|A(\sigma)\|_M^2$$

*is an exact potential of the precise game at traffic $\hat{\xi}$. Every maximiser lies in* $\mathrm{ICNE}_E(\delta)$ *for all $\delta \geq 0$, so $\delta_0^E = \delta_0^M = \delta_0^{ID} = 0$, and* $\mathrm{ICNE}_\Gamma(\delta) \neq \emptyset$ *for all $\delta \geq 0$.*

*Sketch.* For a unilateral change of carrier $i$ from contribution $a$ to $a'$ with rival aggregate $R$, the quadratic terms give $\frac{1}{2}\|a'\|_M^2 - \frac{1}{2}\|a\|_M^2 - \frac{1}{2}\|R + a'\|_M^2 + \frac{1}{2}\|R + a\|_M^2 = -\langle a' - a, MR\rangle$, which equals the change of the coupling term when $\varphi_i = a_i$. A maximiser of an exact potential is a pure equilibrium [45], and the centre of every credal box is the realised aggregate. For $\Gamma$, $G^\Gamma_\delta$ is again of type D1.

*5.4. Credal absorption*

Let $x^*$ be a mixed equilibrium of the precise game at traffic $\hat{\xi} \in \Xi$ (it exists by 86; under affine coupling, mixed equilibria are exactly the profiles in which every plan in the support is a best response to the mean rival aggregate, Proposition 41). Write $B_j$ for carrier $j$'s best-response set at $x^*$, $\bar{a}_j = \sum_p x_j^*(p) a_j(p)$, $\bar{A}^*_{-i} = \sum_{j \neq i} \bar{a}_j$, and define the spreads $s_{jk} = \max_{p,p' \in B_j} |a_{jk}(p) - a_{jk}(p')|$ and $r_{jk} = \max_{p \in B_j} |a_{jk}(p) - \bar{a}_{jk}|$, with $\bar{\sigma} = \max_{j,k} s_{jk}/w_k$, $\bar{\rho} = \max_{j,k} r_{jk}/w_k$, $d_L = \dim \operatorname{span} \bigcup_j (a_j(B_j) - a_j(B_j))$ and $k^\sharp = \min\{N, d_L\}$. Let $\sigma_{\max} = \max_{i,k} \operatorname{range}_{p \in \mathcal{P}_i} a_{ik}(p)/w_k$, so that $\bar{\rho} \leq \bar{\sigma} \leq \sigma_{\max}$. The *anchored radius* of a pure profile $p \in \prod_j B_j$ is $\hat{\delta}(p) = \max_i \|A_{-i}(p) - \bar{A}^*_{-i}\|_{w,\infty}$.

**Theorem 27** (Credal absorption)**.** *Assume A1–A5 and let $x^*$ be a mixed equilibrium at $\hat{\xi} \in \Xi$.*

*(a)* Absorption. *For every $p \in \prod_j B_j$ and every $\delta \geq \hat{\delta}(p)$, $p \in \mathrm{ICNE}_E(\delta) \subseteq \mathrm{ICNE}_M(\delta) \subseteq \mathrm{ICNE}_{ID}(\delta)$.*

*(b)* Radius. $\min_{p \in \prod_j B_j} \hat{\delta}(p) \leq \min\left\{ \max_i \max_k \sum_{j \neq i} r_{jk}/w_k,\ k^\sharp \bar{\sigma} + \bar{\rho},\ \bar{\sigma}\sqrt{k^\sharp \ln(2d)/2} + \bar{\rho} \right\}$.

*(c)* Uniform existence. $\delta_0^{ID} \leq \delta_0^M \leq \delta_0^E \leq \sigma_{\max} \min\{N - 1,\ d + 1,\ 1 + \sqrt{d \ln(2d)/2}\}$.

*(d)* Construction. *A profile achieving the second bound is obtained from $x^*$ by one linear program (a vertex of the polytope of Shapley–Folkman decompositions) followed by an arbitrary choice for at most $k^\sharp$ carriers; one achieving the third bound by independent sampling of these carriers, with success probability at least $1 - 2d\exp(-2t^2/(k^\sharp\bar{\sigma}^2))$ per draw for target $t + \bar{\rho}$; one achieving the first bound is any $p \in \prod_j B_j$.*

*Sketch.* Every carrier in $p \in \prod_j B_j$ plays a best response to the *anchor* $\bar{A}^*_{-i}$, which is nonnegative. If $\delta \geq \hat{\delta}(p)$, the anchor lies in the clipped box around the realised aggregate $A_{-i}(p)$, so the Dirac law at the anchor belongs to the credal set and witnesses E-admissibility. The radius bounds follow from the decomposition $A_{-i}(p) - \bar{A}^*_{-i} = \sum_{j\in J}(a_j(p_j) - y_j) - (a_i(p_i) - \bar{a}_i)$, where $J$ is the set of at most $k^\sharp$ carriers left mixed at a vertex of the Shapley–Folkman polytope (Theorem 39); the sum is bounded deterministically by $k^\sharp\bar{\sigma}$ or, after independent sampling, by Hoeffding's inequality [87] and a union bound. The own term $a_i(p_i) - \bar{a}_i$ cannot be removed (it is the "+1" in the bound of 14).

**Corollary 28** ($\varepsilon$-Nash in the precise game; cf. 14)**.** *A profile $p$ as in Theorem 27(a) is an $\varepsilon$-Nash equilibrium of the precise game at any $\mu \in \Xi$ with $\varepsilon_i \leq \hat{\delta}(p)L_i(p) + \gamma\sum_z \varpi_z T_{iz}(p)$, where $L_i(p) = \max_{p'}\sum_k w_k M_k|\varphi_{ik}(p') - \varphi_{ik}(p_i)|$ and $T_{iz}(p) = \max_{p'}|t_{iz}(p') - t_{iz}(p_i)|$.*

*Remark* 29 (Novelty and regime). Corollary 28 restates, in our notation, the approximate equilibrium of Liu et al. [14], who also use a Shapley–Folkman argument on the generators of a convexified equilibrium (their Lemma 5.6, Proposition 2.4, Theorem 2.7 and Propositions 2.8/Lemma 5.7); we claim no novelty for it. What Theorem 27 adds is an explicit belief box and the reading that, under E-admissibility, the rounding error is absorbed by the belief width, so that the rounded profile is an *exact* imprecise equilibrium. Part (a) follows in two lines from their Lemma 5.6 (every generator point is a best reply to the convexified rivals' aggregate). The content of the result lies in the explicit threshold, in the attainment of its $N - 1$ term at $N = 2$ (Proposition 31), in its failure for $\Gamma$-maximin (Remark 30), and in the contrast with the complexity result below. The terms $d + 1$ and $1 + \sqrt{d\ln(2d)/2}$ do not grow with $N$ but are smaller than $N - 1$ only when $N > d$; the $N$-uniform bound is not shown to be attained. In the city instances $N \leq 6$ and $d = 2Z \geq 84$, so $k^\sharp = N$, the Shapley–Folkman step selects nothing, and the operative bound is the support-selection bound $(N - 1)\bar{\rho}$, which grows with $N$. Shapley–Folkman matters only when the number of players exceeds the dimension, for example with per-vehicle or crowdsourced players.

*Remark* 30 ($\Gamma$-maximin). By Lemma 25, $\mathrm{ICNE}_\Gamma(\delta)$ is the pure-equilibrium set of a *precise* game, so absorption does not apply to it, and under D2 that set can be empty (Proposition 31 at $\delta = 0$).

### *5.5. Tightness*

**Proposition 31** (The support-selection bound is attained)**.** *Let $N = 2$, one zone, D2, $M = \gamma c = 1$ and weight $w > 0$. Carrier 1 has plans $A_1 : (n, \tau, \pi) = (1, 4, 6)$ and $B_1 : (3, 1, 0)$; carrier 2 has $X : (1, 1, 0)$ and $Y : (3, 4, 6)$; payoffs are $\pi - \tau \cdot n_{\mathrm{rival}}$. Then the game has no pure Nash equilibrium, $\delta_0^E = \delta_0^M = 1/w$, and the bound $(N - 1)\bar{\rho}$ of Theorem 27(b) equals $1/w$.*

In each of the four pure profiles exactly one carrier has a strict incentive to deviate at the realised count, and it becomes indifferent only at a belief that differs from the realised count by one vehicle.

**Proposition 32** (Linear dependence on the number of zones is necessary)**.** *For every $Z \geq 1$ and $N \geq 2$ there is a D2 game with $Z$ zones in which every pure profile has $\max_i \varepsilon_i \geq 3\lceil Z/2 \rceil$, while every profile of Theorem 27(a) satisfies $\hat{\delta}(p) L_i(p) \leq 3Z$ $(w = 1)$.*

*5.6. Complexity under vehicle-count congestion*

**Theorem 33** (NP-completeness under D2)**.** *Consider the problem of deciding $\mathrm{ICNE}_C(\delta) \neq \emptyset$ for a game given by explicit plan lists with rational data, $M$, $w$, a degenerate traffic box and a rational $\delta \geq 0$. For each $C \in \{E, M, ID, \Gamma\}$ the problem is in NP. For every digraph $D$ with maximum out-degree $\Delta^+ \geq 1$ one constructs in polynomial time a D2 instance with two plans per carrier and one zone per arc such that, for all $\delta \in [0, 1/(2\Delta^+ + 1))$, $\mathrm{ICNE}_E(\delta) = \mathrm{ICNE}_M(\delta) = \mathrm{ICNE}_{ID}(\delta) = \mathrm{ICNE}_\Gamma(\delta)$ is the set of profiles that play "in" exactly on a kernel of $D$. Hence the problem is strongly NP-complete for each criterion, already at $\delta = 0$, and for every fixed $\delta \in [0, 1/5)$.*

*Sketch.* Deciding whether a digraph has a kernel (an independent set that absorbs every other vertex) is NP-complete [88], also for planar digraphs with in- and out-degree at most two [89]. Each vertex becomes a carrier with an "out" plan of zero data and an "in" plan that puts one vehicle on each out-arc zone and two vehicles on each in-arc zone, with exposure one on out-arcs and a small exposure on in-arcs. The gain of "in" is positive if no out-neighbour plays "in" and at most $-1$ otherwise, and the sign is constant on every credal box of width below $1/(2\Delta^+ + 1)$, so all four criteria select the realised strict best response.

**Corollary 34** (Inapproximability of the existence threshold)**.** *Unless* P = NP*, no polynomial algorithm approximates $\delta_0^E$ of a D2 instance within any fixed factor, or within an additive error below $1/5$.*

Pure-equilibrium existence is known to be NP-complete in polymatrix games and integer programming games [15, 60]; Theorem 33 is a direct kernel reduction specialised to the rank-one D2 structure and extended to four credal criteria and a range of widths. Together with Proposition 26 it gives a complexity split between the two congestion regimes: charging or modelling congestion by exposure (vehicle-hours in a zone) yields a potential game with guaranteed pure imprecise equilibria, while counting vehicles does not. Under D2 the mixed equilibrium that Theorem 27 rounds is itself hard to compute: every bimatrix game embeds as a D2 instance with $N = 2$, so the step is PPAD-hard [90, 91] (Proposition 42).

*5.7. Efficiency*

**Lemma 35** (Imprecise equilibria are approximate equilibria)**.** *If $\sigma \in \mathrm{ICNE}_C(\delta)$ and $\mu \in \Xi$, then $\sigma$ is an $\varepsilon$-Nash equilibrium of the precise game with $\varepsilon_i \leq \delta L_i(\sigma) + \gamma \sum_z \varpi_z T_{iz}(\sigma)$ for $C \in \{E, M\}$ and $\varepsilon_i \leq 2\delta H_i + 2\gamma \Theta_i$ for $C = ID$, with $H_i = \max_p \sum_k w_k M_k \varphi_{ik}(p)$ and $\Theta_i = \max_p \sum_z \varpi_z t_{iz}(p)$.*

**Proposition 36** (Monotone efficiency curves)**.** *For $C \in \{E, M, ID\}$ and $W > 0$,* $\mathrm{PoA}_C(\delta) = W^*/\min_{\mathrm{ICNE}_C(\delta)} W$ *is nondecreasing and* $\mathrm{PoS}_C(\delta) = W^*/\max_{\mathrm{ICNE}_C(\delta)} W$ *nonincreasing in $\delta$, and both are step functions with at most $\prod_i |\mathcal{P}_i|$ steps. For $C = \Gamma$ neither monotonicity holds in general.*

**Proposition 37** (Price of stability via the potential, D1)**.** *Under D1 with zero charges and $\mu = \hat{\mu}$, let $\sigma^*$ maximise $W$. Every profile $\sigma$ with $\Psi_{\hat{\mu}}(\sigma) \geq \max \Psi_{\hat{\mu}} - \eta$ satisfies $W(\sigma) \geq W^* - \sum_z \left[\frac{b_z}{2} \sum_i q_{iz}^{*2} + \frac{\gamma c_z}{2}(1 - \frac{1}{N}) T_z(\sigma)^2\right] - \eta$. In particular a potential maximiser, which lies in every* $\mathrm{ICNE}_E(\delta)$*, bounds* $\mathrm{PoS}_C(\delta)$ *for all $\delta$ and $C \in \{E, M, ID\}$.*

A smoothness-based price-of-anarchy bound also transfers to imprecise equilibria through Lemma 35 (Proposition 43), but it is conditional: we do not have smoothness constants for the Cournot–congestion game with arbitrary private routing costs. The price-of-anarchy values in Section 7 are therefore empirical.

*5.8. Exact thresholds and the rounding algorithm*

For any pure profile $p$, the exact E-threshold of carrier $i$,

$$A_i^E(p) = \min\{\delta : p_i \text{ is E-admissible at width } \delta\},$$

is the value of a linear program over the box, so $p \in \mathrm{ICNE}_E(\delta)$ if and only if $\delta \geq \max_i A_i^E(p)$ (Theorem 44). Theorem 27 and this threshold computation define an algorithm with four phases: relax (a mixed equilibrium by Frank–Wolfe on the concave relaxed potential under D1, or by a double oracle under D2; 92), sparsify (a vertex of the Shapley–Folkman polytope), round, and compute the exact width. Under D2 its output satisfies the bounds of Theorem 27(b) (Theorem 45), but its relaxation phase is PPAD-hard in general. Under D1 the relaxation-and-rounding technique is classical [93, 94, 95]. The full rounding algorithm was not run on the city games, which use the D1 engine of Section 4; the exact thresholds of Theorem 44 were used in the tests of Section 7.7.

*Remark* 38 (Nonaffine congestion). With a power congestion term the expected payoff under mixed rival strategies is no longer a function of the mean aggregate; Lemma 23 and the exact credal statements fail, although relaxed equilibria still exist. We do not treat this case.

*5.9. What is machine-checked*

The repository contains a Lean 4 development with Mathlib (`lean/CCU`, seven modules). `lake build CCU` succeeds, no imported file contains an unproved step, and the theorems listed below depend only on the axioms `propext`, `Classical.choice` and `Quot.sound`. "Full" means the paper's statement for arbitrary finite index sets over the reals; "core" means a finite or algebraic part of the theorem.

- *Credal results* (LeanTargets, Radius): E-admissibility implies maximality and $\mathrm{ICNE}_E(\delta)$ is monotone in $\delta$ (full); the core of Theorem 27(a); the Shapley–Folkman lemma for finite sets (Theorem 39, full); the exact potential identity under D1 (Proposition 26, full); the kernel equivalence of Theorem 33 (core, no hardness); the arithmetic cores of Propositions 31 and 32; and, per coordinate, the first, decomposition and second terms of Theorem 27(b), and the $N - 1$ term of (c), given the rounding data.

- *DIP*: Proposition 6(a),(b) and Lemma 7 (full), the convexified identity, Theorem 9 over an abstract convex set (full), Theorem 10(a) (full, without the charge box) and the counting core of part (b).
- *Marginal tariff*: Theorem 16(a)–(c) and the tariff identity with the revenue formula (full), including termination of improvement dynamics.
- *ARP*: the strong-monotonicity inequality behind Theorem 17 (core), and the plain and damped contraction of Theorem 18 from that hypothesis (full); the damped constant proved, $1-\theta$ for $0<\theta\le 2/(2+\varrho)$, is sharper than the paper's.
- *BRL*: the budget identity for a finite level grid, the budget split of the inf-convolution, and the completion window and functional-dominance validity with the slope conditions taken as hypotheses (core), and the comparison with quantile-vector dominance.

Not formalised: the asymmetric-lot bound of DIP (Theorem 8) and the LP and $\varepsilon$ parts of Theorem 10; Theorem 17 beyond the monotonicity inequality (conjugates, the proximal form) and the pure-game theorems of ARP; probability boxes, upper previsions, quantile functions and the exactness proof of BRL; the Hoeffding term, the linear-programming vertex construction and the NP-completeness statement itself (Mathlib has no framework for polynomial-time reductions); and all computational results. Lean checks the stated cores, not the full theorems.

## 6. Data and calibration

*Road networks and zones.* For each of nine cities (New Delhi, Cairo, Moscow, Buenos Aires, New York, Mumbai, London, Paris, Berlin) we use an OpenStreetMap extract [96] distributed by BBBike (replication timestamp 16 October 2026). The drivable major-road network (motorway to tertiary, with links) is kept as an undirected graph with class-based free-flow speeds and lane counts (OSM `lanes` tag where present, class defaults otherwise). Population comes from the GHS-POP R2023A grid, epoch 2025, at 30 arc-seconds [97]. Zones are blocks of $4\times 4$ raster cells (about 3.7 km) inside the extract's bounding box with at least one inhabitant; zone-to-zone free-flow times are shortest paths between population-weighted centroids plus a fixed access time. Points of interest from the same extracts give a depot score per zone used to place carrier depots. Table F.10 (Appendix F) summarises the instances; $Z$ ranges from 42 to 593.

*Operating parameters.* Per-city parameters were collected from public web sources on 23 October 2026 and graded A (official or primary statistic), B (industry report or country-level proxy) or C (rough estimate or proxy from elsewhere). Table F.21 in Appendix F lists the values used; the repository file `data/parameters/SOURCES.md` gives source, URL, grade and note for each number. Congestion levels are those of the TomTom Traffic Index 2025 [98]; volume-delay curves use the BPR form with $\alpha=0.15$, $\beta=4$ [99], linearised at the observed congestion level.

*Weak numbers.* Parcels per capita are country-level figures. Driver wages are minimum-wage floors or job-site averages, not courier payroll. Drops per vehicle and van capacities outside the United States are proxies. Delivery fees in Cairo, Moscow and Buenos Aires are consumer

or reference rates, not carrier revenue per parcel, and the European fees are regulator revenue per parcel. Moscow is absent from the 2025 TomTom index and uses a 2020 value. Market shares are available only for Paris and Berlin; elsewhere carriers have equal shares. The sensitivity of the equilibria to the five weakest parameters is in Table F.22.

*Calibration rules and simplifications.* Each carrier serves 22 customer zones sampled around its depot with weight proportional to population times a gravity decay in travel time, so carriers overlap in some zones. A lot is 10–50% of a vehicle's daily drops, proportional to the zone's parcel demand. Inverse demand has intercept $\alpha_z = 1.5\times$ fee and slope $b_z = \alpha_z/(2Q_z^{\text{full}})$. The congestion slope $c_z$ is the derivative of the BPR curve at the TomTom level, with capacity from lane-kilometres, scaled by a slice multiplicity $m_z$ (the ratio of the city's real delivery vehicle-hours to the modelled ones) so that the slice faces the congestion of the whole city. Travel-time multipliers of nine sectors follow a lognormal law matching the TomTom all-day and peak congestion; each carrier observes 30 days, from which DKW boxes at confidence $1-\alpha$ are built. The route loss (3) uses the vehicle-hour cost (wage plus fuel) per hour, twice that as fixed cost, three times that per overtime hour, and the shift length as deadline. The simplifications are:

(S1) all-or-nothing lots: a carrier serves a customer zone fully or not at all, which makes the Cournot game lumpy;

(S2) the integer routing step: the main game experiments use branch-and-price with a time limit per call; some earlier experiments use price-and-branch (Section 7);

(S3) vehicle-hours are proportional to the lot, so within a zone a congestion charge on hours and a delivery charge are collinear;

(S4) linear congestion (D1);

(S5) oracle exactness at city scale is checked against an independent oracle, not proved.

Further proxies are the sampled customer zones instead of whole-city demand, equal market shares in seven cities, the slice multiplicity, and a *broad* social cost that adds the delay imposed on background traffic (population times vehicle ownership times one hour per day, valued at the wage, times the added slowdown, divided by $m_z$). The broad objective is reported but not used for conclusions.

*Instances as a benchmark.* The pipeline (extract download, zones, travel times, depots, parameters with grades, p-box construction, game instances with seeds) is in the repository and regenerates all instances from the raw extracts. The instances are small (22 customer zones per carrier, four to six carriers) so that exact oracles are possible; they are intended as a common test bed for algorithms on multi-carrier routing games, not as validated city models.

## 7. Computational study

*Research questions.*

**RQ1** Is BRL pricing exact, what does its completion window save, and how far does branch-and-price scale? (Section 7.1)

**RQ2** Why probability boxes: what does the p-box criterion cost and buy out of sample, compared with sample-average, Wasserstein and CVaR criteria? (Section 7.2)

**RQ3** How do equilibria computed with an exact oracle differ from those with price-and-branch, and how do the engine's components compare? (Section 7.3)

**RQ4** What do the equilibria, the grid leader, the price of anarchy and the imprecision effects look like on the nine cities, and how stable are they over seeds? (Section 7.4)

**RQ5** How do DIP and ARP compare with the grid, and with the social optimum, in oracle calls and gain? (Section 7.5)

**RQ6** Are the findings robust to the carriers' criterion? (Section 7.6)

**RQ7** Do the theoretical bounds hold on the city equilibria, and does vehicle-count congestion break pure equilibria in practice? (Section 7.7)

*Protocol and seeds.* All numbers are generated from the result files of the repository by `make_tables.py`. A seed redraws the depots, the customer zones of each carrier, the travel-time sectors and the 30 observed days, hence the probability boxes; calibration inputs are the same in every seed. Seeds 0–2 were run first for all nine cities. The exact-oracle study was then extended to seeds 3–9, which used 55 046 oracle calls, of which 1.1% were not proven optimal. Coverage is as follows.

- New Delhi, Buenos Aires, London: ten seeds for everything, including the full 63-point exact grid and ARP.
- Cairo, Moscow, New York, Mumbai, Paris, Berlin: 9 seeds (0–8) for the unpriced equilibrium, the social optimum, the price of anarchy and DIP. Grid-derived quantities (congestion-only, delivery-only and combined gains, delivery share, priced gap, DIP against the grid) still have three seeds and a reduced exact grid (7–17 screened points per seed), so they are lower bounds (marked $^*$). ARP was run for Mumbai (five seeds) and not for the other five.
- Imprecision sweeps (three seeds): all 12 widths for New Delhi, Buenos Aires, Mumbai, London; 6 widths for Cairo, Moscow, New York, Paris, Berlin.
- Still computed with price-and-branch: the engine comparison (Table F.9), the ZMC checks, the theory tests (Section 7.7), the tiny-game verification, the DIP comparisons for Berlin seed 2, Moscow seed 2, New York seed 2 (price-and-branch grids), the social-optimum benchmark of the broad objective in New York, Moscow and Berlin, and the p-box reference runs of the criterion study (Section 7.6).

Means over seeds are given with percentile bootstrap 95% confidence intervals (10,000 resamples of the seeds). They describe the spread over sampled slices of a city, not the uncertainty in the calibration inputs, which Table F.22 examines separately. Of the 41,469 oracle calls of the seeds 0–2 study, 99.17% were proven optimal, and the remaining gap of an unproven call was at most 14 USD. Every reported equilibrium is a pure Nash equilibrium of the game with the exact oracle up to unproven calls (Section 7.8).

Table 1: BRL pricing on random instances (nine sectors, $L = 33$ levels, capacity 6, two instances per size). Time in seconds (labels kept); TO: 60 s limit. The risk-neutral labelling solves a different (expected-value) problem and is shown for reference; SAA scenarios optimise the sample-average objective.

| $n$ | risk-neutral | BRL | sector ($K$) | quantile ($L$) | SAA scenarios |
|---|---|---|---|---|---|
| 10 | 0.0050 (19) | **0.0040 (16)** | TO | 2.2 (4946) | 5.4 (7904) |
| 10 | 0.017 (36) | **0.0080 (16)** | TO | 2.1 (4391) | 4.7 (7715) |
| 15 | 0.17 (264) | **0.019 (48)** | TO | TO | TO |
| 15 | 0.029 (69) | **0.010 (24)** | TO | TO | TO |
| 20 | 0.39 (714) | **0.049 (97)** | TO | TO | TO |
| 20 | 1.4 (1730) | **0.047 (89)** | TO | TO | TO |
| 25 | 0.46 (650) | **0.059 (94)** | TO | TO | TO |
| 25 | 0.49 (693) | **0.065 (109)** | TO | TO | TO |

### *7.1. BRL and branch-and-price*

*Exactness.* The upper-prevision formula agrees with a linear program over all joint laws on 40 random instances to a relative error of $4.5 \cdot 10^{-14}$; the comonotone worst case exceeds the value under independent worst marginals by 4.5% on average and up to 14.1%. BRL pricing equals complete enumeration on every test instance, including 18 of 18 sub-instances built from the nine cities. Fixing one risk level for all routes is suboptimal in 9 of 20 small exact comparisons, by up to 6.42% (Proposition 49).

*Label counts.* Table 1 compares labelling rules on random instances ($L = 33$ levels, nine sectors): BRL takes 0.004–0.065 seconds, while sector dominance times out from the smallest size and quantile-vector and scenario labelling from $n = 15$. These random instances are favourable to BRL, and the reductions on real data are smaller (Table 2): with the duals of the root column generation, the window rule keeps 1.8–2.9 times fewer labels than the same stop-loss comparison without the window and 2.8–6.3 times fewer than quantile-vector dominance on the metro instances (22–80 customer zones), and 1.6–14 and 2.7–23 times fewer on the public benchmark instances (below). The ratios shrink as $n$ grows and are lower bounds where a rule hit its comparison cap. Earlier statements of a ten- to hundred-fold reduction came from the random instances only and should not be read as general. We have not compared the completion window with resource-extension-function or function-label dominance [83, 84].

*Branch-and-price and scale.* On the nine metro instances with 22 customer zones, branch-and-price proved optimality in 9 of nine cities, in at most 18 minutes on one core (Moscow, 7487 nodes; Table F.17). Against complete enumeration on 800 random formulations, branch-and-price returned the optimum in 800 of 800, while price-and-branch was strictly suboptimal in 39 of them, by up to 7.3%. The scale study (Table F.11) shows where the exact method stops: optimality was proven for Cairo, London and New York at 30 zones, for London at 40 zones (and for New York at 40 zones with the device configuration), and for Solomon R101 with 25 customers; it was not proven at London with 60 or 80 zones, New Delhi with 60, or Augerat A-n32-k5 (gap 0.25% with 63 047 columns and no cuts). Heuristic-first pricing gave a median 2.2-fold speed-up with an identical LP bound; the completion bound gave none. The method should therefore be read as exact up to about 40 customers per carrier, not beyond.

Table 2: Median number of labels kept by three dominance rules priced once with the duals of the root column generation (metro instances, and Solomon and Augerat instances with a soft duration limit; CG: column generation converged within 150 s). Ratios are medians of per-instance ratios; capped runs give lower bounds.

| family | $n$ | instances | labels: window (BRL) | no window | quantile-vector | no window / window | quantile / window |
|---|---|---|---|---|---|---|---|
| metro | 22 | 9 | 2 070 | 6 453 | 12 692 | 2.6 | 4.2 |
| metro | 30 | 9 | 4 187 | 9 744 | 26 166 | 2.9 | 6.3 |
| metro | 40 | 9 | 7 918 | 21 585 | 49 295 | 2.0 | 4.6 |
| metro | 60 | 8 | 15 801 | 38 138 | 83 375 | 2.2 | 3.8 |
| metro | 80 | 8 | 24 475 | 52 598 | 87 976 | 1.8 | 2.8 |
| public | 25 | 3 | 4 101 | 46 140 | 75 556 | 9.3 | 20 |
| public | 31 | 1 | 3 744 | 51 972 | 87 375 | 14 | 23 |
| public | 44 | 1 | 6 612 | 55 074 | 107 450 | 8.3 | 16 |
| public | 50 | 3 | 24 283 | 69 094 | 89 324 | 2.8 | 3.7 |
| public | 59 | 1 | 13 986 | 108 840 | 140 020 | 7.8 | 10 |
| public | 79 | 1 | 7 845 | 26 199 | 114 320 | 3.3 | 15 |
| public | 100 | 3 | 39 104 | 53 370 | 101 530 | 1.6 | 2.7 |

*Public benchmark instances.* We also ran BRL on Solomon (25, 50, 100 customers) and Augerat A and B instances, treated as capacitated routing with a soft duration limit and four sectors with lognormal common-factor uncertainty. This is an emulation: it is not a reproduction of published branch-and-cut results, and the instances of Zhang et al. and Adulyasak and Jaillet could not be obtained. Over 51 cells (instance, number of training days, draw), the BRL solve was proven optimal in 59% of cases, against 35% for the deterministic plan, 20% for SAA, 3.9% for the Wasserstein criterion and 0% for mean-CVaR (90 s limit, same branch-and-price); BRL was proven on Solomon-25 in a median of 3.8 s (27 cells) but in 0 of 15 Augerat cells. Scenario labelling keeps 61 123–98 930 labels where BRL keeps 13 175–15 455. After a local-search polish of the best plan of any arm, all arms are within 0.17% of the best known plan under their own criterion.

On held-out days from the same law, however, BRL does not beat the sample-average plan (Table 3): its plans cost 0.30–5.0% more on the mean and 1.2–2.8% more at the 95th percentile after polishing. Only in a stress test, with a law shifted away from the training days and ten training days, is its 95th percentile 2.3% lower and its $\mathrm{CVaR}_{0.9}$ 2.5% lower, at 3.3% more on the mean; the advantage vanishes at 30 or more days. BRL's advantage is thus proof speed and label counts against scenario and quantile labelling at about 25 customers, not out-of-sample cost.

*Conservatism.* On the metro instances the robust optimum is not the best plan under the calibrated law: evaluated under it, the branch-and-price plan is worse than the sample-average plan in Buenos Aires (+7.0%), New York (+4.8%), Berlin (+6.8%), and its worst-case cost lies between 10% and 45% above its expected cost under the law. The next subsection places this in context.

### *7.2. Why probability boxes*

A natural question is why to use DKW probability boxes with unknown dependence instead of sample-average approximation (SAA), a type-1 Wasserstein ball or CVaR. We compared the criteria out of sample on the nine metro instances (seed 0, 22 customer zones), with $n = 10, 20, 30, 60$ training days, 20 independent draws, and a held-out set of 20,000 days

Table 3: Out-of-sample comparison on the public instances (17 cells per row), polished plans, relative to the sample-average plan with the same number of training days. Held-out: 2000 days from the training law; shifted law: excess congestion ×1.5, volatility ×1.25, no re-solving.

| $N$ days | plan | held-out mean % | 95th pct % | $\text{CVaR}_{0.9}$ % | shifted law: mean % | shifted law: 95th pct % |
|---|---|---|---|---|---|---|
| 10 | BRL | 5.0 | 2.8 | 2.0 | 3.3 | -2.3 |
| 10 | SAA | 0 | 0 | 0 | 0 | 0 |
| 10 | Wasserstein | 0.036 | 0.012 | 0.076 | 0.065 | 0.17 |
| 10 | mean-CVaR | 0.77 | -0.45 | -0.55 | 0.21 | -1.0 |
| 10 | deterministic | 0.13 | 2.8 | 2.7 | 0.97 | 2.2 |
| 30 | BRL | 0.42 | 1.6 | 1.5 | 0.62 | 0.21 |
| 30 | SAA | 0 | 0 | 0 | 0 | 0 |
| 30 | Wasserstein | -0.087 | -0.041 | 0.051 | -0.015 | 0.23 |
| 30 | mean-CVaR | 0.11 | 0.040 | 0.14 | 0.14 | 0.29 |
| 30 | deterministic | 0.056 | 2.0 | 1.9 | 0.56 | 0.89 |
| 60 | BRL | 0.30 | 1.2 | 0.97 | 0.40 | -0.073 |
| 60 | SAA | 0 | 0 | 0 | 0 | 0 |
| 60 | Wasserstein | -0.033 | 0.27 | 0.25 | 0.091 | 0.32 |
| 60 | mean-CVaR | 0.16 | -0.062 | -0.17 | 0.021 | -0.33 |
| 60 | deterministic | 0.35 | 3.0 | 2.8 | 1.1 | 2.0 |

Table 4: Out-of-sample comparison of criteria on the nine metro instances (mean over 20 draws and cities). Regret: expected cost of the plan above the best known plan for the true law, %. Other columns at $n = 20$: probability that at least one route runs overtime on a day; $\text{CVaR}_{0.95}$ of the daily plan cost above the best known expected cost; criterion value minus realised expected cost.

| Criterion | regret % $n = 10$ | $n = 20$ | $n = 60$ | P(overtime day), $n = 20$ | $\text{CVaR}_{0.95}$ above best, % ($n = 20$) | criterion minus true cost, % ($n = 20$) |
|---|---|---|---|---|---|---|
| SAA | 0.38 | 0.27 | 0.16 | 0.15 | 38.8 | -0.086 |
| Wasserstein, cross-validated radius | 0.43 | 0.26 | 0.15 | 0.14 | 39.1 | 3.2 |
| Wasserstein, theory radius | 1.5 | 1.1 | 0.51 | 0.16 | 41.1 | 9.5 |
| $\text{CVaR}_{0.9}$ | 2.4 | 2.5 | 2.9 | 0.063 | 34.6 | 19 |
| p-box, unknown dependence, 0.80 | 6.0 | 4.6 | 1.6 | 0.069 | 33.9 | 29 |
| p-box, unknown dependence, 0.95 | 7.9 | 5.1 | 2.6 | 0.070 | 34.5 | 35 |
| p-box, unknown dependence, 0.99 | 8.9 | 6.3 | 4.1 | 0.077 | 35.8 | 40 |
| p-box, independent, 0.95 | 5.9 | 3.8 | 1.6 | 0.053 | 31.9 | 29 |
| comonotone, empirical marginals (no band) | 0.31 | 0.090 | 0.051 | 0.13 | 37.5 | 5.3 |

of the instance's law. The planner is a heuristic (Clarke–Wright, local search, ruin and recreate, and a set-partitioning step over all evaluated routes), validated against branch-and-price: optimal in 8 of 9 instances, at most 0.10% above in the others (Paris). Table 4 gives the results.

*The p-box loses in expected cost.* At every $n \leq 60$ its regret is 1.6–8.9% (for confidence 0.80 to 0.99, falling with $n$), against 0.27% for SAA at $n = 20$ and 0.16% at $n = 60$. Wasserstein with a cross-validated radius reproduces SAA (0.26% and 0.15%); at the theory radius it costs 1.1% and 0.51%; $\text{CVaR}_{0.9}$ costs 2.5% and 2.9%. The cost comes from the DKW band, not from the unknown-dependence worst case: the comonotone criterion with empirical marginals and no band has regret 0.090% and 0.051%.

*It buys tail protection and a bound.* At $n = 20$ the probability of an overtime day is 0.15 for SAA and 0.069 for the p-box with confidence 0.80 ($\text{CVaR}_{0.9}$: 0.063), and the $\text{CVaR}_{0.95}$ of the daily cost is 3.6–5.0 percentage points lower than under SAA at every $n$. Under a common additive congestion shock at test time, the p-box plan with confidence 0.80 has the same expected cost as the SAA plan at a shock of about 0.50 ($n = 60$) to 0.60 ($n = 20$) multiplier units (multipliers of the instances are about 1.4–1.6) and is cheaper beyond (2.7% at a shock

of 1.0, $n = 60$). Its criterion value is an upper bound on the true expected cost whenever the DKW band holds: the criterion exceeds the realised cost by 20–40% for the p-box, by 3.2% for the cross-validated Wasserstein criterion and by -0.086% for SAA at $n = 20$. This is the conservatism referred to above.

*What the criterion provides.* P-boxes do not minimise expected cost. They provide a guaranteed upper bound that shrinks with the sample size, and tail protection, at an expected-cost price that is 1.6–8.9% for $n \leq 60$. Which criterion the carriers use matters for the game results; Section 7.6 quantifies that.

### *7.3. Equilibrium engine and exactness of the oracle*

*Effect of the inexact integer step.* The seeds 0–2 experiments were first computed with price-and-branch. Tested against a fresh exact oracle, those base equilibria were exact in New Delhi, Buenos Aires, Mumbai, London in every seed but not in 10 of the 27 city-seeds, where a carrier could gain at most 4.0 USD per day with a plan the pooled integer program had not generated. City-mean welfare changes by at most 0.14% outside Buenos Aires and by -1.4% in Buenos Aires, where seed 0 moves to a different equilibrium (-5.4%); single seeds elsewhere change by up to 0.53% with both signs. The exact oracle changes which equilibrium is reached rather than improving welfare systematically; it takes three to twelve times longer.

*Engine components.* On the city games (Table F.9, price-and-branch oracle) ZMC uses 36–65 calls, about ten per carrier, and its potential gap coincides with those of Frank–Wolfe and column generation with eight rounds to three digits in most cities. Plain best response needs 8–17 calls and is as good as ZMC (within $10^{-3}$) in Cairo, Moscow, New York, Paris, Berlin and worse in New Delhi, Buenos Aires, Mumbai, London. ZMC therefore does not beat best response on these games: only 32–89 coordinates are active, so the dimension that favours ZMC in the synthetic instances is absent. Its dimension independence is not tested at city scale.

### *7.4. Unpriced equilibria and the price of anarchy*

The cities fall into two groups (Table F.12, seeds 0–2, exact oracle). In Cairo, Moscow, New York, Paris, Berlin the carriers serve at least 85% of customer demand; in New Delhi, Buenos Aires, Mumbai, London they serve 18–45%, because the delivery fee is low relative to the vehicle-hour cost or the shift is short. The added slowdown caused by delivery vehicles is 0.77–17.6% (Moscow highest).

*Price of anarchy over ten seeds.* The social optimum is approximated by column generation on the convexified welfare problem followed by rounding and polish, and the equilibria considered are those found by the engine and by best response from the empty and full-service profiles and from three random price vectors. The price of anarchy is therefore *empirical*: a ratio of the rounded optimum to the equilibria found, not a bound over all equilibria. Table 5 gives the ten-seed results. The price of anarchy is at least 1.10 in some seed only in New Delhi, Buenos Aires, London among the equilibria of the engine; its mean is 1.53 in London, 1.38 in Buenos Aires and 1.18 in New Delhi, with ranges 1.25–1.94, 1.28–1.46 and 1.09–1.31. In the other cities it is at most 1.06 in mean (Mumbai's worst equilibrium found has mean 1.10 [1.08, 1.13]). The ordering of the means London > Buenos Aires > New Delhi holds, but the ordering holds seed by seed only in 7 of 10 seeds: London is below Buenos Aires in seeds 4,

Table 5: Ten-seed summary (exact oracle). $n_{\text{eq}}$, $n_{\text{grid}}$: seeds for the equilibrium quantities and for grid-derived quantities; *: reduced exact grid, grid quantities are lower bounds. PoA of the engine's equilibrium, mean with bootstrap 95% CI. Gains are % of |social cost| of the best exact grid point with the congestion charge only or the delivery charge only: mean (maximum over seeds). Delivery share: delivery-only gain over the combined gain per seed. Priced gap: shortfall of the equilibrium under the grid's best charges from the rounded optimum, %.

| City | $n_{\text{eq}}$ | $n_{\text{grid}}$ | PoA mean [95% CI] | cong. only % mean (max) | deliv. only % mean (max) | delivery share [95% CI] | priced gap % mean (max) |
|---|---|---|---|---|---|---|---|
| New Delhi | 10 | 10 | 1.18 [1.14, 1.23] | 0.32 (2.9) | 12.1 (21.2) | 0.814 [0.687, 0.920] | 2.9 (6.7) |
| Cairo* | 9 | 3 | 1.00 [1.00, 1.00] | 0 (0) | 0 (0) | – | 0 (0) |
| Moscow* | 9 | 3 | 1.00 [1.00, 1.00] | 0 (0) | 0.215 (0.387) | – | 0.0041 (0.012) |
| Buenos Aires | 10 | 10 | 1.38 [1.33, 1.42] | 2.4 (6.5) | 34.6 (41.9) | 0.978 [0.963, 0.991] | 1.8 (3.2) |
| New York* | 9 | 3 | 1.00 [1.00, 1.00] | 0.0045 (0.013) | 0.352 (0.460) | – | 0 (0) |
| Mumbai* | 9 | 3 | 1.06 [1.04, 1.07] | 0 (0) | 2.09 (3.31) | 0.877 [0.464, 1.38] | 1.0 (1.5) |
| London | 10 | 10 | 1.53 [1.41, 1.66] | 1.1 (5.4) | 39.8 (69.7) | 0.816 [0.752, 0.880] | 3.4 (7.4) |
| Paris* | 9 | 3 | 1.03 [1.03, 1.04] | 0.041 (0.12) | 3.47 (4.72) | 0.970 [0.910, 1.00] | 0.16 (0.23) |
| Berlin* | 9 | 3 | 1.02 [1.02, 1.03] | 0 (0) | 1.85 (2.17) | 0.943 [0.904, 1.00] | 0.19 (0.46) |

6, 7, while Buenos Aires exceeds New Delhi in 10 of 10. With the grid's best charges the equilibrium is much closer to the optimum, with a priced gap of at most 7.4% in any seed. The relaxation bound exceeds the rounded optimum by up to 25% (London), so the true price of anarchy could be larger. The three-seed statement that the ordering held in every seed did not survive ten seeds. Table F.14 (Appendix) gives the seeds 0–2 detail.

*Imprecision.* Two kinds of imprecision are varied (seeds 0–2; Table F.15, Figure F.2). Widening the travel-time probability boxes from $\varepsilon = 0$ to $\varepsilon = 0.41$ acts through the BRL route costs: deliveries fall by 0–78% and welfare by 2.7–82% at the widest box. Cairo loses no deliveries, Moscow and New York a few per cent, Paris and Berlin somewhat more, Mumbai more, and New Delhi, Buenos Aires and London about half or more; the low-service magnitudes vary by seed (London: -60.0 / -56.1 / -78.2% in seeds 0, 1, 2). Widening the credal width about rivals under Γ-maximin from $\delta = 0$ to $\delta = 0.80$ lowers deliveries by 31–100% and welfare by 22–100%. Deliveries are not always monotone in the width: 3 of 54 series contain a rise, all in Buenos Aires; Proposition 36 does not apply to Γ-maximin or to a single equilibrium that can switch. Imprecision acts like a tax on activity; because vehicle-hours are proportional to deliveries (S3), the model cannot separate a congestion effect from a delivery effect. The imprecision sweeps were not extended to ten seeds.

### *7.5. Leader methods: grid, DIP and ARP*

*The grid.* The baseline leader chooses a congestion charge per vehicle-hour, $t_z = \vartheta \gamma c_z X_z^{\text{NE}}$ with $\vartheta \in \{0, 0.5, 1, 2, 4, 8, 16\}$, and a per-delivery charge $s \in \{-0.6, \ldots, 0.6\}$ times the fee (negative: subsidy), a two-parameter family of 63 points with one warm-started equilibrium each. Table 5 and Table F.13 (Appendix, seeds 0–2) give the reduction of social cost for the best exact point using only the congestion charge, only the delivery charge, or both.

*Congestion charge alone.* The three-seed reading that a congestion charge alone gains less than 1.7% everywhere is false at ten seeds: it reaches at least 1.7% in 4 of 10 seeds in Buenos Aires (maximum 6.5%), 2 of 10 in London (maximum 5.4%) and 1 of 10 in New Delhi (maximum 2.9%). In the seeds 0–2 exact run the Buenos Aires seed-0 case (6.51%) arose because the exact oracle selected a worse unpriced equilibrium (welfare 252.2 against 266.7 USD with price-and-branch). What survives is the comparison: the mean congestion-only gain is 0.32–2.4% in the three full-grid cities, against 12.1–39.8% for the delivery charge alone.

Table 6: Leader methods, mean over seeds (exact oracle). DIP / grid: DIP gain over the best exact grid point's gain; calls: DIP's oracle calls as a percentage of a 63-point grid at the same calls per point. Gain shares against the social optimum (SO): best grid point, DIP, ARP flat; ARP / DIP gain ratio; $n_{\mathrm{ARP}}$ seeds with an ARP run. *: reduced exact grid (lower bounds for the grid column). Shares are not reported where the optimum's gain is below 0.5% of |social cost|.

| City | DIP / grid | DIP calls % of 63-pt grid | grid / SO | DIP / SO | ARP / SO | ARP / DIP | $n_{\mathrm{ARP}}$ |
|---|---|---|---|---|---|---|---|
| New Delhi | 1.23 | 5.69 | 0.815 | 1.00 | 0.933 | 0.933 | 10 |
| Cairo* | – | 2.33 | – | – | – | – | 0 |
| Moscow* | – | 2.53 | – | – | – | – | 0 |
| Buenos Aires | 1.07 | 6.85 | 0.938 | 1.00 | 0.995 | 0.995 | 10 |
| New York* | – | 2.73 | – | 1.00 | – | – | 0 |
| Mumbai* | 1.91 | 6.13 | 0.711 | 1.05 | 0.947 | 0.884 | 5 |
| London | 1.14 | 8.33 | 0.905 | 1.02 | 0.926 | 0.912 | 10 |
| Paris* | 1.04 | 4.69 | 0.959 | 1.00 | – | – | 0 |
| Berlin* | 1.05 | 3.55 | 0.949 | 1.00 | – | – | 0 |

*Per-delivery instrument.* In every case with a material gain the best point has a subsidy. The delivery charge alone achieves 0.81 [0.69, 0.92] of the combined gain in New Delhi, 0.98 [0.96, 0.99] in Buenos Aires and 0.82 [0.75, 0.88] in London (ratio of the per-seed shares; bootstrap 95% CI), and below one half only in New Delhi seed 0. Mumbai's share is not established (3 seeds, CI 0.46–1.4). The gain is material only where the unpriced equilibrium under-delivers; in Cairo, Moscow and New York it is at most 0.46%. In London the combined gain ranges widely over seeds. The mechanism visible in the model is under-provision caused by fixed route costs and market power, which a subsidy corrects, while the congestion externality among carriers is small; within a zone the two instruments are collinear (S3). Subsidies cost the city money; since charges are transfers in the welfare definition, this fiscal cost is not part of the social cost. With the broad objective the leader's best response in New Delhi, Mumbai, London is to stop deliveries, an artifact of the proxy that supports no conclusion (Table F.23, Appendix).

### *7.5.1. DIP against the grid and against the social optimum*

DIP was run with the exact oracle on all nine cities, seeds 0–8 or 0–9 (Table 6). Calls count every routing-oracle query of each method with the same counter. DIP searches a richer family than the grid (one charge per zone against two parameters), so a gain share above one against the grid is not a like-for-like win; restricted to the grid's family, DIP reaches on average 1.0 of the grid gain in the seeds 0–2 runs, with the worst case 0.50 (Mumbai, seed 0, a reduced grid). Table 6 therefore also reports every method's gain as a share of the gain of the rounded social optimum.

*Gain and cost.* At ten seeds DIP reaches 1.23, 1.07 and 1.14 of the best grid point's gain in New Delhi, Buenos Aires and London, with 5.7%, 6.8% and 8.3% of a 63-point grid's calls (seeds 0–2, full-grid cities: 104–135% of the gain with 4.9–9.4% of the calls, 105–221 calls against 1 921–2 419). Against the social optimum, the best grid point reaches 0.815, 0.938 and 0.905 of its gain, and DIP 1.00, 1.00 and 1.02 (values above one occur where DIP's multi-start target is better than the benchmark). The grid's shortfall is the price of its two-parameter family; DIP's near-optimality is by construction the quality of its social target. In Cairo no

Table 7: Leader methods on four cities, seeds 0–2 (exact oracle). $\varrho$: contraction constant of Theorem 18. ARP: flat rule, status and rounds (a cycle is a repeated charge and market state). Gains are percentages of the best exact grid point's gain; tariff+BR: marginal tariff with the carriers' own best responses; tariff+SO: tariff with the potential solver. Calls: oracle calls of the carriers during the method. Mumbai's grid is reduced (gains relative to a lower bound).

| City | seed | $\varrho$ | ARP status / rounds | ARP % | calls | tariff+BR % | calls | tariff+SO % | calls | DIP % | calls | grid calls |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| New Delhi | 0 | 0.40 | cycle / 5 | 90.8 | 53 | 98.7 | 11 | 92.5 | 51 | 123 | 134 | 2404 |
| New Delhi | 1 | 0.19 | fixed point / 3 | 126 | 27 | 135 | 10 | 127 | 51 | 135 | 119 | 2418 |
| New Delhi | 2 | 0.40 | cycle / 6 | 105 | 60 | 103 | 9 | 114 | 48 | 114 | 132 | 2419 |
| Buenos Aires | 0 | 0.029 | fixed point / 3 | 108 | 24 | 108 | 9 | 108 | 40 | 109 | 121 | 1921 |
| Buenos Aires | 1 | 0.016 | fixed point / 3 | 105 | 22 | 105 | 10 | 105 | 40 | 105 | 122 | 1950 |
| Buenos Aires | 2 | 0.11 | fixed point / 3 | 108 | 23 | 108 | 8 | 108 | 39 | 108 | 105 | 1943 |
| Mumbai | 0 | 0.20 | fixed point / 7 | 293 | 65 | 293 | 9 | 263 | 51 | 325 | 139 | 586 |
| Mumbai | 1 | 0.094 | fixed point / 3 | 111 | 30 | 113 | 10 | 108 | 50 | 120 | 169 | 544 |
| Mumbai | 2 | 0.20 | cycle / 5 | 128 | 47 | 111 | 10 | 118 | 49 | 128 | 141 | 537 |
| London | 0 | 0.10 | cycle / 5 | 102 | 51 | 104 | 14 | 100 | 49 | 104 | 170 | 2376 |
| London | 1 | 0.10 | cycle / 5 | 120 | 47 | 122 | 9 | 131 | 50 | 131 | 221 | 2356 |
| London | 2 | 0.10 | cycle / 7 | 103 | 67 | 105 | 11 | 101 | 50 | 106 | 206 | 2378 |

charge improves on the unpriced equilibrium. In all 27 seeds 0–2 runs the cutting-plane loop ended with charges under which the target is an exact pure equilibrium (fresh-oracle regret at most $9.1 \cdot 10^{-13}$ USD); the tangent charge alone sufficed in 18. The uniform subsidy is 0.37–0.50 of the fee and the zone charges follow the competitor-count rule of Proposition 15.

*Qualifiers.* The per-run detail for seeds 0–2 is in Table F.16 and Figure F.5 (Appendix F). Against the reduced exact grids, DIP uses more than 10% of their calls in most runs (Cairo 8.5–12%, Moscow 13%, New York 23%, Mumbai 24–31%, Paris 21–23%, Berlin 15–17%). Wall-clock time is not always lower (median ratio 0.22; London seed 0: 247 s against 157 s). Selection is optimistic: under DIP's charges the grid's protocol reaches the target in 21 of 27 runs, and in 210 enumerated tiny games the equilibrium under DIP's charges was unique in only 35% of the games. In the tiny games DIP reached the exact leader optimum over all zone charges in 93% (100% with the exact social optimum as target), so each miss was a miss of the social solver. With $\Gamma$-maximin carriers, the shifted charges of Theorem 14 kept the target an exact equilibrium in every run at $\delta = 0.2$ and 0.8 (largest regret $9.1 \cdot 10^{-13}$ USD), but at $\delta = 0.8$ the required uniform subsidy is 0.98–3.6 times the fee, which is not a realistic policy.

### *7.5.2. ARP: charges from observed aggregates*

ARP was run with the exact oracle for ten seeds in New Delhi, Buenos Aires and London and five in Mumbai. The carriers best-respond day to day from their current plans after each posted charge, and all their oracle calls are counted; the status-quo equilibrium (40–60 calls) is common to all methods and not counted, as for DIP. Table 7 gives the seeds 0–2 runs in detail.

*City runs.* Over the seeds 0–2 runs, ARP reaches 90.8% to 293% of the grid's gain (mean 125%) after 3–7 posted charges and 22–67 calls, against 104–325% (mean 134%) for DIP with 105–221 calls; in the three full-grid cities the figures are 90.8–126% (ARP) and 104–135% (DIP). It matches DIP within two percentage points in 5 of 12 runs; in the other 7 it is below DIP by up to 32 points of the grid's gain (the largest gaps occur where the iteration cycles or the grid is reduced). The clear miss is New Delhi seed 0 (90.8% against DIP's 123%), where

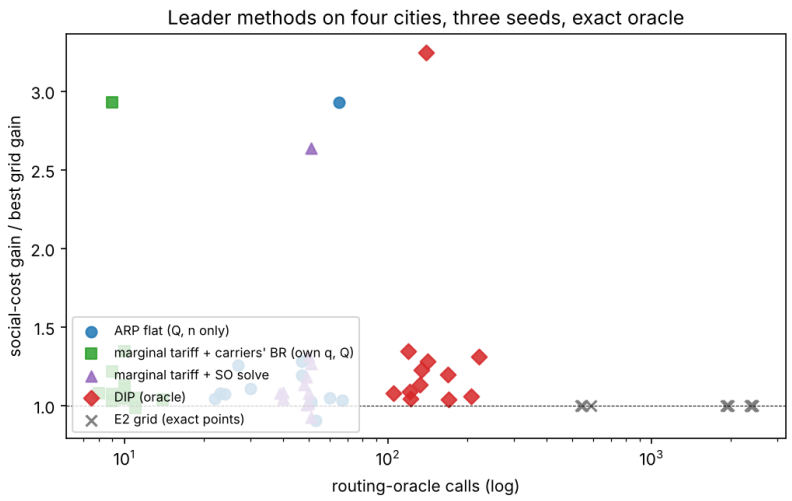



Figure 1: Leader methods on four cities and three seeds (exact oracle): gain relative to the best exact grid point against routing-oracle calls (log scale).

the iteration cycles between two equilibria and $\varrho = 0.40$, the largest of the sample. The flat rule reached an exact fixed point in 6 runs and cycled in 6; successive averages do not remove the cycles. The relative change of the charge vector falls to 0.37–0.63 of its initial value after the first round and to 0.044–0.24 after the second. Over ten seeds in the three full-grid cities, ARP's gain is 0.933, 0.995 and 0.926 of the social optimum's gain (Table 6), that is 0.93, 0.99 and 0.91 of DIP's gain; it reached an exact fixed point in 15 of 30 runs.

The marginal tariff with the carriers' own best responses reaches 98.7–293% (mean 125%) with 8–14 calls and no city-side computation, and never raised the social cost along the way (12 of 12 runs), as Theorem 16 says; with the potential solver it reaches 92.5–263% with 39–51 calls. Every returned profile is an exact pure equilibrium of the posted charges (regret at most $1.1 \cdot 10^{-13}$ USD for ARP and $1.4 \cdot 10^{-14}$ USD for the tariff). ARP's wall-clock time is 1.9–41 s, a median of 0.041 of DIP's.

*Tiny games and contraction constants.* On 210 exhaustively enumerated games (136 with a positive leader gain; Table F.18, Appendix) the flat rule reached a fixed point in 166 games and cycled in 44, almost all at strong congestion; its mean share of the exact optimal gain is 0.72 / 0.64 / 0.23 at congestion multipliers 1, 20, 60 (tariff 0.93 / 0.79 / 0.58, DIP 1.00 / 0.96 / 0.93). Its failures are entry into zones that the optimum leaves unserved, attracted by the subsidy based on the city-mean lot, and two-charge cycles. Knowing the lots does not help (share 0.63): the limit is what the aggregate says about which profile is socially good (Theorem 19(c)). Under the tariff the pure equilibria are exactly the unilateral social optima (the best tariff equilibrium is the social optimum in 210 of 210 games). On 48 convexified tiny games with $\varrho$ from 0.0062 to 0.81 the Lipschitz bound of Theorem 17, the plain-step bound and the damped-step bound of Theorem 18 (tight, observed maximum 1.00 of the bound) held in all games. ARP is not better than DIP in solution quality; it is better in the information it needs and in cost.

Table 8: Game results under four carrier criteria, three seeds, mean. PoA with the true expected routing cost and with the carriers' own (model) cost. DIP gain as % of |social cost| under each cost. Best uniform per-delivery charge on the coarse grid as a fraction of the fee. Weighted DIP charge: $u_z$/fee weighted by deliveries.

| City | carriers' criterion | served share | PoA (true cost) | PoA (model cost) | DIP gain %, true cost | DIP gain %, model cost | best uniform charge / fee, true cost | same, model cost | DIP charge $u$/fee (weighted) |
|---|---|---|---|---|---|---|---|---|---|
| New Delhi | p-box | 0.26 | 1.35 | 1.18 | 34.0 | 18.2 | -0.53 | -0.20 | -0.17 |
| New Delhi | SAA | 0.48 | 1.19 | 1.21 | 18.4 | 20.7 | -0.17 | -0.23 | -0.18 |
| New Delhi | Wass. CV | 0.46 | 1.19 | 1.20 | 18.5 | 19.4 | -0.20 | -0.20 | -0.18 |
| New Delhi | Wass. theory | 0.41 | 1.25 | 1.20 | 24.6 | 19.2 | -0.27 | -0.20 | -0.18 |
| Buenos Aires | p-box | 0.37 | 1.49 | 1.36 | 51.2 | 35.9 | -0.77 | -0.30 | -0.24 |
| Buenos Aires | SAA | 0.67 | 1.16 | 1.19 | 16.0 | 18.5 | -0.30 | -0.30 | -0.25 |
| Buenos Aires | Wass. CV | 0.58 | 1.27 | 1.23 | 26.6 | 23.5 | -0.37 | -0.23 | -0.25 |
| Buenos Aires | Wass. theory | 0.57 | 1.31 | 1.28 | 30.4 | 27.9 | -0.30 | -0.40 | -0.25 |
| London | p-box | 0.18 | 2.06 | 1.66 | 108 | 65.8 | -0.80 | -0.30 | -0.25 |
| London | SAA | 0.38 | 1.52 | 1.55 | 51.1 | 53.2 | -0.33 | -0.33 | -0.26 |
| London | Wass. CV | 0.32 | 1.69 | 1.65 | 68.0 | 63.8 | -0.47 | -0.40 | -0.26 |
| London | Wass. theory | 0.31 | 1.75 | 1.66 | 71.8 | 65.4 | -0.40 | -0.30 | -0.26 |

### *7.6. Robustness to the carriers' criterion*

All game results above describe carriers who plan against the p-box worst-case route cost. To test how much this matters we recomputed the unpriced equilibrium, the price of anarchy, DIP and a coarse 16-point grid for carriers who plan with the sample-average cost of the same 30 days, with a Wasserstein ball with cross-validated radius, or with the theory radius, for all nine cities and 3 seeds. The SAA and Wasserstein carriers use a heuristic oracle without an optimality proof (equilibria verified against an independent fresh pool, regret at most $5 \cdot 10^{-13}$); the p-box reference uses price-and-branch; the coarse grid is easier to match than the 63-point grid; the cross-validated Wasserstein criterion was run for one seed only in Paris and Berlin. Table 8 gives the three low-service cities.

*What is robust.* Under all four criteria: the delivery charge alone achieves 0.88–1.0 of the best grid gain (0.74–1.00 for the theory radius); the congestion charge alone is small, though marginal for SAA in Buenos Aires (3.3% against 18% for the delivery charge); the price of anarchy is at least 1.10 only in London, Buenos Aires and New Delhi (at most 1.06 elsewhere with SAA carriers); DIP's per-delivery charge follows $-0.375\,\text{fee}/n_z$ (slopes 0.361–0.379, correlation at least 0.984 in absolute value); DIP matches the coarse-grid gain (ratio at least 0.99) with 5.0–10% of the calls of a 63-point grid (21–43% of the coarse grid's); the optimal charge is a subsidy in every city; and cities with complete service are unchanged.

*What is not robust.* The levels. With SAA carriers the unpriced served share roughly doubles (London 0.18 to 0.38, Buenos Aires 0.37 to 0.67, New Delhi 0.26 to 0.48) and the price of anarchy measured with the true cost falls (London 2.06 to 1.52, Buenos Aires 1.49 to 1.16, New Delhi 1.35 to 1.19); the model-cost price of anarchy moves little. The ordering of the price of anarchy is not preserved: with SAA carriers it is 1.16 and 1.19 in Buenos Aires and New Delhi. The leader's gain under the true cost falls by half or more (London 108% to 51.1%, Buenos Aires 51.2% to 16.0%, New Delhi 34.0% to 18.4%), and the best uniform charge for the true-cost objective shrinks by 58–69% (p-box 0.80, 0.77, 0.53 of the fee, beyond the $-0.6$ end of the grid axis, against 0.33, 0.30, 0.17 for SAA), while DIP's zone charges themselves are almost unchanged. *All levels of the price of anarchy, served share and gain in this paper describe carriers planning against a guaranteed upper bound; the orderings and the policy directions are robust, the levels are not.*

*7.7. Theory against data*

The city games are of type D1, so the engine's profile is an exact pure equilibrium with E-threshold zero and Theorem 27 holds trivially for it. To test the theorems non-trivially we computed the exact thresholds of Theorem 44 on mixed equilibria with belief weights equal to the realised aggregate (price-and-branch oracle; each threshold is relaxed by the largest improvement observed at a verified equilibrium). Table F.19 and Figure F.6 are in Appendix F.

*Credal absorption.* Non-trivial mixed equilibria were constructed from pairs of distinct pure equilibria by solving the two-plan indifference conditions and checking that every support plan is a best response over all plans. This succeeded in 14 of the 27 city-seed games; Cairo, Moscow have none, because almost all demand is served. On all 274 tested profiles no statement of Theorem 27 was violated (0 violations). The bound is loose: the largest empirical threshold is 0.28–0.60 times the anchored bound (median 0.45). As Remark 29 states, on these games $N \leq 6 < d$, so the $(N-1)\bar{\rho}$ term is the smallest of the three in every case; the $N$-free terms are not tested by the city data. Without the best-response polish, the rounded profile of the first engine phase is E-admissible at a finite width in only 10 of 27 games. Limits: mixed equilibria were found only with two-plan supports, and thresholds are exact only up to the oracle gap.

$\Gamma$-*maximin.* In all 108 (city, seed, $\delta$) cases the surcharge equilibrium differs from the precise equilibrium, and in 106 of them the precise equilibrium is not $\Gamma$-admissible. The share of demand served drops by 0–0.082 at $\delta = 0.1$ and by 0.17–0.77 at $\delta = 1$.

*Vehicle-count congestion.* The gadgets behave as proved: Proposition 31 has no pure equilibrium and threshold exactly $1/w$ (3 of 3 weights), and the kernel construction reproduces the kernel sets for all four criteria on 223 of 223 digraphs. Among random bilinear D2 games with three players, 6 of 400 have no pure equilibrium, against 0 of 150 D1 controls. Small routing-integrated D2 games always had a pure equilibrium (0 of 540), and on the city games of Cairo, Mumbai and Paris with a vehicle-count externality scaled by 1, 10, 100, 1 000, 10 000, all 180 best-response runs converged. Non-existence required adversarial plan menus; the D2 calibration is an assumption introduced for this test.

*7.8. Verification*

On 40 tiny games solved by exhaustive enumeration, the engine's profile is a pure equilibrium in every case and reaches the global potential minimum in 40 of 40 (best response 35, Frank–Wolfe with rounding 38, column generation 39; Table F.25). At city scale every base equilibrium, every verified grid best point (27 city-seeds) and every equilibrium of the price-of-anarchy experiment has regret at most $9.1 \cdot 10^{-13}$ USD against an independent fresh exact oracle. Because some oracle calls were not proven optimal, the guaranteed regret bound is nonzero for 10 profiles, at most 4.84 USD. These checks rely on the same branch-and-price method as the oracle; the pricer was checked against enumeration (Section 7.1).

## 8. Managerial implications

The model is stylised, several parameters are weak (Section 6) and the effects below are qualitative. They indicate what to test with local data, not what to do in any particular city.

*Where pricing matters.* In the model, the efficiency loss from competition is concentrated where service is incomplete. Where carriers already serve almost all demand (Cairo, Moscow, New York, Paris, Berlin), the price of anarchy is close to one and charges change little; kerb space and background traffic are the more relevant levers there. Where service is partial, the instrument with the largest effect is a per-delivery subsidy, and in every material case the best flat charge was a subsidy; it achieves 0.82–0.98 of the gain of a two-instrument charge in the three cities with a full grid. A congestion charge on vehicle-hours alone gains much less (mean 0.32–2.4% against 12.1–39.8%), but not nothing: in 7 of 30 city-seeds it reached at least 1.7%. The size of the gain depends strongly on the sampled slice, so the direction is better supported than the size. The ordering of the cities by price of anarchy holds in 7 of 10 seeds only. A subsidy has a fiscal cost that the welfare criterion with transfers does not count, access stays unequal in the low-service cities, and an objective that counts delay to other road users can reverse the recommendation.

*How a city could compute charges.* A city does not need to search a grid. With the carriers' routing oracle, DIP gives zone charges at 5.7–8.3% of a 63-point grid's calls, with a readable structure (a subsidy of roughly 37.5% of the fee divided by the number of carriers in the zone, corrected for congestion). Without the oracle, ARP needs only the zone totals after each posted charge, and the marginal tariff needs billing on own deliveries and the zone total. The price of observation-only pricing is selection risk: the flat rule can cycle or settle in a worse local optimum, and the tariff stops at a unilateral social optimum. A pilot with a few posted charges is a cheap check. Zone-specific charges are harder to administer than a flat subsidy.

*Levels depend on how carriers plan.* All levels in this paper describe carriers who plan against a guaranteed upper bound on route cost. With sample-average carriers, service at the unpriced equilibrium roughly doubles, the price of anarchy measured with the true cost falls, the leader's gain falls by half or more, and the subsidy that is best for the true-cost objective shrinks by 58–69%; the sign of the subsidy, the dominance of the delivery charge and the structure of DIP's charges do not change. A city should therefore calibrate the level of any charge on observed carrier behaviour, and treat our magnitudes as indicative only.

*For carriers and for modelling.* Wider travel-time boxes and wider beliefs about rivals reduce the volume a carrier finds worth serving. A carrier with enough data to estimate the law should consider that the p-box criterion costs expected cost (1.6–8.9% at up to 60 training days) in exchange for tail protection and a bound. For a modeller, exposure-based congestion gives a potential game with guaranteed pure equilibria, whereas vehicle-count congestion can make existence NP-complete; on our instances that worst case did not occur.

## 9. Limitations

- *Model.* Lots are all-or-nothing (S1), so part of the measured inefficiency may be an artifact of indivisibility. Congestion is linear and, within a zone, proportional to deliveries (S3, S4). Demand is sampled, market shares are equal in seven cities, and the congestion coupling is scaled by a slice multiplicity. Most operating parameters are country-level or rough; delivery fee, drops per vehicle and the number of carriers move welfare by tens of per cent over their plausible ranges (Table F.22), and the sign of the best per-delivery instrument depends on the fee-to-cost ratio.

- *Exactness and coverage.* 344 of 41,469 oracle calls were not proven optimal; for 10 profiles the guaranteed regret bound is positive (at most 4.84 USD). The grid is exact in full only for New Delhi, Buenos Aires, London, so the grid's gains in the other six cities are lower bounds; imprecision sweeps in Cairo, Moscow, New York, Paris, Berlin use three exact widths per sweep; the engine comparison, ZMC checks, theory tests and tiny-game verification use price-and-branch, whose equilibria were not exact in 10 of 27 city-seeds. Branch-and-price was tested only on instances with 22 customer zones.
- *Carriers' criterion and BRL.* All levels describe carriers who plan against the p-box upper bound; the robustness study used a heuristic oracle for sample-average and Wasserstein carriers, a coarse 16-point grid, three seeds, and price-and-branch for the p-box reference, and the leader's true-cost gain and best charge depend on the criterion (Section 7.6). The out-of-sample comparison of criteria uses one slice per city and a heuristic planner validated against branch-and-price in nine instances. BRL does not beat sample-average or Wasserstein plans on held-out cost, was not compared with resource-extension-function dominance, and was benchmarked on public instances only by an emulation (soft duration limits, four sectors), not by reproducing published results.
- *Leader methods.* DIP is no better than its social target, its selection is optimistic, its closed forms need D1 and one hours-per-delivery ratio, and the required subsidies under strong ambiguity aversion are unrealistic. ARP was run on four cities, can cycle, can settle in a worse local optimum, and has no guarantee for the pure game unless its fixed point is reached; Theorem 19(b) is conditional on potential minimisation. Neither method has a worst-equilibrium guarantee. The comparison with the grid is on a two-parameter family against zone-specific charges.
- *Theory.* The credal results assume affine coupling. The absorption tests rely on two-plan mixed equilibria and thresholds exact up to the oracle gap; the full rounding algorithm was not run on real data; D2 was tested only with a calibration introduced for the test. The Lean formalisation covers only the finite and algebraic cores listed in Section 5.9.
- *Evidence.* Seeds vary the sampled slice, not the calibration inputs. Ten seeds exist only for New Delhi, Buenos Aires, London; the other cities have 9 seeds for equilibria and three seeds with a reduced grid for grid-derived quantities (lower bounds), the imprecision sweeps and the criterion study use three seeds, and ARP was run in four cities. Several readings from the earlier three-seed analysis did not survive ten seeds: a congestion-only gain below 1.7% everywhere, and the price-of-anarchy ordering London > Buenos Aires > New Delhi in every seed. Earlier price-and-branch readings also failed (the role of the congestion charge in Mumbai, a price of stability always within 0.05 of the price of anarchy). Price-of-anarchy values come from at most six equilibria per city and seed. The broad social cost is driven by a proxy and supports no conclusion. Novelty searches were web and index searches, partly at the level of abstracts; the closest neighbours (14, 55, 59, 77, 37) should be compared in full text.

Open questions include the complexity of finding a pure E-admissible equilibrium above the guaranteed width (we conjecture PPAD-hardness), smoothness constants for the Cournot–congestion game, an absorption result for nonaffine congestion, worst-equilibrium guarantees for the leader, and carrier-level data that would replace our proxies.

## 10. Conclusion

A city that prices competing delivery carriers faces NP-hard best responses under probability-box travel times, a congestion game among the carriers, and a bilevel search over charges. BRL prices routes exactly under the dependence-robust criterion and, with branch-and-price, proved optimal robust plans on all nine metro instances with 22 customer zones; its window rule keeps fewer labels than weaker rules, but the gain shrinks on real data, the plans do not beat sample-average plans out of sample, and exact solution stops at about 40 customers. DIP made its target an exact equilibrium in every run and reached 1.07–1.23 times the best grid point's gain with 5.7–8.3% of its calls in the three cities with ten seeds, partly because its charges are zone-specific; against the social optimum the grid reaches 0.815–0.938 and DIP 1.00–1.02. ARP, which sees only zone totals, reached 0.933–0.995 of the social optimum's gain, contracts at a rate independent of instance size in the convexified game, and in the pure game can cycle or settle in a worse local optimum. Pure equilibria exist under exposure-based congestion and deciding existence is strongly NP-complete under vehicle-count congestion; the credal absorption width has an $(N-1)$ term that is attained at $N = 2$ and binds on the city instances, and the theorem is close to earlier work on approximate equilibria. On the nine cities, the price of anarchy is near one where service is complete and larger, with a wide spread over seeds and an ordering that holds in 7 of 10 seeds, where it is partial; a per-delivery subsidy captures most of the grid leader's gain; a congestion charge alone captures much less but sometimes more than 1.7%. The levels of these quantities describe carriers who plan against a guaranteed upper bound; with sample-average carriers service roughly doubles and the leader's gain falls by half or more, while orderings and policy directions persist. These findings come from stylised slices with weak parameters, and the novelty of each component is limited as stated in Section 1.

### Data and code availability

Code, processed data, parameter sources, result files, figures and the scripts that generate the tables and numbers of this paper are available at `https://github.com/vinhqdang/Competing-Carriers-under-Uncertainty`. Raw OpenStreetMap extracts are distributed under the Open Database License; GHS-POP is distributed by the European Commission Joint Research Centre.

### Declaration of competing interest

The author declares no competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.


### Funding

This research received no specific grant from any funding agency in the public, commercial or not-for-profit sectors.

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## Appendix A. Proofs for the equilibrium theory

The proofs in this appendix are taken from the theory notes in the repository (file `theory/main.tex`), with the notation of Section 3.

### *Appendix A.1. Classical tools*

**Theorem 39** (Shapley–Folkman, constructive vertex form; 49)**.** *Let $S_1, \dots, S_N \subset \mathbb{R}^d$ be finite and nonempty, $L = \operatorname{span} \bigcup_i (S_i - S_i)$ and $d_L = \dim L$. Index the elements of $S_i$ by $\mathcal{P}_i$ through an onto map $a_i$. For $\bar{x} \in \sum_i \operatorname{conv} S_i$, the polytope $\Lambda(\bar{x}) = \{\lambda \in \prod_i \Delta(\mathcal{P}_i) : \sum_i \sum_p \lambda_i(p) a_i(p) = \bar{x}\}$ is nonempty, and at every vertex $\lambda$ the set $J(\lambda) = \{i : |\operatorname{supp} \lambda_i| \geq 2\}$ has $|J(\lambda)| \leq \min\{N, d_L\}$.*

*Proof.* Nonemptiness: $\bar{x} = \sum_i y_i$ with $y_i \in \operatorname{conv} S_i$, and each $y_i$ is a convex combination of points $a_i(p)$. $\Lambda(\bar{x})$ is a bounded polyhedron in standard form $\{\lambda \geq 0 : H\lambda = h\}$ with columns $c_{i,p} = (a_i(p), e_i) \in \mathbb{R}^d \times \mathbb{R}^N$ and $h = (\bar{x}, \mathbf{1})$. A feasible $\lambda$ is a vertex if and only if the columns $\{c_{i,p} : \lambda_i(p) > 0\}$ are linearly independent. Fix a vertex and pick $p_i^0 \in \operatorname{supp} \lambda_i$ for each $i$. The family $\{c_{i,p_i^0}\}_i \cup \{c_{i,p} - c_{i,p_i^0} : p \in \operatorname{supp} \lambda_i \setminus \{p_i^0\}\}$ is obtained from the positive columns by an invertible (unitriangular) transformation, so it is linearly independent. Its second part consists of the vectors $(a_i(p) - a_i(p_i^0), 0) \in L \times \{0\}$, which number $\sum_i (|\operatorname{supp} \lambda_i| - 1) \geq |J(\lambda)|$. Linearly independent vectors in $L \times \{0\}$ number at most $d_L$. Hence $|J| \leq d_L$, and trivially $|J| \leq N$. □

*Proof of Lemma 23.* (a) $u_i(p; m, \xi)$ is affine in $(m, \xi)$ with fixed coefficients, and $P$ has finite first moments; linearity of the integral gives the identity. (b) Into: by definition. Onto: for $k = (m, \xi) \in K_i^\delta(\sigma)$, the Dirac law at $(m, \xi)$ is supported in $\mathbb{R}^d_{\geq 0} \times \mathbb{R}^Z_{\geq 0}$ because $m \geq 0$ and $\xi \in \Xi \subset \mathbb{R}^Z_{\geq 0}$; its means are $(m, \xi)$, so it belongs to $\mathcal{M}_i^\delta(\sigma)$. (c) Each criterion depends on $P$ only through the numbers $\mathbb{E}_P[u(p)]$ and $\mathbb{E}_P[u(p') - u(p)]$, which are affine functions of the mean by (a). By (b), ranging over $P$ is the same as ranging over $k \in K_i^\delta(\sigma)$, a product of compact intervals; continuous functions attain their extrema on it. □

*Proof of Lemma 24.* For a finite $\mathcal{P}$ and the compact convex box $K$ with continuous affine payoffs: $E \subseteq M$. If $p$ is optimal at $k_0$, then for every $p'$, $\bar{u}(p'; k_0) - \bar{u}(p; k_0) \leq 0$, so $\min_k[\bar{u}(p') - \bar{u}(p)] \leq 0$. $M \subseteq ID$. If $p$ is maximal, then for each $p'$ there is $k$ with $\bar{u}(p; k) \geq \bar{u}(p'; k)$; hence $\max \bar{u}(p; \cdot) \geq \bar{u}(p; k) \geq \bar{u}(p'; k) \geq \min \bar{u}(p'; \cdot)$. $\Gamma \subseteq M$. If $\min_k[\bar{u}(p'; k) - \bar{u}(p; k)] > 0$, then $\min \bar{u}(p') > \min \bar{u}(p)$, so $p$ is not $\Gamma$-maximin. These inclusions hold carrier by carrier and give the inclusions between equilibrium sets. *Monotonicity.* $K^\delta(c) \subseteq K^{\delta'}(c)$ for every $c$, and the realised centre $A_{-i}(\sigma)$ does not depend on $\delta$. E: a witness belief stays admissible. M: a dominance strict on the larger set is strict on the smaller one, so non-domination on the smaller set implies it on the larger one. ID: $\overline{\mathbb{E}}$ increases and every $\underline{\mathbb{E}}$ decreases as the set grows. □

**Example 40** ($\Gamma$-equilibria are not nested)**.** Take one carrier with a fixed external aggregate $c = (1, 2)$, $d = 2$, $M = I$, $w = (2, 1)$, no traffic term, and two plans with $\pi = 0$, $\varphi(A) = (1, 0)$, $\varphi(B) = (0, 1)$. The lower values are $-(1 + 2\delta)$ for $A$ and $-(2 + \delta)$ for $B$, so $\mathrm{ICNE}_\Gamma(0) = \{A\}$ and $\mathrm{ICNE}_\Gamma(2) = \{B\}$.

*Proof of Lemma 25.* A minimum of affine functions of the mixture $x$ is concave, and

$$\min_k \textstyle\sum_p x(p)\bar{u}_i(p; k) \geq \sum_p x(p) \min_k \bar{u}_i(p; k)$$

because the left side picks one $k$ for all $p$. With $\varphi, t \geq 0$ and $M \geq 0$, $\bar{u}_i(p; m, \xi)$ is coordinatewise nonincreasing in $(m, \xi)$. The box $K^\delta(c)$ has the coordinatewise maximum $c + \delta w$, which lies in it (clipping only removes points below 0), and $\Xi$ has the maximum $\overline{\mu}$. So $k^+ = (c + \delta w, \overline{\mu})$ minimises every $\bar{u}_i(p; \cdot)$ and the lower expectation is linear in the mixture. Substituting $k^+$ with $c = A_{-i}(\sigma)$ gives

$$\bar{u}_i(p; k^+) = \pi_i^\Gamma(p) - \langle \varphi_i(p), M A_{-i}(\sigma) \rangle,$$

the payoff of $G_\delta^\Gamma$; $\Gamma$-maximin is then best response in $G_\delta^\Gamma$. □

**Proposition 41** (Relaxed equilibria are mixed equilibria)**.** *Under Assumption A2, $x$ is a mixed Nash equilibrium of the finite game with payoffs $\bar{u}_i(\sigma_i; A_{-i}(\sigma), \hat{\xi})$ if and only if* $\operatorname{supp} x_i \subseteq \arg\max_p \bar{u}_i(p; \bar{A}_{-i}(x), \hat{\xi})$ *for all $i$. Such $x$ exist.*

*Proof.* Under the product law $x_{-i}$, the expected payoff of a pure plan $p$ is $\mathbb{E}[\bar{u}_i(p; A_{-i}, \hat{\xi})] = \bar{u}_i(p; \mathbb{E}A_{-i}, \hat{\xi})$ because $\bar{u}_i$ is affine in $A_{-i}$, and $\mathbb{E}A_{-i} = \bar{A}_{-i}(x)$ by linearity. A mixed profile is a Nash equilibrium if and only if each $x_i$ is supported on pure best responses to $x_{-i}$. Existence follows from Nash [86]. □

*Proof of Proposition 26.* For a unilateral change $p \to p'$ of carrier $i$, put $a = a_i(p)$, $a' = a_i(p')$ and $R = A_{-i}$. Then $\Psi(p') - \Psi(p) = \pi_i(p') - \pi_i(p) + \frac{1}{2}\|a'\|_M^2 - \frac{1}{2}\|a\|_M^2 - \frac{1}{2}\|R+a'\|_M^2 + \frac{1}{2}\|R+a\|_M^2 - \gamma\langle t_i(p') - t_i(p), \hat{\xi}\rangle$, which equals $\pi_i(p') - \pi_i(p) - \langle a' - a, MR\rangle - \gamma\langle t_i(p') - t_i(p), \hat{\xi}\rangle = u_i(p') - u_i(p)$, using $\varphi_i = a_i$. A maximiser $\sigma$ is a pure equilibrium [45]. The point $(A_{-i}(\sigma), \hat{\xi})$ lies in $K_i^\delta(\sigma)$ for every $\delta$, so $\sigma \in \mathrm{ICNE}_E(\delta)$. For $\Gamma$, Lemma 25 shows that $\mathrm{ICNE}_\Gamma(\delta)$ is the equilibrium set of $G_\delta^\Gamma$, which is again D1 with modified private profits and therefore has a potential and a pure equilibrium. □

*Proof of Theorem 27. (a)* Fix $i$ and $\delta \geq \hat{\delta}(p)$. The anchor $\bar{A}^*_{-i}$ is a sum of convex combinations of nonnegative vectors, hence nonnegative. By definition of $\hat{\delta}(p)$, $|\bar{A}^*_{-i,k} - A_{-i,k}(p)| \leq \hat{\delta}(p) w_k \leq \delta w_k$ for all $k$, so $(\bar{A}^*_{-i}, \hat{\xi}) \in K_i^\delta(p)$. By Lemma 23(b) the Dirac law at this point belongs to $\mathcal{M}_i^\delta(p)$; under it the expected payoff is $\bar{u}_i(\cdot; \bar{A}^*_{-i}, \hat{\xi})$, which $p_i \in B_i$ maximises. Hence $p_i$ is E-admissible for every $i$, and the inclusions follow from Lemma 24.

*(b), first term.* For any $p \in \prod_j B_j$, $A_{-i}(p) - \bar{A}^*_{-i} = \sum_{j \neq i}(a_j(p_j) - \bar{a}_j)$ and $|a_{jk}(p_j) - \bar{a}_{jk}| \leq r_{jk}$.

*(b), second term.* $\bar{A}^* \in \sum_j \operatorname{conv} a_j(B_j)$ because $\operatorname{supp} x_j^* \subseteq B_j$. Apply Theorem 39 with $\mathcal{P}_j$ replaced by $B_j$: a vertex $\lambda$ of $\Lambda(\bar{A}^*)$ exists and can be found by linear programming. Let $J = J(\lambda)$, $|J| \leq k^\sharp$, and $y_j = \sum_p \lambda_j(p) a_j(p)$. For $j \notin J$, $\lambda_j$ is a point mass at some $p_j \in B_j$ and $a_j(p_j) = y_j$; for $j \in J$ choose any $p_j \in \operatorname{supp} \lambda_j \subseteq B_j$. Since $\sum_j y_j = \bar{A}^* = \sum_j \bar{a}_j$,

$$A_{-i}(p) - \bar{A}^*_{-i} = \textstyle\sum_{j \in J}(a_j(p_j) - y_j) - (a_i(p_i) - \bar{a}_i). \tag{A.1}$$

For $j \in J$ both $a_{jk}(p_j)$ and $y_{jk}$ lie in $[\min_{B_j} a_{jk}, \max_{B_j} a_{jk}]$, so the sum is at most $k^\sharp \bar{\sigma} w_k$ in coordinate $k$, and the last term is at most $\bar{\rho} w_k$.

*(b), third term.* If $J = \emptyset$ or $\bar{\sigma} = 0$, the sum vanishes. Otherwise draw $p_j \sim \lambda_j$ independently for $j \in J$. For each $k$ the variables $(a_{jk}(p_j) - y_{jk})/w_k$, $j \in J$, are independent, have mean zero and range in an interval of length at most $\bar{\sigma}$. By Hoeffding's inequality and a union bound over $k$, $\Pr(\|E\|_{w,\infty} \geq t) \leq 2d \exp(-2t^2/(k^\sharp \bar{\sigma}^2))$, where $E$ is the sum in (A.1). This is below one for $t > t_0 = \bar{\sigma}\sqrt{k^\sharp \ln(2d)/2}$; outcomes are finitely many, so some outcome has $\|E\|_{w,\infty} \leq t_0$, and $\hat{\delta}(p) \leq t_0 + \bar{\rho}$ for it.

*(c)* Relaxed equilibria exist (Proposition 41). Use $r_{jk} \leq s_{jk}$, $\bar{\rho} \leq \bar{\sigma} \leq \sigma_{\max}$, $k^\sharp \leq d$ and $\sum_{j \neq i} r_{jk}/w_k \leq (N-1)\sigma_{\max}$; the order of thresholds follows from Lemma 24. *(d)* This was shown in (b); linear programming finds a vertex in time polynomial in $\sum_j |B_j|$ and $d$. □

*Proof of Corollary 28.* Let $m^* = \bar{A}^*_{-i}$ and $m = A_{-i}(p)$. For any $p'$, $\bar{u}_i(p'; m, \mu) - \bar{u}_i(p_i; m, \mu) = [\bar{u}_i(p'; m^*, \hat{\xi}) - \bar{u}_i(p_i; m^*, \hat{\xi})] - \langle \varphi_i(p') - \varphi_i(p_i), M(m - m^*)\rangle - \gamma\langle t_i(p') - t_i(p_i), \mu - \hat{\xi}\rangle$. The bracket is nonpositive because $p_i \in B_i$; each term $|M_k(\varphi_{ik}(p') - \varphi_{ik}(p_i))(m_k - m_k^*)|$ is at most $w_k M_k |\Delta\varphi_{ik}| \hat{\delta}(p)$, and $|\mu_z - \hat{\xi}_z| \leq \varpi_z$. □

*Proof of Proposition 31.* For carrier 1, $\bar{u}(A_1; n) - \bar{u}(B_1; n) = 6 - 3n$, so $A_1$ is optimal iff $n \leq 2$; for carrier 2, $\bar{u}(Y; n) - \bar{u}(X; n) = 6 - 3n$, so $Y$ is optimal iff $n \leq 2$. With two plans and a convex belief set, E-admissibility and maximality coincide: some $n$ in the box makes the plan weakly better. In $(A_1, X)$ carrier 2 sees $n = 1$ and needs a belief $\geq 2$; in $(A_1, Y)$ carrier 1 sees $n = 3$ and needs a belief $\leq 2$; in $(B_1, Y)$ carrier 2 sees $n = 3$ and needs $\leq 2$; in $(B_1, X)$ carrier 1 sees $n = 1$ and needs $\geq 2$. Each requirement holds iff $\delta w \geq 1$, and the other carrier

best-responds at the realised count. Hence $\mathrm{ICNE}_E(\delta) = \mathrm{ICNE}_M(\delta) = \emptyset$ for $\delta < 1/w$ and both equal the full profile set at $\delta = 1/w$; there is no pure Nash equilibrium. The unique mixed equilibrium makes each carrier indifferent, so the rival's mean count is 2, both mix $\frac{1}{2}$–$\frac{1}{2}$, $r_j = 1$, $\bar{\rho} = 1/w$ and $(N-1)\bar{\rho} = 1/w$. □

*Proof of Proposition 32.* Take two carriers with plan sets $\{A_1, B_1\}^Z$ and $\{X, Y\}^Z$, the one-zone data of Proposition 31 in each zone and no cross-zone interaction, and $N-2$ carriers with a single null plan. Payoffs are sums over zones and plan sets are products, so a carrier's regret is the sum of its zone regrets. In each zone and each of the four zone profiles, exactly one of the two carriers has regret 3 and the other 0 (e.g. in $(A_1, X)$ carrier 2 gains $2-(-1) = 3$ by switching to $Y$). So $\varepsilon_1 + \varepsilon_2 = 3Z$ with both multiples of 3, and $\max \geq 3\lceil Z/2 \rceil$. For the upper bound, the zone-wise mixed equilibrium has $\bar{\rho} = 1$, every $p \in \prod B_j$ has $\hat{\delta}(p) \leq 1$, and $L_i(p) \leq \sum_z 1 \cdot 1 \cdot |\Delta\tau_z| = 3Z$. □

*Proof of Theorem 33. Membership.* The witness is a profile $\sigma$. With $K$ the clipped box and $\Xi$ a point: E-admissibility of $\sigma_i$ is feasibility of the linear system $\{m \in K : \bar{u}_i(\sigma_i; m) \geq \bar{u}_i(p'; m)\ \forall p'\}$; maximality requires $\max_{m \in K}[\bar{u}_i(\sigma_i; m) - \bar{u}_i(p'; m)] \geq 0$ for each $p'$, a linear function maximised over a box coordinatewise; ID and $\Gamma$ require maxima and minima of $\bar{u}_i(p; \cdot)$ over the box, also coordinatewise.

*Construction.* Let $V(D) = [n]$ and $\Delta^- = \max(1, \max \text{ in-degree})$. Create one zone per arc $e = (u, v)$, using only its vehicle coordinate, with $M_e = \gamma c_e = 1$, $w_e = 1$ and no deliveries. Set $\eta = 1/(2\Delta^- + 2)$. Carrier $i$ has the plan *out* with all data 0, and the plan *in* with $\pi_i = 1$ and, for each out-arc $e = (i, j)$, vehicles $n_{ie} = 1$ and exposure $\tau_{ie} = 1$; for each in-arc $e = (h, i)$, vehicles $n_{ie} = 2$ and exposure $\tau_{ie} = \eta$; zero elsewhere. This is D2, since $\varphi = \tau \neq a = n$.

*Payoff gap.* The gain of *in* at belief $m$ is $D_i(m) = 1 - \sum_{e=(i,j)} m_e - \eta \sum_{e=(h,i)} m_e$. At the realised aggregate of $\sigma^S$, $m_e = 2 \cdot \mathbf{1}[j \in S]$ for $e = (i, j)$ and $m_e = \mathbf{1}[h \in S]$ for $e = (h, i)$, so $D_i^{\mathrm{real}} = 1 - 2o_i - \eta h_i$ with $o_i = |N^+(i) \cap S|$ and $h_i = |N^-(i) \cap S| \leq \Delta^-$. If $o_i = 0$, $D_i^{\mathrm{real}} \geq 1 - \Delta^-/(2\Delta^- + 2) > \frac{1}{2}$; if $o_i \geq 1$, $D_i^{\mathrm{real}} \leq -1$.

*Robustness of the sign.* Over $K^\delta$, $|D_i(m) - D_i^{\mathrm{real}}| \leq \delta(\deg^+(i) + \eta \deg^-(i)) < \delta(\Delta^+ + \frac{1}{2})$, which is below $\frac{1}{2}$ for $\delta < 1/(2\Delta^+ + 1)$; clipping only shrinks the box. So $D_i$ has the sign of $D_i^{\mathrm{real}}$ on the whole box.

*All criteria select the realised strict best response.* If $D_i > 0$ on the box, *in* is the unique optimum at every belief, hence the only E-admissible plan; *out* is strictly dominated everywhere, so *in* is the only maximal plan; and $\min_K \bar{u}(\mathrm{in}) > 0 = \max_K \bar{u}(\mathrm{out})$, so *out* is interval-dominated and *in* is the unique $\Gamma$-maximin plan. The case $D_i < 0$ is symmetric. So $\sigma$ is admissible under every criterion iff each carrier plays *in* exactly when $o_i = 0$.

*Equilibria are kernels.* With $S = \{i : \sigma_i = \mathrm{in}\}$, the condition reads $i \in S \Rightarrow N^+(i) \cap S = \emptyset$ (independence) and $i \notin S \Rightarrow N^+(i) \cap S \neq \emptyset$ (absorption). NP-hardness follows from Chvátal [88]; all numbers are polynomially bounded, so hardness is strong; with $\Delta^+ \leq 2$ [89] the threshold is 1/5. □

*Proof of Corollary 34.* On the instances of Theorem 33, $\delta_0^E = 0$ if $D$ has a kernel; otherwise $\mathrm{ICNE}_E(\delta) = \emptyset$ for all $\delta < 1/(2\Delta^+ + 1)$, so $\delta_0^E \geq 1/(2\Delta^+ + 1)$. A multiplicative approximation outputs 0 in the first case and a positive number in the second; an additive one separates the cases when its error is below $1/(2\Delta^+ + 1)$. □

**Proposition 42** (The relaxation step is PPAD-hard under D2)**.** *Every bimatrix game $(R, C) \in \mathbb{Q}^{m\times n} \times \mathbb{Q}^{m\times n}$ is, in polynomial time, the precise game of a D2 instance with $N = 2$ and $d = m + n$. Hence computing a mixed equilibrium under D2 is PPAD-hard.*

*Proof.* Let $M = I_{m+n}$, no traffic term, $\bar{R} = \max R$ and $\bar{C} = \max C$. Carrier 1 has plans $p \in [m]$ with $a_1(p) = e_p$, $\varphi_1(p) = \sum_q (\bar{R} - R_{pq}) e_{m+q} \geq 0$ and $\pi_1(p) = \bar{R}$; carrier 2 has plans $q \in [n]$ with $a_2(q) = e_{m+q}$, $\varphi_2(q) = \sum_p (\bar{C} - C_{pq}) e_p \geq 0$ and $\pi_2(q) = \bar{C}$. Then $u_1(p; a_2(q)) = R_{pq}$ and $u_2 = C_{pq}$. By Proposition 41, relaxed equilibria are exactly the Nash equilibria of $(R, C)$, and computing these is PPAD-complete [90, 91]. □

*Proof of Lemma 35.* Write $k^t = (A_{-i}(\sigma), \mu) \in K_i^\delta(\sigma)$ and $\Delta(p', k) = \bar{u}_i(p'; k) - \bar{u}_i(\sigma_i; k)$, affine in $k$. For any $k$ in the (unclipped) box, $|\Delta(p', k) - \Delta(p', k^t)| \leq \delta \sum_k w_k M_k |\Delta\varphi_{ik}| + \gamma \sum_z \varpi_z |\Delta t_{iz}|$. *E:* take the witness $k$, at which $\Delta(p', k) \leq 0$. *M:* for each $p'$, maximality and compactness give $k_{p'}$ with $\Delta(p', k_{p'}) \leq 0$. *ID:* $\bar{u}_i(p'; k^t) - \bar{u}_i(\sigma_i; k^t) \leq [\bar{u}_i(p'; k^t) - \min_K \bar{u}_i(p')] + [\max_K \bar{u}_i(\sigma_i) - \bar{u}_i(\sigma_i; k^t)]$, since $\max_K \bar{u}_i(\sigma_i) \geq \min_K \bar{u}_i(p')$, and each bracket is at most $\delta H_i + \gamma \Theta_i$. □

*Proof of Proposition 36.* The minimum over a growing finite set does not increase and the maximum does not decrease (Lemma 24). Each profile enters $\mathrm{ICNE}_C(\delta)$ at a single threshold and never leaves, so the sets change at most at $\prod_i |\mathcal{P}_i|$ values. For $\Gamma$ see Example 40. □

**Proposition 43** (Price of anarchy under smoothness; conditional)**.** *Assume $W(\sigma) \geq \sum_i U_i(\sigma)$ for all $\sigma$, $W^* > 0$, and that the precise game is $(\lambda, \mu)$-smooth: $\sum_i U_i(\sigma_i^*, \sigma_{-i}) \geq \lambda W(\sigma^*) - \mu W(\sigma)$ for all $\sigma$ and an optimal $\sigma^*$. For $C \in \{E, M, ID\}$ let $\mathcal{E}(\delta) = \max_{\sigma \in \mathrm{ICNE}_C(\delta)} \sum_i \varepsilon_i(\sigma)$ from Lemma 35. If $\lambda W^* > \mathcal{E}(\delta)$, then $\mathrm{PoA}_C(\delta) \leq (1 + \mu)/(\lambda - \mathcal{E}(\delta)/W^*)$.*

*Proof.* For $\sigma \in \mathrm{ICNE}_C(\delta)$: $W(\sigma) \geq \sum_i U_i(\sigma) \geq \sum_i [U_i(\sigma_i^*, \sigma_{-i}) - \varepsilon_i] \geq \lambda W^* - \mu W(\sigma) - \mathcal{E}(\delta)$. Rearrange. □

*Proof of Proposition 37.* With $\pi_i$ as in Section 3 and $x = \tau$, direct expansion gives $W - \Psi_{\hat{\mu}} = \sum_z [\frac{b_z}{2} \sum_i q_{iz}^2 - \frac{\gamma c_z}{2}(T_z^2 - \sum_i \tau_{iz}^2)]$; the $\alpha$, $\hat{\mu}$ and $C_i$ terms cancel. By Cauchy–Schwarz, $\sum_i \tau_{iz}^2 \geq T_z^2/N$, and $\sum q^2 \geq 0$, so $W(\sigma) \geq \Psi_{\hat{\mu}}(\sigma) - \sum_z \frac{\gamma c_z}{2}(1 - \frac{1}{N}) T_z(\sigma)^2$. Next, $\Psi_{\hat{\mu}}(\sigma) \geq \Psi_{\hat{\mu}}(\sigma^*) - \eta$, and $\Psi_{\hat{\mu}}(\sigma^*) \geq W^* - \sum_z \frac{b_z}{2} \sum_i q_{iz}^{*2}$ because $T^2 \geq \sum \tau^2$ for $\tau \geq 0$. Membership of a potential maximiser in every $\mathrm{ICNE}_E(\delta)$ is Proposition 26, extended to M and ID by Lemma 24. □

**Theorem 44** (Exact thresholds and regret brackets)**.** *Let $p$ be a pure profile. (a) The exact E-threshold of carrier $i$ is the optimal value of the linear program $A_i^E(p) = \min\{\delta \geq 0 : m \geq 0,\ \xi \in \Xi,\ |m_k - A_{-i,k}(p)| \leq \delta w_k\ \forall k,\ \bar{u}_i(p_i; m, \xi) \geq \bar{u}_i(p'; m, \xi)\ \forall p' \in \mathcal{P}_i\}$ ($+\infty$ if infeasible), and $p \in \mathrm{ICNE}_E(\delta)$ iff $\delta \geq \max_i A_i^E(p)$. With an exact oracle and implicit $\mathcal{P}_i$, Kelley's cutting-plane method on this program terminates finitely. (b) For anchors $m^i \geq 0$, $\xi^i \in \Xi$ with $p_i \in \arg\max \bar{u}_i(\cdot; m^i, \xi^i)$, $A_i^E(p) \leq \|A_{-i}(p) - m^i\|_{w,\infty}$. (c) For a heuristic plan $h_i$ and a valid bound $\mathcal{U}_i \geq \max_{p'} \bar{u}_i(p'; m, \xi)$, the regret of $p_i$ lies in $[\bar{u}_i(h_i) - \bar{u}_i(p_i),\ \mathcal{U}_i - \bar{u}_i(p_i)]$.*

*Proof.* (a) By Lemma 23, $p_i$ is E-admissible at width $\delta$ iff the system with that $\delta$ is feasible; feasibility is upward closed in $\delta$ (Lemma 24) and the infimum of an LP is attained when finite. Kelley's method adds the oracle's best plan at the current $(m, \xi)$ as a violated constraint; a plan already present cannot be violated, so cuts are distinct and finitely many. (b) The anchor is feasible with $\delta = \|A_{-i}(p) - m^i\|_{w,\infty}$. (c) Immediate from the definitions. □

**Theorem 45** (Rounding guarantee under D2)**.** *Under D2, if the relaxation phase returns a mixed equilibrium $x$, the sparsify phase (a feasibility LP over $\Lambda(\bar{A}(x))$ with plan sets $\operatorname{supp} x_j$) and the round phase return $p \in \prod_j B_j(x)$ with $p \in \mathrm{ICNE}_E(\delta)$ for all $\delta \geq \hat{\delta}(p)$; $\hat{\delta}(p)$ satisfies the second bound of Theorem 27(b) for any choice in the round phase and the third for a successful sample. Sparsify and round run in time polynomial in $\sum_j |\operatorname{supp} x_j|$ and $d$.*

*Proof.* The sparsify LP's vertices have $|J| \leq k^\sharp$ by Theorem 39. The rest is Theorem 27(a), (b) and (d). □

The double-oracle relaxation [92] stops, with exact restricted solves and exact oracles, after at most $\sum_i |\mathcal{P}_i|$ rounds at a relaxed equilibrium: at termination every support plan attains the restricted maximum and no plan exceeds it, and each non-terminal round adds a new plan.

## Appendix B. Proofs for BRL

*Proof of Theorem 1.* Write $S = \sum_{a \in r} X_a$ (arcs counted with multiplicity). Fix $j$ and $\Theta = \Theta_j$.

*Step 1 (pointwise inequality).* For reals $x_a$, $d_a$ with $\sum_a d_a \leq \Theta$, $\left(\sum_a x_a - \Theta\right)^+ \leq \left(\sum_a (x_a - d_a)\right)^+ \leq \sum_a (x_a - d_a)^+$, because $t \mapsto t^+$ is nondecreasing and subadditive.

*Step 2 (upper bound).* Let $P \in \mathcal{M}$ with marginals $F_a$. Taking expectations, $\mathbb{E}_P (S - \Theta)^+ \leq \sum_a \mathbb{E}_{F_a} (X_a - d_a)^+$. Since $F_a \geq \underline{F}_a$ pointwise, $F_a$ is first-order dominated by $\underline{F}_a$, and $x \mapsto (x - d)^+$ is nondecreasing, so $\mathbb{E}_{F_a} (X_a - d_a)^+ \leq \overline{\Pi}_a(d_a)$; likewise $\mathbb{E}_{F_a} X_a \leq \bar{x}_a$. Hence $\mathbb{E}_P \phi(S) \leq c_0 + \gamma \sum_a \bar{x}_a + \sum_j \rho_j \sum_a \overline{\Pi}_a(d_a^j)$ for every $P$ and admissible $d^j$.

*Step 3 (attainment).* Let $U$ be uniform on $(0,1)$, $\bar{X}_a = \overline{Q}_a(U)$ and $\bar{S} = \Sigma(U)$ with $\Sigma(u) = \sum_a \overline{Q}_a(u)$; then $\bar{X}_a \sim \underline{F}_a$ and the law of $(\bar{X}_a)$ belongs to $\mathcal{M}$. Let $u^* = \sup\{u \in (0,1] : \Sigma(u) \leq \Theta\}$ ($\sup \emptyset = 0$). If $u^* = 1$, put $d_a = \overline{Q}_a(1)$; then $\sum_a d_a \leq \Theta$ and both sides of Step 1 vanish almost surely. If $u^* < 1$, put $\ell_a = \overline{Q}_a(u^*)$ (or $\overline{Q}_a(0^+)$ if $u^* = 0$) and $h_a = \overline{Q}_a(u^{*+})$; then $\sum_a \ell_a \leq \Theta$ when $u^* > 0$ and $\sum_a h_a \geq \Theta$. If $u^* = 0$ choose $d_a \leq \ell_a$ with $\sum_a d_a = \Theta$; otherwise choose $d_a \in [\ell_a, h_a]$ with $\sum_a d_a = \Theta$. On $\{U \leq u^*\}$ all $\bar{X}_a \leq d_a$ and both sides of Step 1 are 0; on $\{U > u^*\}$ all $\bar{X}_a \geq d_a$ and $\sum_a \left(\bar{X}_a - d_a\right)^+ = \bar{S} - \Theta = \left(\bar{S} - \Theta\right)^+$. So Step 1 holds with equality almost surely, and the comonotone law attains the bound of Step 2; the minima are attained.

*Step 4.* $\mathbb{E}\phi(\bar{S}) = \int_0^1 \phi(\Sigma(u))du$ because $\bar{S} = \Sigma(U)$.

*Step 5 (integrality).* With integer quantiles and $\Theta$, the numbers $\ell_a$, $h_a$, $\overline{Q}_a(1)$ are integers and integer vectors with the required properties exist in each case of Step 3. □

*Proof of Theorem 2.* Substitute the budget form of Theorem 1 into the objective and merge the two minimisations. □

**Lemma 46** (Concatenation is inf-convolution)**.** *For a label $p$ and a completion $q$, $\overline{\mathbb{E}}[\phi(T_{p \oplus q})] - c_0 = f_{p \oplus q}(T_{\max}) = \min_D \{f_p(D) + f_q(T_{\max} - D)\}$.*

*Proof.* The stop-loss terms are separable over arcs. Split the budget vector into the arcs of $p$ and of $q$, put $D = \sum_{a \in p} d_a$, minimise first over the arcs of $p$ with budget $D$ and then over those of $q$ with budget $T_{\max} - D$, and apply Theorem 1 to each part. The mean terms add. □

**Lemma 47** (Completion window)**.** *Let $C_{\min}(v) \le S_{q,1}$ for every completion $q$ of labels ending at $v$. For every completion $q$, the minimum in Lemma 46 is attained at some $D^* \le \overline{D}(p)$; if $D^* < \underline{D}(p)$ then $D^* \le S_{p,1}$.*

*Proof.* Let $\Phi_q(D) = f_p(D) + f_q(T_{\max} - D)$. The map $D \mapsto f_q(T_{\max} - D)$ is convex and nondecreasing with slope in $[0, \rho]$, equal to $\rho$ when $T_{\max} - D < S_{q,1}$; the slope of $f_p$ lies in $[-\rho, 0]$ and is 0 for $D > S_{p,L}$. (i) On $D > T_{\max} - S_{q,1}$, $\Phi_q$ has slope $\ge 0$, so a minimiser exists with $D \le T_{\max} - S_{q,1} \le T_{\max} - C_{\min}(v)$. (ii) On $D > S_{p,L}$, $\Phi_q$ has slope $\ge 0$, so decreasing $D$ to $S_{p,L}$ does not increase it. Applying (i) and then (ii) gives $D^* \le \overline{D}(p)$. The last claim is immediate from $\underline{D}(p) \le S_{p,1}$. □

*Proof of Theorem 4. Validity.* Let $q$ be a completion of $p_2$; since $\mathcal{V}(p_1) \subseteq \mathcal{V}(p_2)$ and $\ell(p_1) \le \ell(p_2)$, it is a completion of $p_1$. By Lemma 47 there is a minimiser $D^* \le \overline{D}(p_2)$ for $p_2 \oplus q$. Let $g$ be the difference in Definition 3. If $D^* \in \mathcal{W}(p_2)$, $g(D^*) \le 0$. Otherwise $D^* < \underline{D}(p_2) \le S_{p_2,1}$; on $(-\infty, S_{p_2,1}]$, $f_{p_2}$ has slope $-\rho$ and $f_{p_1}$ slope $\ge -\rho$, so $g$ is nondecreasing there and $g(D^*) \le g(\underline{D}(p_2)) \le 0$. In both cases, by Lemma 46, $f_{p_1 \oplus q}(T_{\max}) - y(p_1) \le f_{p_1}(D^*) + f_q(T_{\max} - D^*) - y(p_1) \le f_{p_2 \oplus q}(T_{\max}) - y(p_2)$.

*Exactness.* Call a label processed if it is popped while not deleted. Let OPT be the optimum and $\Omega$ the set of pairs $(o, q)$ with $o$ a created label, $q$ a completion of $o$ and reduced cost of $o \oplus q$ equal to OPT. The pair (depot label, optimal route) is in $\Omega$ and the depot label is processed. Among pairs in $\Omega$ with $o$ processed choose one with the fewest arcs $|q|$. If $|q| = 1$, closing $o$ records a route of value OPT. If $|q| \ge 2$, let $(v, j)$ be the first arc of $q$, $q'$ the rest and $o_1 = o \oplus (v, j)$, which is created when $o$ is processed; $(o_1, q') \in \Omega$. If $o_1$ is processed, $|q'| < |q|$ contradicts minimality. Otherwise $o_1$ was discarded or deleted because a live label $o_2$ functionally dominates it; by validity $(o_2, q') \in \Omega$, and if $o_2$ is not processed it was deleted later by some $o_3$, with $(o_3, q') \in \Omega$ by transitivity. The moments at which these labels cease to be live strictly increase and labels are finitely many, so the chain ends at a processed label with $(o_k, q') \in \Omega$ and $|q'| < |q|$, a contradiction. □

**Proposition 48** (Comparison with vector dominance)**.** *Assume $\mathcal{V}(p_1) \subseteq \mathcal{V}(p_2)$, $\ell(p_1) \le \ell(p_2)$ and $y(p_1) \ge y(p_2)$. (a) If $S_{p_1} \le S_{p_2}$ componentwise, $p_1$ functionally dominates $p_2$. (b) In the sector model $X_a = \sum_k \tau_{a,k} \xi_k$ with probability boxes on the multipliers, $\tau_{p_1} \le \tau_{p_2}$ componentwise implies $S_{p_1} \le S_{p_2}$. (c) Neither converse holds.*

*Proof.* (a) $f$ is nondecreasing in each $S_l$. (b) The upper quantiles of the multipliers are nonnegative. (c) With $L = 2$, $\omega = (\frac{1}{2}, \frac{1}{2})$, $\gamma = 0$, $\rho = 1$ and equal duals: $S_{p_1} = (1, 1)$ and $S_{p_2} = (0, 2.2)$ give $f_{p_1} \le f_{p_2}$ on $\mathbb{R}$ (values $(1, 1.1)$, $(0, 0.6)$, $(0, 0)$ at $D = 0, 1, 2.2$, constant difference $-0.1$ below 0) although $S_{p_1,1} > S_{p_2,1}$. □

**Proposition 49** (A single risk level is not exact)**.** *For every $u$, $\min_{\text{plan}} \sum_r \overline{\mathbb{E}}[\phi(T_r)] \ge \max_u \min_{\text{plan}} \sum_r [c_0 + \gamma \bar{x}_r + \rho(\mathcal{H}_r(u) - (1-u)T_{\max})]$, where $\mathcal{H}_r(u) = \int_u^1 \sum_{a \in r} \overline{Q}_a(v)\, dv$; the inequality can be strict and the best fixed-$u$ plan need not be optimal.*

*Proof.* The maximum over $u$ inside each route is at least its value at a common $u$, and $\min \max \ge \max \min$. Strictness is shown by exact computations by enumeration and set partitioning (Section 7.1); structurally, the optimal level of a route is its worst-case on-time level, which differs between routes of the same plan. □

## Appendix C. Zone-market clearing (ZMC)

ZMC is the first phase of the equilibrium engine (Section 4.3). It is stated here with its guarantees and proofs; Remark 5 gives the negative empirical result.

ZMC targets the D1 game in cost form. Let $c_i(p) = -\pi_i(p) + \gamma\langle t_i(p), \hat{\xi}\rangle$ (charges included), $\tilde{c}_i = c_i - \frac{1}{2}\|a_i\|_M^2$, and assume $M_k > 0$ on the active coordinates. Then $\Phi(\sigma) = \sum_i \tilde{c}_i(\sigma_i) + \frac{1}{2}\|A(\sigma)\|_M^2 = -\Psi_{\hat{\xi}}(\sigma)$ is an exact potential (Proposition 26). For a common price $\lambda \in \mathbb{R}^d$, the pricing oracle returns $O_i(\lambda) = \arg\min_p\{\tilde{c}_i(p) + \langle\lambda, a_i(p)\rangle\}$, which is the oracle of Assumption A3 with $Q = M$. Let $V_i(\lambda)$ be its value, $\mathcal{D}(\lambda) = \sum_i V_i(\lambda) - \frac{1}{2}\|\lambda\|_{M^{-1}}^2$ the price dual of the convexified potential, $\lambda^*$ its unique maximiser, and $\bar{\Phi}^* = \max\mathcal{D} \le \min\Phi$ (Lemma 53).

**Theorem 50** (Clearing identity)**.** *Let $\sigma_i \in O_i(\lambda)$ for all $i$ and let $e = M^{-1}\lambda - A(\sigma)$ be the clearing error (supply minus demand at price $\lambda$). Then (a) $\Phi(\sigma) - \mathcal{D}(\lambda) = \frac{1}{2}\|e\|_M^2$, so $\Phi(\sigma) - \min\Phi \le \frac{1}{2}\|e\|_M^2$; (b) carrier $i$'s Nash regret is at most $\frac{1}{2}\sum_{k\in\mathcal{Z}_i} M_k e_k^2$, where $\mathcal{Z}_i$ is the set of coordinates in which its plans differ; (c) $\sigma_i$ is an exact best response to the belief $m_i = M^{-1}\lambda - a_i(\sigma_i) = A_{-i}(\sigma) + e$, hence E-admissible at every width $\delta \ge \max_{k\in\mathcal{Z}_i}|e_k|/w_k$ (if $m_{ik} \ge 0$ on $\mathcal{Z}_i$); (d) $e = 0$ implies that $\sigma$ minimises $\Phi$ and is a pure Nash equilibrium; (e) conversely, exact clearing is attainable if and only if the convexification of the potential is tight at a pure profile.*

Part (a) is Fenchel–Young equality and a standard duality-gap fact. The game-specific content is in (b) and (c): the own-contribution term of the absorption bound cancels, because under D1 the own quadratic term turns a pricing-optimal plan into a best response to the anchor $m_i$. Finding a good pure equilibrium thus becomes a price search: find one $\lambda$ at which the carriers' pricing answers nearly clear every zone. Because $\mathcal{D}$ is concave, the aggregate response is antitone in $\lambda$, and along each coordinate the sign of the residual $s_k = A_k(\lambda) - \lambda_k/M_k$ localises $\lambda_k^*$ up to $M_k\chi_k$, where $\chi_k$ is the cross-zone variation: how far other coordinates' prices can move the response in coordinate $k$ at a fixed own price.

**Theorem 51** (Bracket invariant and dimension-free rate)**.** *Start from brackets $[0, M_k A_k^{\max}]$ and, after each round, move the lower (upper) end of bracket $k$ to $\lambda_k$ when $s_k > 0$ ($s_k < 0$). Then $\operatorname{dist}(\lambda_k^*, [l_k, u_k]) \le M_k\chi_k$ after every round. If the bracket midpoint is queried at least every other round, after $T$ rounds ($NT$ oracle calls, independent of $d$) every point of the bracket satisfies $|\lambda_k - \lambda_k^*| \le M_k A_k^{\max} 2^{-\lceil T/2\rceil} + M_k\chi_k$ for all $k$. If plan sets are separable across coordinates, $\chi = 0$.*

The floor $M_k\chi_k$ is genuine when zones are complementary (Example 54). Algorithm 4 alternates midpoint steps (for the rate) with secant steps on isotonic fits of each coordinate's response (for speed), then rounds over the collected plans and finishes with best-response polish.

**Theorem 52** (Guarantee of ZMC)**.** *With exact oracles, ZMC uses $NT$ pricing calls before the polish; the polish stops after finitely many best-response calls at a pure Nash equilibrium $\sigma$ with $\Phi(\sigma) \le \bar{\Phi}^* + \frac{1}{2}\min_t\|e^t\|_M^2$; and $\sigma \in \mathrm{ICNE}_C(\delta)$ for all $\delta \ge 0$ and $C \in \{E, M, ID\}$. Run on the surcharged game of Lemma 25, it returns an exact $\Gamma$-maximin equilibrium at width $\delta$.*

Charges translate the supply line: under charges $t$ the equilibrium gross price is 1-Lipschitz in $t$ and convexified aggregates respond monotonically (Proposition 55), so brackets and plan pools warm-start the next evaluation when a city leader searches over charges.

**Algorithm 4** ZMC: zone-market clearing for pure equilibria of the D1 game

**Require:** pricing and best-response oracles; rounds $T$

$l \leftarrow 0$, $u \leftarrow MA^{\max}$, $\lambda \leftarrow (l+u)/2$; plan pools $\mathcal{Q}_i \leftarrow \emptyset$

**for** $t = 0, \dots, T-1$ **do**

  for all $i$ in parallel: $\sigma_i^t \leftarrow O_i(\lambda)$; $\mathcal{Q}_i \leftarrow \mathcal{Q}_i \cup \{\sigma_i^t\}$ ▷ $N$ routing solves

  $s \leftarrow A(\sigma^t) - M^{-1}\lambda$; update every bracket by the sign of $s_k$ (Theorem 51)

  **if** $t$ odd: $\lambda \leftarrow (l+u)/2$; **else** for each $k$: secant step on an isotonic fit of $(\lambda_k, A_k)$, clipped to $[l_k, u_k]$

$\sigma^{\text{price}} \leftarrow$ the pricing profile with the smallest $\frac{1}{2}\|e^t\|_M^2$

$\sigma^{\text{round}} \leftarrow$ restricted master over the pools and derandomised rounding (no oracle calls)

$\sigma \leftarrow$ the better of the two; sequential best responses until no carrier improves

**return** $\sigma$

*Prior art and limits.* Price decomposition with a pricing oracle is classical (Dantzig–Wolfe, column generation and bundle methods; 100); learning separable piecewise-linear concave approximations from sampled slopes is SPAR [101]; Walrasian prices with indivisible goods are computed from demand oracles by Paes Leme and Wong [102]; and Duguet et al. [103] relate the regret of price-allocation pairs to duality and integrality gaps in combinatorial markets, which is the closest conceptual neighbour of Theorem 50(a)–(b). We claim novelty only for the own-term cancellation with credal anchors (Theorem 50(b),(c),(e)), the sign-bracket invariant with the cross-variation floor (Theorem 51) and the method they define. ZMC has no D2 version, does not bound the clearing error of pure answers a priori, and spends $NT$ calls even on easy games.

*Proofs*

**Lemma 53** (Price duality)**.** *Let* $\bar{\Phi}(x) = \sum_i \sum_p x_i(p)\tilde{c}_i(p) + \frac{1}{2}\|\bar{A}(x)\|_M^2$ *on* $\prod_i \Delta(\mathcal{P}_i)$, $\bar{\Phi}^* = \min \bar{\Phi}$ *and* $\mathcal{B} = \prod_k [0, M_k A_k^{\max}]$ *with* $A_k^{\max} = \sum_i \max_p a_{ik}(p)$. *(a)* $\mathcal{D}$ *is strongly concave with a unique maximiser* $\lambda^*$*, and* $\max \mathcal{D} = \bar{\Phi}^* \le \min_\sigma \Phi(\sigma)$. *(b) All minimisers of* $\bar{\Phi}$ *share the aggregate* $\bar{A}^*$*, and* $\lambda^* = M\bar{A}^* \in \mathcal{B}$. *(c)* $M^{-1}\lambda^* \in R(\lambda^*)$*, where* $R(\lambda) = \partial^+ \sum_i V_i(\lambda)$.

*Proof.* (a) $V = \sum_i V_i$ is concave and finite and $\frac{1}{2}\|\lambda\|_{M^{-1}}^2$ strongly convex, so $\mathcal{D}$ is strongly concave, continuous and coercive, with a unique maximiser. For $L(x, \lambda) = \sum_i \langle x_i, \tilde{c}_i \rangle + \langle \lambda, \bar{A}(x) \rangle - \frac{1}{2}\|\lambda\|_{M^{-1}}^2$, affine in $x$ on a compact convex set and concave in $\lambda$, Sion's theorem gives $\min_x \sup_\lambda L = \sup_\lambda \min_x L$. For fixed $x$ the supremum is at $\lambda = M\bar{A}(x)$ with value $\bar{\Phi}(x)$; for fixed $\lambda$, $\min_x L = \mathcal{D}(\lambda)$ because a linear function on a product of simplices is minimised at a vertex carrier by carrier. Pure profiles are vertices, where $\bar{\Phi} = \Phi$. (b) Two minimisers with different aggregates would contradict strict convexity of $\frac{1}{2}\|\cdot\|_M^2$. A minimiser $x^*$ and $\lambda^*$ form a saddle point, and the unique maximiser of $L(x^*, \cdot)$ is $M\bar{A}(x^*)$. Bounds on $\bar{A}^*$ follow from nonnegativity. (c) This is $0 \in \partial^+ \mathcal{D}(\lambda^*) = R(\lambda^*) - M^{-1}\lambda^*$, by the sum rule for concave functions [104]. □

*Proof of Theorem 50.* For any pure $\sigma$ and $\lambda$, with $A = A(\sigma)$ and $e = M^{-1}\lambda - A$,

$$\begin{aligned} \sum_i [\tilde{c}_i(\sigma_i) + \langle \lambda, a_i(\sigma_i) \rangle] - \tfrac{1}{2}\|\lambda\|_{M^{-1}}^2 &= \textstyle\sum_i \tilde{c}_i(\sigma_i) + \langle M(A+e), A \rangle - \tfrac{1}{2}\|A+e\|_M^2 \\ &= \Phi(\sigma) - \tfrac{1}{2}\|e\|_M^2. \end{aligned} \tag{$*$}$$

(a) If $\sigma_i \in O_i(\lambda)$ the left side is $\mathcal{D}(\lambda) \le \bar{\Phi}^* \le \min \Phi$. (b) Fix $i$, $a = a_i(\sigma_i)$, $a' = a_i(p)$, $\Delta = a' - a$ (zero outside $\mathcal{Z}_i$). Since $MA_{-i} = \lambda - M(a+e)$, the change in carrier $i$'s cost $c_i + \langle a_i, MA_{-i}\rangle$ equals $[\tilde{c}_i(p) + \langle \lambda, a'\rangle] - [\tilde{c}_i(\sigma_i) + \langle \lambda, a\rangle] + \frac{1}{2}\|\Delta\|_M^2 - \langle \Delta, Me\rangle \ge -\frac{1}{2}\sum_{k\in\mathcal{Z}_i} M_k e_k^2$, minimising coordinatewise at $\Delta_k = e_k$. (c) For any $p$, $c_i(p) + \langle a', Mm_i\rangle = [\tilde{c}_i(p) + \langle \lambda, a'\rangle] + \frac{1}{2}\|a' - a\|_M^2 - \frac{1}{2}\|a\|_M^2$, minimised by $\sigma_i$. Coordinates outside $\mathcal{Z}_i$ shift all plans' costs equally, so the belief can be replaced by one equal to $A_{-i}$ there; within $\mathcal{Z}_i$ it deviates from $A_{-i}$ by $|e_k| \le \delta w_k$, and under D1 the Dirac law at it lies in the credal set (Lemma 23). (d) If $e = 0$, $\Phi(\sigma) = \mathcal{D}(\lambda) \le \bar{\Phi}^* \le \min \Phi \le \Phi(\sigma)$; a global minimiser of an exact potential is a pure equilibrium, and uniqueness gives $\lambda = \lambda^*$. (e) If $\Phi(\sigma^\circ) = \bar{\Phi}^*$, Lemma 53(b) gives $\lambda^* = MA(\sigma^\circ)$, so $e = 0$, and $(*)$ shows that $\sum_i \{\tilde{c}_i(\sigma_i^\circ) + \langle \lambda^*, a_i(\sigma_i^\circ)\rangle - V_i(\lambda^*)\} = 0$, a sum of nonnegative terms. □

*Proof of Theorem 51.* Let $\chi_k = \sup\{\sup_{r\in R(\lambda)} \inf_{r'\in R(\lambda')} |r_k - r'_k| : \lambda, \lambda' \in \mathcal{B},\ \lambda_k = \lambda'_k\}$. *Key inequality.* Let a query $\lambda \in \mathcal{B}$ have $\lambda_k = \mu$ and response $r \in R(\lambda)$. Suppose $s_k > 0$ but $\mu - \lambda_k^* > M_k\chi_k$. Let $\lambda'$ equal $\lambda^*$ except $\lambda'_k = \mu$; then $\lambda' \in \mathcal{B}$. With $r^* = M^{-1}\lambda^* \in R(\lambda^*)$, antitonicity of $R$ gives $r'_k \le \lambda_k^*/M_k$ for every $r' \in R(\lambda')$, and the definition of $\chi_k$ gives $r' \in R(\lambda')$ with $r_k \le r'_k + \chi_k + \eta$ for any $\eta > 0$. Hence $s_k = r_k - \mu/M_k \le -(\mu - \lambda_k^* - M_k\chi_k)/M_k + \eta < 0$ for small $\eta$, a contradiction. The case $s_k < 0$ is symmetric, and $s_k = 0$ combines both. *Induction.* Initially $\lambda_k^* \in [0, M_k A_k^{\max}]$. If $s_k > 0$ the new bracket is $[\mu, u_k]$; if $\lambda_k^* \ge \mu$ the distance is unchanged, and otherwise it is $\mu - \lambda_k^* \le M_k\chi_k$. The other cases are symmetric. *Rate.* Midpoint rounds halve every bracket and other rounds never enlarge it, so after $T$ rounds with at least $\lceil T/2\rceil$ midpoints the width is at most $M_k A_k^{\max} 2^{-\lceil T/2\rceil}$. If plan sets are separable, $V_i$ is separable and the projection of $R(\lambda)$ on coordinate $k$ depends on $\lambda_k$ only, so $\chi_k = 0$. □

**Example 54** (The floor is genuine)**.** Let $d = 2$, $M = I$ and one carrier with plans 0 ($\tilde{c} = 0$, $a = (0,0)$) and $b$ ($\tilde{c} = -1.5$, $a = (1,1)$). Then $\lambda^* = (0.75, 0.75)$, $\chi_1 = \chi_2 = 1$, and the queries $(0.7, 0.9)$, $(0.6, 1)$, $(0.5, 1)$ all return plan 0 (tie at the last resolved towards 0), so the bracket update sets $u_1 = 0.5$ and excludes $\lambda_1^*$ by 0.25, within the floor but positive.

*Proof of Theorem 52.* Each round makes one pricing call per carrier. The polish is sequential best-response dynamics in a finite exact-potential game; every accepted change strictly decreases $\Phi$, which takes finitely many values, so it stops at a pure Nash equilibrium. The polish does not increase $\Phi$, and the start has $\Phi \le \Phi(\sigma^{\text{price}}) = \mathcal{D}(\lambda^{t^\star}) + \frac{1}{2}\|e^{t^\star}\|_M^2 \le \bar{\Phi}^* + \frac{1}{2}\min_t \|e^t\|_M^2$ by Theorem 50(a) and Lemma 53(a). An exact pure equilibrium makes each plan optimal at the centre belief, which lies in every credal box; the $\Gamma$ case is Lemma 25. □

**Proposition 55** (Lipschitz dependence on charges)**.** *With charges $t$ on the contributions, let $\mu^*(t) = \lambda^*(t) + t$. Then $\|\mu^*(t) - \mu^*(t')\|_{M^{-1}} \le \|t - t'\|_{M^{-1}}$ and $\langle \bar{A}^*(t) - \bar{A}^*(t'), t - t'\rangle \le 0$.*

*Proof.* With charges, $V^t(\lambda) = V(\lambda + t)$, so the equilibrium condition reads $r = M^{-1}(\mu - t) \in R(\mu)$. For $r' = M^{-1}(\mu' - t') \in R(\mu')$, antitonicity gives $\langle r - r', \mu - \mu'\rangle \le 0$, i.e. $\|\mu - \mu'\|_{M^{-1}}^2 \le \langle M^{-1}(t - t'), \mu - \mu'\rangle \le \|t - t'\|_{M^{-1}}\|\mu - \mu'\|_{M^{-1}}$. For the second claim, $\langle r - r', t - t'\rangle = \langle r - r', (\mu - \mu') - M(r - r')\rangle \le -\|r - r'\|_M^2 \le 0$. □

## Appendix D. Proofs for dual implementation pricing

Notation as in Section 4.4; $m_z^S = b_z + 2\kappa_z$ is the social slope, and $q_{iz}^2 = d_{iz}q_{iz}$ is used throughout. The proofs are taken from the theory notes of the repository, file `theory/new_algorithm_C/theory.tex`.

*Proof of Proposition 6.* (a) $\tau_{iz} = rq_{iz}$. (b) $s + rt_z = u_z$ and $t_z \geq 0$ because $s \leq u_z$; zones without lots carry no plan. (c) Substitution of $s_z \equiv s$ and $t_z = \vartheta g_z X_z^{\mathrm{NE}}$. □

*Proof of Lemma 7.* With $q_{iz}^2 = d_{iz}q_{iz}$, $X_z = rQ_z$ and $m_z^S - m_z = \kappa_z$,

$$\Phi_u - \mathrm{SC} = \sum_z \left[\tfrac{m_z}{2}\textstyle\sum_i d_{iz}q_{iz} - \tfrac{\kappa_z}{2}Q_z^2 + u_z Q_z\right].$$

Insert $u_z = \kappa_z Q_z^\circ - \frac{m_z}{2}\bar{d}_z + v_z$ and complete the square: $-\frac{\kappa_z}{2}Q_z^2 + \kappa_z Q_z^\circ Q_z = -\frac{\kappa_z}{2}(Q_z - Q_z^\circ)^2 + \frac{\kappa_z}{2}(Q_z^\circ)^2$, and $\frac{m_z}{2}\sum_i d_{iz}q_{iz} - \frac{m_z}{2}\bar{d}_z Q_z = \sum_i \frac{m_z}{2}(d_{iz} - \bar{d}_z)q_{iz}$. □

**Proposition 56** (Carrier-specific charges; classical)**.** *Let $\sigma^*$ be a pure profile and $U_{iz} = \kappa_z Q_{-i,z}^* - \frac{b_z}{2}d_{iz}$. Under carrier-specific per-delivery charges $U_i$, for all $i$ and $p, p'$: $J_i^U(p;\sigma_{-i}^*) - J_i^U(p';\sigma_{-i}^*) = \mathrm{SC}(p,\sigma_{-i}^*) - \mathrm{SC}(p',\sigma_{-i}^*)$. Hence every profile that no single carrier can improve socially is a pure equilibrium.*

*Proof.* $\mathrm{SC}(p,\sigma_{-i}^*) = R_i(p) + \sum_z q_{iz}(-\alpha_z + m_z^S Q_{-i,z}^* + \frac{m_z^S}{2}d_{iz}) + \text{const}$ and $J_i^U(p) = R_i(p) + \sum_z q_{iz}(-\alpha_z + m_z d_{iz} + m_z Q_{-i,z}^* + U_{iz})$; the brackets coincide because $m_z + \kappa_z = m_z^S$ and $m_z - \frac{b_z}{2} = \frac{m_z^S}{2}$. □

*Convexified game.* With mixtures $x_i \in \Delta(\mathcal{P}_i)$, $\bar{q}_i = \mathbb{E}_{x_i} q_i$ and $\bar{Q} = \sum_i \bar{q}_i$, the convexified potential and social cost are $\overline{\Phi}_u(x) = \sum_i \mathbb{E}_{x_i}[R_i + \sum_z q_{iz}(-\alpha_z + u_z + \frac{m_z}{2}d_{iz})] + \sum_z \frac{m_z}{2}\bar{Q}_z^2$ and $\overline{\mathrm{SC}}(x) = \sum_i \mathbb{E}_{x_i}[R_i - \sum_z \alpha_z q_{iz}] + \sum_z \frac{m_z^S}{2}\bar{Q}_z^2$. Both are convex, agree with $\Phi_u$ and SC at pure profiles, and the identity of Lemma 7 holds for them with $Q, q$ replaced by $\bar{Q}, \bar{q}$.

*Proof of Theorem 8.* $\overline{\mathrm{SC}}$ is linear in $x$ plus $\sum_z \frac{m_z^S}{2}\bar{Q}_z^2$ with $\bar{Q}$ linear in $x$, so $\overline{\mathrm{SC}}(x) - \overline{\mathrm{SC}}(x^*) = \langle \nabla\overline{\mathrm{SC}}(x^*), x - x^*\rangle + \sum_z \frac{m_z^S}{2}(\bar{Q}_z - \bar{Q}_z^*)^2$, and the inner product is non-negative by optimality of $x^*$. Hence $\Delta := \overline{\mathrm{SC}}(x^u) - \overline{\mathrm{SC}}(x^*) \geq \sum_z \frac{m_z^S}{2}(\Delta\bar{Q}_z)^2$. By the identity with $Q^\circ = \bar{Q}^*$, $v = 0$: $\overline{\Phi}_u(x) = \overline{\mathrm{SC}}(x) - \sum_z \frac{\kappa_z}{2}(\bar{Q}_z - \bar{Q}_z^*)^2 + L(x) + \text{const}$. Since $x^u$ minimises $\overline{\Phi}_u$, $\Delta \leq \sum_z \frac{\kappa_z}{2}(\Delta\bar{Q}_z)^2 + L(x^*) - L(x^u)$. Combining, $\sum_z \frac{b_z + \kappa_z}{2}(\Delta\bar{Q}_z)^2 \leq L(x^*) - L(x^u)$; then $\sum_z \frac{\kappa_z}{2}(\Delta\bar{Q}_z)^2 \leq \varrho[L(x^*) - L(x^u)]$ and $\Delta \leq (1+\varrho)[L(x^*) - L(x^u)]$. Finally $|L(x^*) - L(x^u)| \leq \sum_{i,z} \frac{m_z}{2}|d_{iz} - \bar{d}_z|\,|\bar{q}_{iz}^* - \bar{q}_{iz}^u| \leq E$ because $\bar{q}_{iz} \in [0, d_{iz}]$. □

*Proof of Theorem 9.* With $\bar{d} = d$, $L$ vanishes on the feasible set, so $E = 0$ in Theorem 8 and every minimiser of $\overline{\Phi}_u$ has aggregate $\bar{Q}^*$ and social cost $\overline{\mathrm{SC}}(x^*)$. Moreover $\overline{\Phi}_u = \overline{\mathrm{SC}} - \sum_z \frac{\kappa_z}{2}(\bar{Q}_z - \bar{Q}_z^*)^2 + \text{const}$; the subtracted term has zero gradient at $x^*$, so $x^*$ satisfies the first-order optimality condition of the convex function $\overline{\Phi}_u$. □

*Proof of Theorem 10.* (a) $J_i^u(p) = J_i^0(p) + \langle u, q_i(p)\rangle$, so constraint $(i,p)$ reads $J_i^u(p) \geq J_i^u(\sigma_i^*)$; all of carrier $i$'s constraints hold iff $\min_p J_i^u(p) \geq J_i^u(\sigma_i^*)$, and the violation $J_i^u(\sigma_i^*) - J_i^u(p)$ is maximised by a best response. (b) If a round adds a cut $(i, b_i)$, the current $u$ violates it

and satisfies every earlier cut (it is LP-feasible), so the cut is new; there are $\sum_i(|\mathcal{P}_i|-1)$ non-trivial constraints. Every cut is a valid constraint of $P(\sigma^*)$, so an infeasible LP shows that $P(\sigma^*)$ is empty. □

*Proof of Proposition 11.* $\sigma$ is an equilibrium under $u$ iff $u \in P(\sigma)$, so both descriptions of $V^\star$ range over the same pairs $(u,\sigma)$. The inequality is an inclusion of feasible sets. If the loop ends with $u \in P(\sigma^*)$, the leader attains $\mathrm{SC}(\sigma^*)$. □

*Proof of Theorem 12.* Best-response dynamics strictly decreases $\Phi_u$ on a finite set, so it ends at a pure equilibrium with $\Phi_u(\sigma^u) \le \Phi_u(\sigma^*)$. Substitute the identity with $Q^\circ = Q^*$ on both sides; the constants cancel and the congestion term vanishes at $\sigma^*$. □

*Proof of Corollary 13.* Count the calls of the three phases; Theorem 10 bounds the rounds, and each improvement of the fallback dynamics costs at most $N$ calls. □

*Proof of Theorem 14.* By Lemma 25, carrier $i$'s $\Gamma$-maximin payoff is its precise payoff minus $\delta\langle a_i(p), Mw\rangle$ (traffic already enters the route loss at its upper expectation). With $a_i = (q_i, rq_i)$, $\delta\langle a_i(p), Mw\rangle = \langle \hat{s}(\delta), q_i(p)\rangle$, so the $\Gamma$-maximin cost under $u$ is the precise cost under $u + \hat{s}(\delta)$, and $u$ implements $\sigma^*$ iff $u + \hat{s}(\delta) \in P(\sigma^*)$. The social cost does not contain the surcharge. The last claim is Lemma 24: an equilibrium of the precise game lies in $\mathrm{ICNE}_C(\delta)$ for $C \in \{E, M, ID\}$ and every $\delta$. □

*Proof of Proposition 15.* The first claim is Theorem 9 with a uniform tangent charge. For the second, $Q_z^{\mathrm{full}} = n_z d_z$ with equal lots, and without congestion ($m_z \approx b_z$ when $\kappa_z \ll b_z$) $u_z = -\frac{b_z}{2} d_z = -\frac{\alpha_z}{4n_z} = -0.375\,\mathrm{fee}/n_z$. The last claim is Proposition 6. □

## Appendix E. Proofs for aggregate-response pricing

Notation as in Section 4.4: $q_{iz}^2 = d_{iz} q_{iz}$, $\kappa_z = g_z r^2$, $m_z = b_z + \kappa_z$, $m_z^S = b_z + 2\kappa_z$, and $\Phi_u$ the exact potential of the game under the per-delivery-equivalent charge $u$. For a reference $\bar{d} \in \mathbb{R}^Z$, $L_{\bar{d}}(\sigma) = \sum_{i,z} \frac{m_z}{2}(d_{iz} - \bar{d}_z) q_{iz}$ and $E_{\bar{d}} = \sum_{i,z} \frac{m_z}{2} |d_{iz} - \bar{d}_z| d_{iz}$. The convexified potential $\overline{\Phi}_u$ and social cost $\overline{\mathrm{SC}}$ are those of Appendix D. The proofs are taken from `theory/new_algorithm_D/theory.tex`.

**Lemma 57.** *$G$ is convex and piecewise linear on $\mathcal{Q}$; $F$ is proper, closed and 1-strongly convex in $\|\cdot\|_m$, i.e. $F - \frac{1}{2}\|\cdot\|_m^2$ is convex. Moreover* $\min_{x\in\mathcal{X}} \overline{\Phi}_u(x) = \min_Q\{F(Q) + \langle u, Q\rangle\}$ *and* $\min_{x\in\mathcal{X}}[\overline{\mathrm{SC}}(x) + L_{\bar{d}}(x)] = \min_Q\{F(Q) + g(Q)\}$ *with* $g(Q) = \sum_z[\frac{\kappa_z}{2} Q_z^2 - \frac{m_z}{2}\bar{d}_z Q_z]$, *and in both cases $x$ is a minimiser on the left if and only if $\bar{Q}(x)$ is a minimiser on the right and $x$ attains the minimum defining $G$ at $Q = \bar{Q}(x)$.*

*Proof.* $G$ is the optimal value of a linear program in $x$ whose right-hand side $Q$ enters through $\bar{Q}(x) = Q$; such a value function is convex and piecewise linear on the set of right-hand sides for which the program is feasible, which is $\mathcal{Q}$. The quadratic part of $F$ is $\frac{1}{2}\|Q\|_m^2$ and the rest is convex, so $F - \frac{1}{2}\|\cdot\|_m^2$ is convex; $F$ is closed because $G$ is continuous on the closed set $\mathcal{Q}$. For the identities, write $\frac{m_z}{2} d_{iz} = \frac{m_z}{2}(d_{iz} - \bar{d}_z) + \frac{m_z}{2}\bar{d}_z$ in $\overline{\Phi}_u$ and $\frac{m_z^S}{2} = \frac{m_z}{2} + \frac{\kappa_z}{2}$ in $\overline{\mathrm{SC}} + L_{\bar{d}}$; in both the objective depends on $x$ only through $\bar{Q}(x)$ plus the quantity minimised in the definition of $G$. □

*Proof of Theorem 17.* By Lemma 57, $F + \langle u, \cdot\rangle$ is strongly convex on the compact set $\mathcal{Q}$, so it has a unique minimiser $Q(u)$, and every convexified equilibrium has $\bar{Q}(x) = Q(u)$. For closed proper strongly convex $F$ the conjugate $F^*$ is finite (the domain is bounded) and differentiable with $\nabla F^*(y) = \arg\max_Q\{\langle y, Q\rangle - F(Q)\}$ [104, Theorem 26.3]; with $y = -u$ this is $Q(u)$. The optimality conditions read $-u_k \in \partial F(Q_k)$; strong convexity gives $\langle(-u_1) - (-u_2), Q_1 - Q_2\rangle \geq \|Q_1 - Q_2\|_m^2$, and by Cauchy–Schwarz in the $m$-inner product $\langle u_2 - u_1, Q_1 - Q_2\rangle \leq \|u_2 - u_1\|_{m^{-1}}\|Q_1 - Q_2\|_m$, which gives the Lipschitz bound.

Let $D(u) = F^*(-u) + g^*(u)$ with $g^*(u) = \sum_z (u_z + \frac{m_z}{2}\bar{d}_z)^2/(2\kappa_z)$ (maximise $\langle u, Q\rangle - g(Q)$ coordinatewise). $D$ is differentiable with $\nabla D(u) = -Q(u) + \kappa^{-1} \odot (u + \frac{m}{2}\bar{d})$, which is observable, and strongly convex because $g^*$ has Hessian $\mathrm{diag}(1/\kappa)$. Fenchel duality [104, Corollary 31.2.1], with attainment because $g$ is finite and continuous, gives $\min_Q\{F + g\} = -\min_u D(u)$. The minimiser $u^*$ is unique and characterised by $\nabla D(u^*) = 0$, i.e. $u^* = \kappa \odot Q(u^*) - \frac{m}{2} \odot \bar{d} =: T(u^*)$; any fixed point of $T$ is such a zero. At the saddle point the primal minimiser is $Q(u^*)$, which minimises $F + g$, i.e. $\overline{\mathrm{SC}} + L_{\bar{d}}$ by Lemma 57. For welfare, let $x^*$ minimise $\overline{\mathrm{SC}}$ and $x^\infty$ be a convexified equilibrium under $u^*$; then $\overline{\mathrm{SC}}(x^\infty) - \overline{\mathrm{SC}}(x^*) \leq L_{\bar{d}}(x^*) - L_{\bar{d}}(x^\infty) \leq \sum_{i,z} \frac{m_z}{2}|d_{iz} - \bar{d}_z||\bar{q}^*_{iz} - \bar{q}^\infty_{iz}| \leq E_{\bar{d}}$, because $\bar{q}_{iz} \in [0, d_{iz}]$; with symmetric lots $L_{\bar{d}} \equiv 0$. □

*Proof of Theorem 18.* Let $K = \mathrm{diag}(\kappa)$ and equip $\mathbb{R}^Z$ with $\langle\cdot,\cdot\rangle_{\kappa^{-1}}$. The gradient of $D$ in this geometry is $K\nabla D(u) = u + \frac{m}{2}\bar{d} - \kappa \odot Q(u)$, so the rule (5) is $u^{t+1} = u^t - \theta K \nabla D(u^t)$, gradient descent with step $\theta$. Since $\kappa_z \leq \varrho m_z$ and $1/m_z \leq \varrho/\kappa_z$ on active zones, the Lipschitz bound of Theorem 17 gives $\|Q(u_1) - Q(u_2)\|_\kappa \leq \sqrt{\varrho}\|Q(u_1) - Q(u_2)\|_m \leq \sqrt{\varrho}\|u_1 - u_2\|_{m^{-1}} \leq \varrho\|u_1 - u_2\|_{\kappa^{-1}}$. Hence $u \mapsto \kappa \odot Q(u)$ is $\varrho$-Lipschitz from $\|\cdot\|_{\kappa^{-1}}$ to itself (note $\|\kappa \odot v\|_{\kappa^{-1}} = \|v\|_\kappa$), $K\nabla D$ is $(1 + \varrho)$-Lipschitz, and $D$ is strongly convex with modulus 1 in this geometry. (a) $T$ is $\varrho$-Lipschitz and $u^* = T(u^*)$. (b) For a $\mu$-strongly convex function with $L$-Lipschitz gradient, gradient descent with step $\theta \leq 2/(\mu + L)$ satisfies $\|u^{t+1} - u^*\|^2 \leq (1 - \frac{2\theta\mu L}{\mu + L})\|u^t - u^*\|^2$ [105, Theorem 2.1.15]; the proof uses only an inequality valid in any inner-product space. With $\mu = 1$, $L = 1 + \varrho$ and $\theta = 2/(2 + \varrho)$ the factor is $1 - 4(1 + \varrho)/(2 + \varrho)^2 = \varrho^2/(2 + \varrho)^2$. □

*Proof of Theorem 19.* (a) Apply Lemma 7 with $Q^\circ = Q(\sigma)$. Since $\sigma$ is an equilibrium, $\Phi_u(\sigma) \leq \Phi_u(\sigma')$ [45]. The term $\sum_z \frac{\kappa_z}{2}(Q_z - Q^\circ_z)^2$ vanishes at $\sigma$ and equals $\sum_z \frac{\kappa_z}{2}\Delta Q_z^2$ at $\sigma'$; rearranging the identity for $\sigma$ and $\sigma'$ gives the inequality. The last bound uses $|q_{iz}(p) - q_{iz}(\sigma_i)| \leq d_{iz}$.

(b) $u^{t+1} = \kappa \odot Q^t - \frac{m}{2}\bar{d} = T(u^t) + \kappa \odot e^t$ and $u^* = T(u^*)$, so the first inequality follows from Theorem 18(a); iterate for the limsup. For the gap bound let $h = F + \langle u^t, \cdot\rangle$, which is 1-strongly convex in $\|\cdot\|_m$ and minimised at $Q(u^t)$, so $h(Q) \geq h(Q(u^t)) + \frac{1}{2}\|Q - Q(u^t)\|_m^2$. A pure profile is a point of $\mathcal{X}$ with $\bar{Q} = Q^t$, so $\overline{\Phi}_{u^t}(\sigma^t) \geq h(Q^t)$ while $\min \overline{\Phi}_{u^t} = h(Q(u^t))$ (Lemma 57); with $\Phi_{u^t}(\sigma^t) = \overline{\Phi}_{u^t}(\sigma^t)$ at a pure profile, $\eta_t \geq \frac{1}{2}\|e^t\|_m^2$ and $\|e^t\|_\kappa \leq \sqrt{\varrho}\|e^t\|_m \leq \sqrt{2\varrho\eta_t}$.

(c) By Theorem 17, $Q = Q(u)$ iff $-u \in \partial F(Q)$. $F$ is polyhedral plus a quadratic on the polytope $\mathcal{Q}$, so $\partial F(Q) \neq \emptyset$ on $\mathcal{Q}$ and the graph of the response map is the graph of $-\partial F$ on $\mathcal{Q}$. A closed proper convex function is determined by its subdifferential up to an additive constant [104, Theorem 24.9]. The convexified optimum minimises $F + g$ with $g$ known, and $u^*$ is the unique zero of $\nabla D$; both are unchanged by a constant shift of $F$. Example 58 shows that the pure optimum is not determined. □

**Example 58.** One zone, $\kappa = 0$, slope $b = m$, intercept $\alpha$, reference $\bar{d} = d$. Instance A: two carriers, each with lot $d$ and plans "stay out" ($R = 0$) and "serve" ($R = r$). Instance B:

one carrier with lot $2d$ and plans "stay out" ($R = 0$) and "serve" ($R = 2r - md^2$), with $r > md^2/2$. In both, $\mathcal{Q} = [0, 2d]$ and $G$ is linear with $G_A(Q) = G_B(Q) = \frac{r}{d}Q$, so $F_A = F_B$ and the response maps coincide. The pure social costs are $\mathrm{SC}_A(Q) = \frac{r}{d}Q - \alpha Q + \frac{m}{2}Q^2$ on $Q \in \{0, d, 2d\}$ and $\mathrm{SC}_B(Q) = (2r - md^2)\mathbf{1}[Q = 2d] - \alpha Q + \frac{m}{2}Q^2$ on $\{0, 2d\}$. With $\alpha = \frac{r}{d} + md$, $Q = d$ is the unique pure optimum of A with value $-\frac{m}{2}d^2$, while $Q = 2d$ is the unique pure optimum of B with value $-md^2$. The optima differ in aggregate and value although nothing observable in the convexified game differs.

**Lemma 59** (Lot estimates)**.** *(a) If lots are symmetric within zones, $Q_z/n_z = \bar{d}_z$ in every served zone. (b) If between two observations $n_z$ changes by exactly one, $|Q_z^t - Q_z^{t-1}|$ is the lot of the carrier that entered or left; if $n_z = 1$, $Q_z$ is the lot of the serving carrier. (c) For a zone never served the observation carries no information about its lots; the bound of Theorem 17 holds for any reference vector, with $E_{\bar{d}}$ for the vector actually used.*

*Proof.* (a) $Q_z = n_z \bar{d}_z$. (b) The other lots in $z$ are unchanged, so the difference of the sums is the changed term. (c) Two instances that differ only in the lots of a never-served zone are indistinguishable; the bound was shown for an arbitrary reference vector. □

*Proof of Proposition 20.* By Lemma 25, the Γ-maximin cost gains the surcharge

$$\delta\langle a_i(p), Mw\rangle = \langle \hat{s}(\delta), q_i(p)\rangle,$$

with $\hat{s}_z(\delta) = \delta\hat{w}_z$, $\hat{w}_z = b_z w_{z,Q} + g_z r\, w_{z,X}$, linear in deliveries; so the response to $u$ is the precise response to $u + \hat{s}$, the map $u \mapsto T(u + \hat{s})$ is $\varrho$-Lipschitz and the rule contracts as before, with a fixed point at which $u + \hat{s}$ is a zero of the dual gradient of $\min_Q F + g + \langle \hat{s}, Q\rangle$. For the loss, let $Q_0$ and $Q_\delta$ minimise $H = F + g$ and $H + \langle \hat{s}, \cdot\rangle$, both 1-strongly convex in $\|\cdot\|_m$. Strong convexity at $Q_\delta$ gives $H(Q_\delta) - H(Q_0) \le \langle \hat{s}, Q_0 - Q_\delta\rangle - \frac{1}{2}\|Q_0 - Q_\delta\|_m^2 \le \|\hat{s}\|_{m^{-1}}\|Q_0 - Q_\delta\|_m - \frac{1}{2}\|Q_0 - Q_\delta\|_m^2 \le \frac{1}{2}\|\hat{s}\|_{m^{-1}}^2$.

For E-admissibility, write $\Delta_z = u_z - u_z^*$ for a posted charge $u$ and a pure equilibrium $\sigma^*$ under $u^*$ with precise beliefs. Carrier $i$'s cost of plan $p$ at charge $u$ and belief $(\mu_Q, \mu_X)$ about rivals is $R_i(p) + \sum_z q_{iz}(p)(-\alpha_z + m_z d_{iz} + b_z\mu_{Q,z} + g_z r\mu_{X,z} + u_z)$. Choose $\mu_{Q,z} = Q_{-i,z} - \Delta_z w_{z,Q}/\hat{w}_z$ and $\mu_{X,z} = X_{-i,z} - \Delta_z w_{z,X}/\hat{w}_z$. Then $b_z\mu_{Q,z} + g_z r\mu_{X,z} + u_z = b_z Q_{-i,z} + g_z r X_{-i,z} + u_z^*$, so every plan's cost at this belief and charge $u$ equals its cost at the realised aggregate and charge $u^*$, where $\sigma_i^*$ is optimal. The belief lies in the box iff $|\Delta_z| \le \delta\hat{w}_z = \hat{s}_z(\delta)$, and it is nonnegative when $\Delta_z \le 0$, or when $Q_{-i,z} \ge \Delta_z w_{z,Q}/\hat{w}_z$ and $rQ_{-i,z} \ge \Delta_z w_{z,X}/\hat{w}_z$ for $\Delta_z > 0$. By Lemma 23, $\sigma_i^*$ is E-admissible. The round count follows from Theorem 18(a) and $|u_z - u_z^*| \le \sqrt{\kappa_z}\|u - u^*\|_{\kappa^{-1}}$. □

*Proof of Theorem 16.* (a) $J_i^T(p; \sigma_{-i}) = R_i(p) + \sum_z q_{iz}(-\alpha_z + m_z d_{iz} + m_z Q_{-i,z} + \kappa_z Q_{-i,z} - \frac{b_z}{2}d_{iz}) = R_i(p) + \sum_z q_{iz}(-\alpha_z + \frac{m_z^S}{2}d_{iz} + m_z^S Q_{-i,z})$, using $q_{iz}^2 = d_{iz}q_{iz}$, $m_z - \frac{b_z}{2} = \frac{m_z^S}{2}$ and $m_z + \kappa_z = m_z^S$. On the other side $\mathrm{SC}(p, \sigma_{-i}) - \mathrm{SC}(\emptyset_i, \sigma_{-i}) = R_i(p) + \sum_z[-\alpha_z q_{iz} + \frac{m_z^S}{2}((q_{iz} + Q_{-i,z})^2 - Q_{-i,z}^2)]$, which is the same expression. The right side differs from $\mathrm{SC}(p, \sigma_{-i})$ by a term independent of $p$, which is the definition of an exact potential. (b) Monderer–Shapley for the potential SC on a finite profile set. (c) Immediate from the first expression in (a). □

## Appendix F. Additional tables

Table F.9: Methods on the nine city games (seed 0, price-and-branch oracle). $d_{\text{eff}}$: active coordinates. Oracle calls and relative potential gap to the convexified optimum, in %, for best response (BR), ZMC with eight rounds, Frank–Wolfe and column generation with eight rounds (FW, CG8) and column generation to convergence (CG); all methods end with the same best-response polish.

| City | $N$ | $Z$ | $d_{\text{eff}}$ | BR | ZMC | FW | CG8 | CG | BR | ZMC | FW | CG8 | CG |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| New Delhi | 5 | 120 | 64 | 14 | 45 | 45 | 50 | 65 | 13.8 | 10.2 | 10.2 | 10.2 | 10.2 |
| Cairo | 4 | 42 | 32 | 8 | 36 | 24 | 16 | 16 | $1.0 \cdot 10^{-5}$ | $1.0 \cdot 10^{-5}$ | $1.0 \cdot 10^{-5}$ | $1.0 \cdot 10^{-5}$ | $1.0 \cdot 10^{-5}$ |
| Moscow | 5 | 160 | 71 | 12 | 45 | 45 | 45 | 60 | 0.220 | 0.220 | 0.220 | 0.220 | 0.220 |
| Buenos Aires | 4 | 208 | 62 | 10 | 39 | 37 | 40 | 91 | 17.9 | 16.3 | 16.6 | 16.0 | 16.0 |
| New York | 4 | 234 | 64 | 8 | 36 | 36 | 36 | 76 | 0.451 | 0.451 | 0.437 | 0.437 | 0.437 |
| Mumbai | 5 | 143 | 55 | 14 | 48 | 48 | 48 | 60 | 4.60 | 1.98 | 1.98 | 1.98 | 1.98 |
| London | 5 | 348 | 89 | 15 | 49 | 49 | 50 | 70 | 20.0 | 18.0 | 18.0 | 18.0 | 18.0 |
| Paris | 5 | 198 | 68 | 13 | 50 | 50 | 50 | 108 | 2.74 | 2.78 | 2.74 | 2.74 | 2.74 |
| Berlin | 6 | 593 | 82 | 17 | 65 | 59 | 59 | 143 | 2.32 | 2.43 | 2.32 | 2.46 | 2.46 |

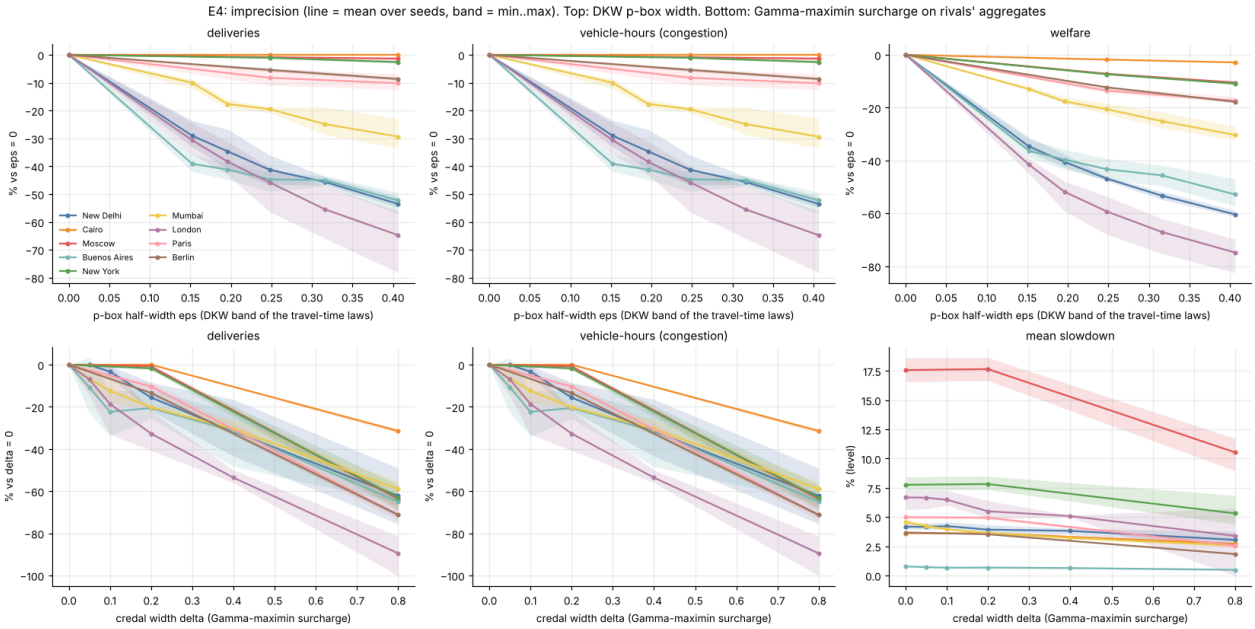



Figure F.2: Imprecision, exact oracle, mean over the seeds (line) with the minimum–maximum band. Top: probability-box half-width (route costs through BRL). Bottom: Γ-maximin credal width.

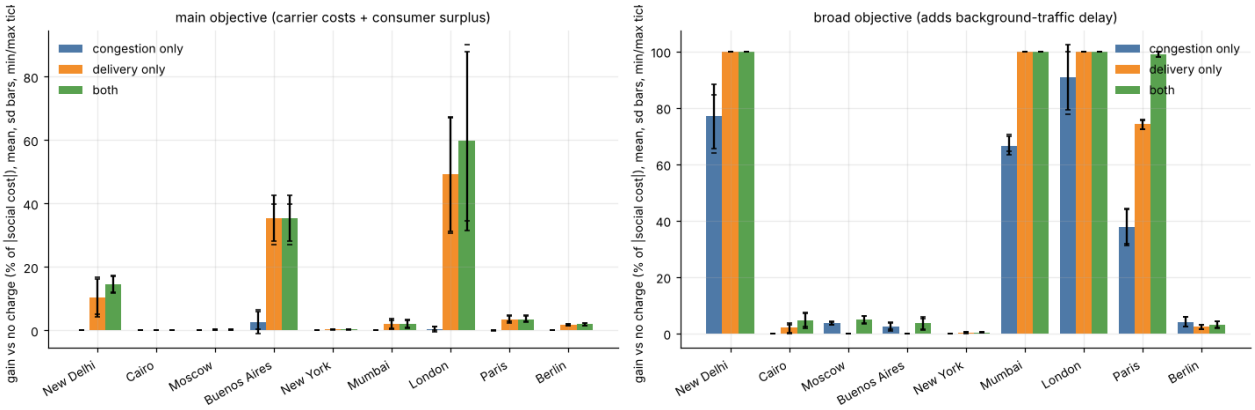



Figure F.3: Gain of the grid leader's best exact charge over no charge, by instrument: mean over the seeds with standard-deviation bars and minimum/maximum ticks. Left: main objective. Right: broad objective with the background-delay proxy (shut-down optima are an artifact). Cairo, Moscow, New York, Mumbai, Paris, Berlin use a reduced exact grid (lower bounds).

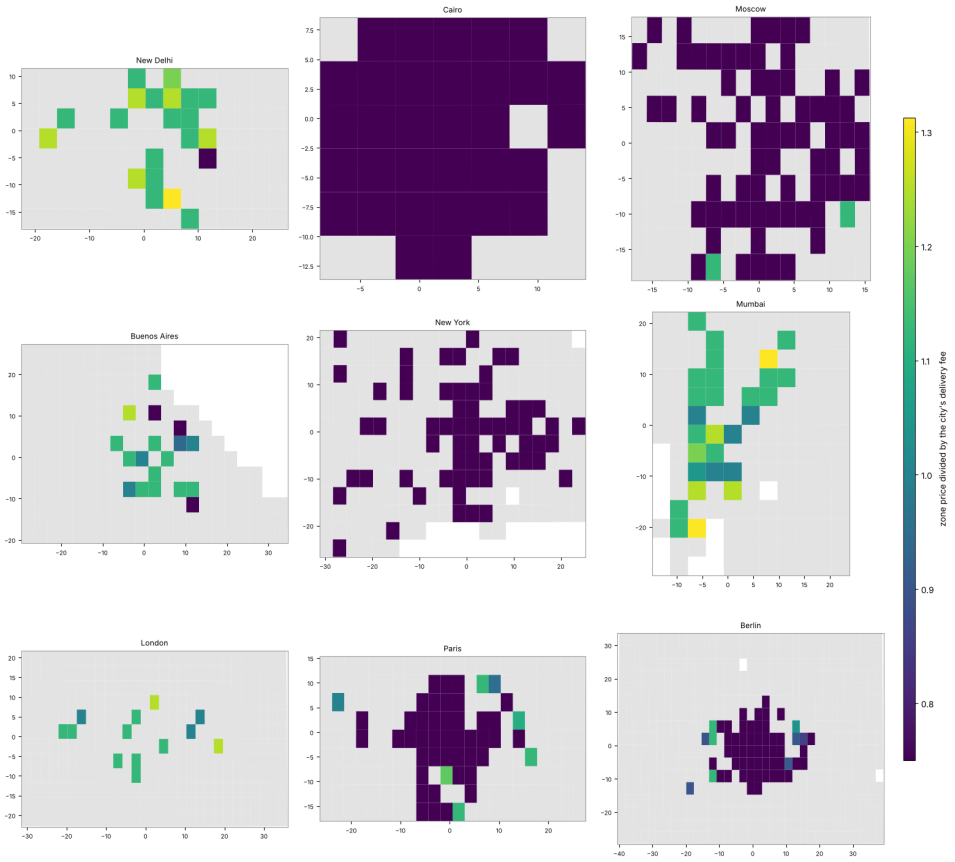



Figure F.4: Unpriced equilibrium zone prices divided by the city's delivery fee, exact oracle, seed 0 (light grey: zones without deliveries).

Table F.10: City instances. $Z$: zones; population from GHS-POP inside the zones; major road and lane kilometres inside the zones; nodes of the major-road graph; $N$: number of carriers.

| City | $Z$ | pop. (M) | major road km | lane-km | graph nodes | $N$ |
|---|---|---|---|---|---|---|
| New Delhi | 120 | 25.0 | 5 556 | 8 590 | 138 542 | 5 |
| Cairo | 42 | 14.9 | 2 649 | 4 498 | 119 027 | 4 |
| Moscow | 160 | 12.6 | 4 475 | 12 325 | 146 234 | 5 |
| Buenos Aires | 208 | 13.3 | 5 764 | 12 032 | 130 526 | 4 |
| New York | 234 | 12.1 | 8 798 | 17 770 | 382 014 | 4 |
| Mumbai | 143 | 22.0 | 4 259 | 7 092 | 109 804 | 5 |
| London | 348 | 11.3 | 7 110 | 13 628 | 350 931 | 5 |
| Paris | 198 | 9.02 | 5 990 | 10 854 | 309 175 | 5 |
| Berlin | 593 | 4.65 | 5 490 | 11 147 | 192 577 | 6 |

Table F.11: Full branch-and-price as the instance grows (1800 s limit). Metro instances with $n$ customer zones, one Solomon and one Augerat instance. Gap: relative gap at the limit; no valid bound: a pricing call hit the comparison cap. Devices: ng-4 start with cycle-driven enlargement, heuristic-first pricing, completion bound.

| instance | $n$ | ng-8, exact pricing | devices (ng-4 start, heuristic-first pricing) |
|---|---|---|---|
| augerat_A/A-n32-k5 | 31 | limit, gap 0.25% | limit, gap 0.30% |
| metro/Cairo | 30 | optimal, 17.2 s | optimal, 20.2 s |
| metro/Cairo | 40 | limit, gap 0.0041% | limit, gap 0.0032% |
| metro/London | 30 | optimal, 12.9 s | optimal, 14.4 s |
| metro/London | 40 | optimal, 559 s | optimal, 234 s |
| metro/London | 60 | limit, no valid bound | limit, no valid bound |
| metro/London | 80 | limit, no valid bound | limit, no valid bound |
| metro/NewDelhi | 60 | limit, no valid bound | limit, no valid bound |
| metro/NewYork | 30 | optimal, 36.0 s | optimal, 14.8 s |
| metro/NewYork | 40 | limit, no valid bound | optimal, 1 387 s |
| solomon25/R101 | 25 | optimal, 8.39 s | optimal, 1.72 s |

Table F.12: Unpriced equilibria with the exact oracle (mean over three seeds; seed standard deviation in parentheses). Served: share of customer demand delivered; slowdown: added mean travel-time slowdown; price: mean zone price (USD per parcel); veh.-h: vehicle-hours per day; welfare $W$ in USD per day of the slice; calls: oracle calls; $\text{regret}^{\text{LP}}$: guaranteed upper bound on any carrier's gain from deviating (USD), nonzero only where an oracle call was not proven optimal; $\eta_{\text{rel}}$: upper bound on the relative potential gap.

| City | $N$ | $Z$ | served | slowdown % | price | veh.-h | welfare | calls | $\text{regret}^{\text{LP}}$ | $\eta_{\text{rel}}$ |
|---|---|---|---|---|---|---|---|---|---|---|
| New Delhi | 5 | 120 | 0.26 (0.01) | 4.2 | 0.803 | 49.1 | 82.7 (7.0) | 48 | $7.1 \cdot 10^{-15}$ | 0.12 |
| Cairo | 4 | 42.0 | 1.0 (0) | 3.6 | 0.885 | 125 | 1 066 (20) | 39 | 0.0020 | $-2.4 \cdot 10^{-4}$ |
| Moscow | 5 | 160 | 0.99 (0.004) | 18 | 2.76 | 170 | 6 756 (154) | 47 | 1.2 | 0.0036 |
| Buenos Aires | 4 | 208 | 0.36 (0.03) | 0.77 | 0.941 | 43.6 | 272 (19) | 39 | $2.1 \cdot 10^{-14}$ | 0.24 |
| New York | 4 | 234 | 0.99 (0.002) | 7.8 | 6.82 | 112 | 20 003 (1 441) | 39 | 4.8 | 0.0050 |
| Mumbai | 5 | 143 | 0.45 (0.02) | 4.6 | 0.751 | 83.1 | 153 (10) | 47 | $1.8 \cdot 10^{-14}$ | 0.046 |
| London | 5 | 348 | 0.18 (0.03) | 6.7 | 4.67 | 24.8 | 570 (101) | 48 | $1.1 \cdot 10^{-13}$ | 0.44 |
| Paris | 5 | 198 | 0.85 (0.04) | 5.0 | 4.66 | 105 | 5 332 (437) | 50 | 0.76 | 0.039 |
| Berlin | 6 | 593 | 0.89 (0.01) | 3.7 | 3.36 | 127 | 8 619 (256) | 58 | 0.90 | 0.029 |

Table F.13: Grid leader, main objective, exact oracle (3 seeds). Gain of the best charge over no charge (% of |social cost|), mean [min, max] over seeds, with the congestion charge only, the delivery charge only, or both. Delivery share: gain with the delivery charge alone divided by the gain with both, per seed, shown where the combined gain is at least 0.5%. Subsidy: seeds in which the best point with both instruments has $s < 0$. Served share and city revenue (USD/day; negative is a net subsidy) at that point. †: only a subset of the grid solved exactly, so gains are lower bounds and shares are ratios of lower bounds.

| City | cong. only | deliv. only | both | deliv. share by seed | subsidy | served | revenue |
|---|---|---|---|---|---|---|---|
| New Delhi | 0 | 10.3 [5.0, 17] | 14.5 [12, 17] | 0.41 / 0.97 / 0.65 | 3/3 | 0.39 [0.36, 0.42] | -54.7 [-64, -48] |
| Cairo† | 0 | 0 | 0 | – | 0/3 | 1.0 | 0 |
| Moscow† | 0 | 0.215 [0.13, 0.39] | 0.215 [0.13, 0.39] | – | 3/3 | 1.00 [1.00, 1.0] | -2 411 [-3 375, -747] |
| Buenos Aires | 2.49 [0.43, 6.5] | 35.4 [27, 40] | 35.4 [27, 40] | 1.0 / 1.0 / 1.0 | 3/3 | 0.74 [0.65, 0.79] | -413 [-528, -226] |
| New York† | 0.00448 [0, 0.013] | 0.352 [0.27, 0.46] | 0.352 [0.27, 0.46] | – | 3/3 | 1.0 | -4 024 [-8 085, -1 891] |
| Mumbai† | 0 | 2.09 [0.37, 3.3] | 2.16 [0.81, 3.3] | 0.46 / 1.4 / 0.79 | 3/3 | 0.55 [0.49, 0.58] | -64.4 [-91, -20] |
| London | 0.446 [0, 1.3] | 49.3 [31, 67] | 59.7 [34, 90] | 0.92 / 0.89 / 0.74 | 3/3 | 0.49 [0.39, 0.54] | -1 359 [-1 803, -808] |
| Paris† | 0.0412 [0, 0.12] | 3.47 [2.8, 4.7] | 3.57 [2.8, 4.7] | 1.0 / 0.91 / 1.0 | 3/3 | 0.95 [0.89, 0.99] | -2 712 [-4 160, -849] |
| Berlin† | 0 | 1.85 [1.6, 2.2] | 1.98 [1.6, 2.4] | 0.93 / 1.0 / 0.90 | 3/3 | 0.94 [0.93, 0.97] | -3 334 [-5 258, -1 921] |

Table F.14: Empirical price of anarchy and stability, main objective, exact oracle (3 seeds). $W_{\mathrm{SO}}$: rounded social optimum (USD/day), mean [min, max]; #NE: distinct equilibria found; PoA of the engine's equilibrium, of the worst equilibrium found, and PoS (best found), per seed; priced gap: shortfall of the equilibrium under the grid's best charges from $W_{\mathrm{SO}}$ (%), per seed.

| City | $W_{\mathrm{SO}}$ | #NE | PoA (ZMC eq.) | PoA (worst NE) | PoS | priced gap % |
|---|---|---|---|---|---|---|
| New Delhi | 97.7 [86, 106] | 4–5 | 1.15 / 1.23 / 1.16 | 1.15 / 1.24 / 1.20 | 1.12 / 1.23 / 1.16 | 2.4 / 4.9 / 1.6 |
| Cairo | 1 066 [1 046, 1 086] | 2–3 | 1.00 / 1.00 / 1.00 | 1.00 / 1.00 / 1.00 | 1.00 / 1.00 / 1.00 | 0 / 0 / 0 |
| Moscow | 6 771 [6 639, 6 929] | 2 | 1.00 / 1.00 / 1.00 | 1.00 / 1.00 / 1.00 | 1.00 / 1.00 / 1.00 | 0.012 / 0 / 0 |
| Buenos Aires | 375 [361, 390] | 3–6 | 1.43 / 1.42 / 1.29 | 1.43 / 1.54 / 1.31 | 1.32 / 1.41 / 1.27 | 2.5 / 1.3 / 1.7 |
| New York | 20 072 [18 604, 21 466] | 1–3 | 1.00 / 1.00 / 1.00 | 1.00 / 1.00 / 1.00 | 1.00 / 1.00 / 1.00 | 0 / 0 / 0 |
| Mumbai | 158 [148, 170] | 4–6 | 1.02 / 1.04 / 1.04 | 1.11 / 1.08 / 1.04 | 1.02 / 1.04 / 1.04 | 1.5 / 1.2 / 0.39 |
| London | 927 [882, 991] | 5–6 | 1.57 / 1.45 / 1.94 | 1.61 / 1.67 / 1.98 | 1.52 / 1.42 / 1.94 | 1.4 / 7.4 / 2.2 |
| Paris | 5 530 [5 023, 5 879] | 3–5 | 1.03 / 1.03 / 1.05 | 1.03 / 1.03 / 1.05 | 1.03 / 1.02 / 1.05 | 0.047 / 0.19 / 0.23 |
| Berlin | 8 800 [8 570, 9 137] | 2–4 | 1.02 / 1.02 / 1.03 | 1.02 / 1.02 / 1.03 | 1.02 / 1.02 / 1.02 | 0.054 / 0.054 / 0.46 |

Table F.15: Imprecision, exact oracle (3 seeds; mean [min, max]). Left block: widest probability box ($\varepsilon = 0.41$) against $\varepsilon = 0$, change in deliveries and welfare (%). Right block: widest Γ-maximin width ($\delta = 0.80$) against $\delta = 0$. Monotone series: (seed, sweep) series in which deliveries never rise with the width. †: three exact widths per sweep.

| City | Δdeliv. % | Δ$W$ % | Δdeliv. % | Δ$W$ % | monotone series |
|---|---|---|---|---|---|
| New Delhi | -53.5 [-55, -51] | -60.4 [-61, -59] | -62.2 [-75, -49] | -51.2 [-65, -39] | 6/6 |
| Cairo† | 0 | -2.88 [-3.0, -2.7] | -31.4 [-33, -31] | -22.8 [-24, -22] | 6/6 |
| Moscow† | -1.35 [-2.1, -0.95] | -10.5 [-11, -9.9] | -63.7 [-68, -62] | -55.1 [-61, -52] | 6/6 |
| Buenos Aires | -52.3 [-57, -49] | -52.9 [-57, -47] | -64.9 [-65, -64] | -54.6 [-56, -52] | 3/6 |
| New York† | -2.62 [-3.0, -2.1] | -10.9 [-11, -11] | -63.2 [-69, -57] | -54.4 [-61, -47] | 6/6 |
| Mumbai | -29.4 [-34, -23] | -30.3 [-32, -27] | -59.0 [-62, -56] | -44.7 [-48, -41] | 6/6 |
| London | -64.8 [-78, -56] | -74.8 [-82, -70] | -89.5 [-100, -81] | -86.5 [-100, -76] | 6/6 |
| Paris† | -10.2 [-12, -6.9] | -17.3 [-18, -16] | -71.3 [-74, -67] | -60.4 [-64, -54] | 6/6 |
| Berlin† | -8.68 [-9.2, -8.1] | -17.8 [-18, -17] | -71.2 [-72, -70] | -59.6 [-61, -58] | 6/6 |

Table F.16: DIP against the grid, per city and seed (0 / 1 / 2). Grid pts: exact grid points available (pb: the 63-point price-and-branch grid). Gain share: DIP gain divided by the best grid gain (n/a: no charge improves on the unpriced equilibrium). Calls: DIP's oracle calls as a percentage of the calls of the grid run, and of a 63-point grid at the same calls per point. Rounds: cutting-plane rounds (1: the tangent charge implemented the target without any cut). Exact: runs in which the target is an exact pure equilibrium under the returned charges. PoA of the grid's best point and of DIP's priced equilibrium.

| City | grid pts | gain share | calls % (grid run) | calls % (63 pts) | rounds | exact | PoA grid | PoA DIP |
|---|---|---|---|---|---|---|---|---|
| New Delhi | 63 / 63 / 63 | 1.23 / 1.35 / 1.14 | 5.6 / 4.9 / 5.5 | 5.6 / 4.9 / 5.5 | 2 / 1 / 2 | 3/3 | 1.02 / 1.05 / 1.02 | 1.00 / 1.00 / 1.00 |
| Cairo | 13 / 17 / 13 | n/a / n/a / n/a | 11 / 8.5 / 12 | 2.3 / 2.3 / 2.4 | 1 / 1 / 1 | 3/3 | 1.00 / 1.00 / 1.00 | 1.00 / 1.00 / 1.00 |
| Moscow | 12 / 10 / 63$^{pb}$ | 1.10 / 1.00 / 1.00 | 13 / 13 / 2.9 | 2.6 / 2.1 / 2.9 | 1 / 1 / 1 | 3/3 | 1.00 / 1.00 / 1.00 | 1.00 / 1.00 / 1.00 |
| Buenos Aires | 63 / 63 / 63 | 1.09 / 1.05 / 1.08 | 6.3 / 6.3 / 5.4 | 6.3 / 6.3 / 5.4 | 1 / 1 / 1 | 3/3 | 1.03 / 1.01 / 1.02 | 1.00 / 1.00 / 1.00 |
| New York | 7 / 7 / 63$^{pb}$ | 0.999 / 0.999 / 1.04 | 24 / 23 / 2.9 | 2.6 / 2.6 / 2.9 | 1 / 1 / 1 | 3/3 | 1.00 / 1.00 / 1.00 | 1.00 / 1.00 / 1.000 |
| Mumbai | 15 / 14 / 14 | 3.25 / 1.20 / 1.28 | 24 / 31 / 26 | 5.6 / 6.9 / 5.8 | 1 / 2 / 2 | 3/3 | 1.02 / 1.00 / 1.00 | 0.997 / 0.997 / 0.995 |
| London | 63 / 63 / 63 | 1.04 / 1.31 / 1.06 | 7.2 / 9.4 / 8.7 | 7.2 / 9.4 / 8.7 | 3 / 3 / 3 | 3/3 | 1.01 / 1.08 / 1.02 | 1.00 / 1.00 / 0.995 |
| Paris | 14 / 13 / 13 | 1.02 / 1.06 / 1.05 | 22 / 23 / 21 | 4.9 / 4.8 / 4.4 | 3 / 1 / 2 | 3/3 | 1.00 / 1.00 / 1.00 | 1.00 / 1.00 / 1.00 |
| Berlin | 13 / 13 / 63$^{pb}$ | 1.03 / 1.03 / 1.13 | 17 / 15 / 4.1 | 3.5 / 3.2 / 4.1 | 1 / 1 / 1 | 3/3 | 1.00 / 1.00 / 1.00 | 1.00 / 1.00 / 1.00 |

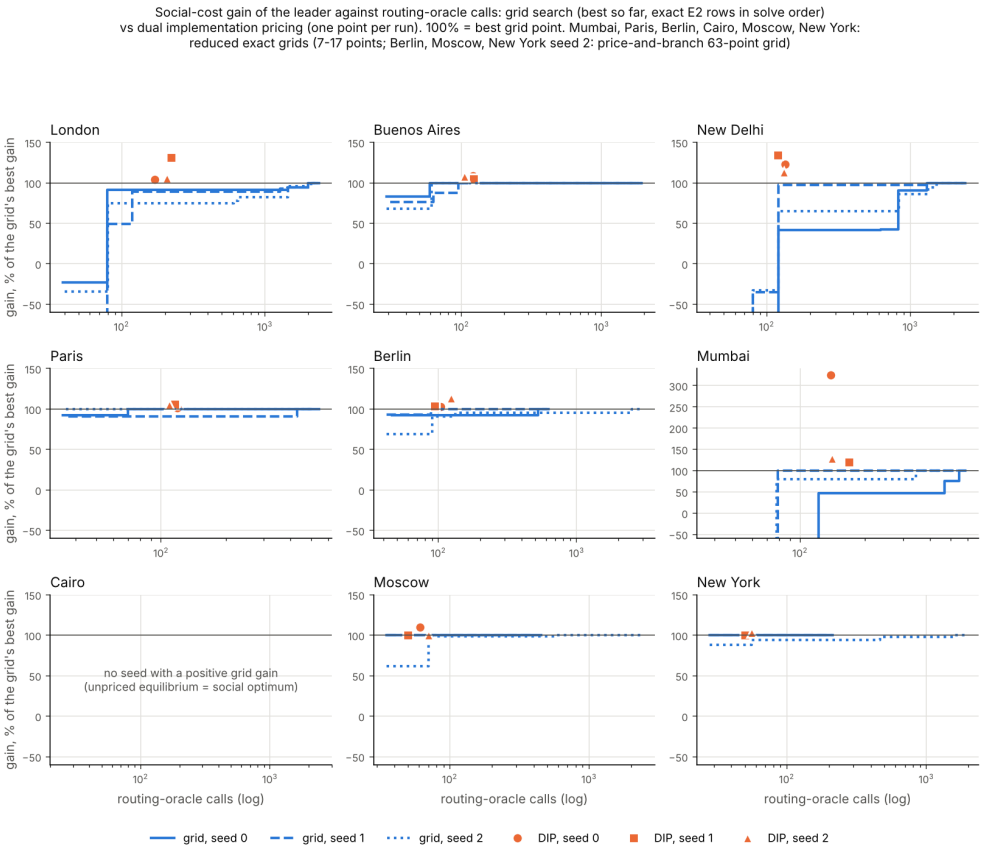



Figure F.5: Leader's social-cost gain against routing-oracle calls: best grid point found so far (lines, exact grid rows in solve order) and DIP (one marker per run); 100% is the best grid point. Six cities have reduced exact grids; for Berlin seed 2, Moscow seed 2, New York seed 2 the grid is the price-and-branch grid.

Table F.17: Branch-and-price with BRL pricing on the nine metro routing instances (seed 0, 22 customer zones, robust objective in driver-hours). Root LP: column-generation bound; P&B plan: price-and-branch plan; B&P optimum and remaining gap; nodes and time on one core; difference of the optimum from the best of three heuristics under the robust objective, and from the sample-average heuristic plan under the calibrated law. *Moscow congestion imputed.

| City | root LP | P&B plan | B&P optimum | gap % | nodes | time s | vs best heur. % | vs SAA, true law % |
|---|---|---|---|---|---|---|---|---|
| New Delhi | 63.03 | 63.18 | 63.18 | 0 | 65 | 1.63 | 0 | 0 |
| Cairo | 40.39 | 40.42 | 40.42 | 0 | 215 | 5.30 | 0 | 0 |
| Moscow* | 42.01 | 43.65 | 43.17 | 0 | 7487 | 1 063 | -1.1 | -0.57 |
| Buenos Aires | 47.29 | 48.37 | 48.13 | 0 | 1659 | 166 | -2.4 | 7.0 |
| New York | 46.10 | 46.10 | 46.10 | 0 | 1 | 8.44 | -10 | 4.8 |
| Mumbai | 53.82 | 54.08 | 53.91 | 0 | 177 | 3.59 | -0.32 | -0.31 |
| London | 51.04 | 52.03 | 51.38 | 0 | 59 | 21.2 | 0 | 0 |
| Paris | 37.73 | 38.32 | 38.25 | 0 | 4305 | 715 | -0.10 | -0.026 |
| Berlin | 51.85 | 52.67 | 52.06 | 0 | 3 | 5.63 | -0.92 | 6.8 |

Table F.18: Tiny games with exhaustive enumeration, games with a positive leader gain. Mean share of the exact optimal gain, share of games in which the optimum is reached, games worse than the unpriced equilibrium, and mean oracle calls. For tariff rows, "worse than unpriced" compares with the potential-minimising unpriced equilibrium of the enumeration.

| Method | city's information | mean share | optimum hit % | worse than unpriced | calls |
|---|---|---|---|---|---|
| ARP flat (plain) | $(Q_z, n_z)$ | 0.56 | 54 | 9 | 13 |
| ARP flat (successive averages) | $(Q_z, n_z)$ | 0.64 | 57 | 8 | 51 |
| ARP flat, lots known | $(Q_z, n_z)$ + lots | 0.63 | 54 | 15 | 12 |
| tariff + carriers' best responses | own $q$, $Q_z$ | 0.80 | 71 | 3 | 5.4 |
| tariff + potential solver | oracle | 0.94 | 89 | 1 | 24 |
| DIP | oracle | 0.97 | 92 | 0 | 39 |
| grid | oracle | 0.83 | 67 | 0 | 1 492 |

Table F.19: Theorems on the city games (three seeds). Mixed: seeds with a non-trivial mixed equilibrium. For those seeds (means): $\bar{\rho}$; the anchored three-term bound of Theorem 27(b); the $N$-uniform bound of Theorem 27(c); the largest empirical E-threshold over the tested profiles; and the ratio of the latter to the anchored bound. Γ-maximin (Lemma 25): drop in the share of demand served from the precise to the surcharge equilibrium at $\delta = 0.1$ and $\delta = 1$, and the maximality threshold of the surcharge equilibrium divided by $\delta$ at $\delta = 1$.

| City | mixed | $\bar{\rho}$ | anchored bd. | $N$-unif. bd. | max $\delta_E$ | ratio | drop $\delta$=0.1 | drop $\delta$=1 | M-thr./$\delta$ |
|---|---|---|---|---|---|---|---|---|---|
| New Delhi | 3/3 | 1.0 | 1.61 | 10.7 | 0.64 | 0.41 | 0.0075 | 0.18 | 0.89 |
| Cairo | 0/3 | – | – | – | – | – | 0 | 0.47 | 0.99 |
| Moscow | 0/3 | – | – | – | – | – | 0.0021 | 0.73 | 0.97 |
| Buenos Aires | 1/3 | 0.96 | 1.76 | 8.80 | 0.83 | 0.47 | 0.082 | 0.31 | 0.93 |
| New York | 1/3 | 0.82 | 1.58 | 3.68 | 0.82 | 0.52 | 0.0090 | 0.71 | 0.98 |
| Mumbai | 2/3 | 0.74 | 1.23 | 10.6 | 0.53 | 0.43 | 0.049 | 0.34 | 0.92 |
| London | 3/3 | 0.85 | 1.56 | 13.6 | 0.82 | 0.53 | 0.033 | 0.17 | 0.76 |
| Paris | 3/3 | 0.88 | 1.39 | 8.43 | 0.69 | 0.49 | 0.036 | 0.74 | 0.96 |
| Berlin | 1/3 | 0.94 | 0.937 | 7.97 | 0.41 | 0.44 | 0.050 | 0.77 | 0.97 |

Table F.20: Coverage of the exact-oracle game experiments per city and seed: exact grid points of the leader experiment (out of 63), price-of-anarchy experiment computed (yes/–), and exact widths of the imprecision experiment (out of 12); – means not computed with the exact oracle. The base equilibria are exact for all city-seeds.

| City | E2 s0 | E3 s0 | E4 s0 | E2 s1 | E3 s1 | E4 s1 | E2 s2 | E3 s2 | E4 s2 |
|---|---|---|---|---|---|---|---|---|---|
| New Delhi | 63/63 | yes | 12/12 | 63/63 | yes | 12/12 | 63/63 | yes | 12/12 |
| Cairo | 13/63 | yes | 6/12 | 17/63 | yes | 6/12 | 13/63 | yes | 6/12 |
| Moscow | 12/63 | yes | 6/12 | 10/63 | yes | 6/12 | 11/63 | yes | 6/12 |
| Buenos Aires | 63/63 | yes | 12/12 | 63/63 | yes | 12/12 | 63/63 | yes | 12/12 |
| New York | 7/63 | yes | 6/12 | 7/63 | yes | 6/12 | 9/63 | yes | 6/12 |
| Mumbai | 15/63 | yes | 12/12 | 14/63 | yes | 12/12 | 14/63 | yes | 12/12 |
| London | 63/63 | yes | 12/12 | 63/63 | yes | 12/12 | 63/63 | yes | 12/12 |
| Paris | 14/63 | yes | 6/12 | 13/63 | yes | 6/12 | 13/63 | yes | 6/12 |
| Berlin | 13/63 | yes | 6/12 | 13/63 | yes | 6/12 | 13/63 | yes | 6/12 |

Table F.21: Calibration per city (base values; ranges and sources in `data/parameters/SOURCES.md`). Parcels per capita are country-level; TomTom congestion is the 2025 city value (Moscow: 2020); drops per vehicle and van capacity are proxies outside the United States; fees and wages are graded C for most cities.

| City | parcels/cap./day | cong. % | drops/veh. | shift h | van cap. | fee USD | wage USD/h | fuel USD/L | $N$ | veh./1000 |
|---|---|---|---|---|---|---|---|---|---|---|
| New Delhi | 0.00964 | 60.2 | 40.0 | 10 | 40.0 | 0.690 | 1.07 | 1.15 | 5 | 400 |
| Cairo | 0.00561 | 40.5 | 40.0 | 8.4 | 40.0 | 1.18 | 0.610 | 0.440 | 4 | 220 |
| Moscow | 0.156 | 54.0 | 70.0 | 8.4 | 126 | 3.67 | 4.53 | 0.900 | 5 | 297 |
| Buenos Aires | 0.0150 | 48.1 | 70.0 | 8.4 | 126 | 0.980 | 1.28 | 1.45 | 4 | 331 |
| New York | 0.184 | 48.8 | 120 | 8.4 | 226 | 9.09 | 22.3 | 1.22 | 4 | 242 |
| Mumbai | 0.00964 | 63.2 | 40.0 | 10 | 40.0 | 0.690 | 1.07 | 1.15 | 5 | 330 |
| London | 0.166 | 51.6 | 72.0 | 7.3 | 126 | 4.21 | 18.8 | 2.34 | 5 | 288 |
| Paris | 0.0650 | 40.0 | 70.0 | 8.4 | 126 | 6.04 | 15.7 | 2.42 | 5 | 575 |
| Berlin | 0.143 | 44.6 | 126 | 8.4 | 206 | 4.40 | 18.8 | 2.39 | 6 | 330 |

Table F.22: Sensitivity of unpriced welfare (% change, low to high value of each weak parameter, seed 0). The scenarios use the exact oracle; the reference base case is the earlier price-and-branch equilibrium of the same seed, so differences of the order of the oracle effect (Section 7.1) are not meaningful.

| City | fee | wage | drops | $N$ | parcels |
|---|---|---|---|---|---|
| New Delhi | -58 to 317 | -53 to 43 | -90 to 294 | -38 to 53 | -9 to 0 |
| Cairo | 19 to 19 | -5 to 2 | -31 to 59 | -25 to 24 | -0 to 0 |
| Moscow | -75 to 48 | -26 to 6 | -43 to 105 | -62 to 21 | -0 to 1 |
| Buenos Aires | -59 to 452 | -48 to -48 | -87 to 474 | -13 to 72 | -1 to 6 |
| New York | -41 to 55 | -8 to 1 | -57 to 9 | -27 to 23 | -0 to 0 |
| Mumbai | -43 to 208 | -36 to 26 | -72 to 169 | -19 to 29 | -5 to -1 |
| London | -95 to 363 | -3 to 28 | -95 to 795 | -26 to 107 | -2 to 4 |
| Paris | -58 to 83 | -4 to 10 | -57 to 149 | -16 to 27 | -0 to 1 |
| Berlin | -58 to 24 | -5 to 3 | -82 to 4 | -3 to -3 | -0 to 2 |

Table F.23: City leader, broad objective with the background-delay proxy (3 seeds). Gains as in Table F.13 (mean [min, max]); combined gain per seed; served share at the best point with both instruments. Shut-down optima (served share 0) are an artifact of the proxy.

| City | cong. only | deliv. only | both | both by seed | served |
|---|---|---|---|---|---|
| New Delhi | 77.1 [64, 85] | 100 | 100 | 100 / 100 / 100 | 0 |
| Cairo† | 0 | 2.06 [0, 3.3] | 4.59 [2.5, 7.5] | 2.47 / 7.50 / 3.81 | 0.67 [0.65, 0.69] |
| Moscow† | 3.67 [3.1, 4.1] | 0.0230 [0, 0.069] | 4.87 [3.6, 6.3] | 3.59 / 4.72 / 6.31 | 0.92 [0.91, 0.93] |
| Buenos Aires | 2.41 [1.4, 4.0] | 0 | 3.71 [1.1, 5.4] | 4.62 / 5.42 / 1.09 | 0.35 [0.29, 0.46] |
| New York† | 0.00636 [0, 0.019] | 0.320 [0.18, 0.52] | 0.344 [0.26, 0.52] | 0.259 / 0.518 / 0.256 | 1.00 [0.99, 1.0] |
| Mumbai† | 66.7 [65, 71] | 100 | 100 | 100 / 100 / 100 | 0 |
| London | 90.9 [78, 100] | 100 | 100 | 100 / 100 / 100 | 0 |
| Paris† | 37.7 [32, 44] | 74.2 [72, 76] | 99.1 [98, 100] | 99.3 / 100 / 98.1 | 0.042 [0, 0.085] |
| Berlin† | 4.07 [2.6, 5.9] | 2.39 [1.5, 3.2] | 3.10 [2.1, 4.5] | 2.13 / 2.66 / 4.50 | 0.74 [0.65, 0.88] |

Table F.24: ZMC on synthetic D1 games with non-separable plan sets (five seeds per row). Entries: mean oracle calls / mean potential gap to the convexified optimum; last column: instances in which ZMC has the smallest gap among FW8, CG8 and multi-start BR.

| $N$ | $Z$ | BR | BR multi-start | FW8 | CG8 | ZMC | CG full | ZMC best |
|---|---|---|---|---|---|---|---|---|
| 4 | 30 | 12 / 5.33 | 45 / 3.22 | 38 / 3.64 | 39 / 3.20 | 39 / 2.77 | 58 / 2.69 | 4/5 |
| 5 | 60 | 21 / 15.8 | 56 / 11.8 | 50 / 7.78 | 51 / 7.80 | 49 / 6.90 | 159 / 6.13 | 4/5 |
| 6 | 120 | 28 / 50.2 | 70 / 38.4 | 61 / 21.4 | 63 / 27.5 | 60 / 17.1 | 365 / 14.2 | 4/5 |
| 6 | 250 | 26 / 91.5 | 71 / 80.2 | 64 / 47.9 | 67 / 55.4 | 60 / 33.4 | 797 / 27.8 | 5/5 |

Table F.25: Tiny games solved by exhaustive enumeration (earlier price-and-branch oracle): share of instances in which the method's profile is a Nash equilibrium of the enumerated game and reaches the global potential minimum, largest independent-oracle regret and mean excess of the potential over the global minimum.

| Method | games | calls | is NE | max regret | global min. | excess |
|---|---|---|---|---|---|---|
| ZMC (6 rounds) | 40 | 17.5 | 1.00 | 0 | 1.00 | 0 |
| BR | 40 | 4.85 | 1.00 | 0 | 0.875 | 0.93 |
| FW (6 rounds) | 40 | 16.7 | 1.00 | 0 | 0.950 | 0.067 |
| CG | 40 | 11.1 | 1.00 | 0 | 0.975 | 0.020 |

Table F.26: Checks of ZMC on the city games (seed 0, earlier price-and-branch oracle): largest residual of the clearing identity (Theorem 50(a)); share of rounds in which its bound holds; inverted brackets; active coordinates; largest relative bracket width after eight rounds; in bisection mode, share of coordinates within the bound of Theorem 51 at $T = 8$, largest normalised error and the bound $2^{-4}$.

| City | identity resid. | bound holds | inverted | active | $\text{width}_8$ | $\text{within}_8$ | $\text{err}_8$ | $\text{bound}_8$ |
|---|---|---|---|---|---|---|---|---|
| New Delhi | $2.8 \cdot 10^{-14}$ | 1.00 | 0 | 128 | 0.0435 | 1.00 | 0.00385 | 0.0625 |
| Cairo | $1.6 \cdot 10^{-13}$ | 0.875 | 0 | 64 | 0.0435 | 1.00 | 0.00385 | 0.0625 |
| Moscow | $8.1 \cdot 10^{-13}$ | 1.00 | 0 | 142 | 0.0435 | 1.00 | 0.00385 | 0.0625 |
| Buenos Aires | $5.7 \cdot 10^{-14}$ | 1.00 | 0 | 124 | 0.0435 | 1.00 | 0.00385 | 0.0625 |
| New York | $3.4 \cdot 10^{-12}$ | 1.00 | 0 | 128 | 0.0435 | 1.00 | 0.00385 | 0.0625 |
| Mumbai | $4.3 \cdot 10^{-14}$ | 1.00 | 0 | 110 | 0.0435 | 1.00 | 0.00385 | 0.0625 |
| London | $1.1 \cdot 10^{-13}$ | 1.00 | 0 | 178 | 0.0435 | 1.00 | 0.00385 | 0.0625 |
| Paris | $1.5 \cdot 10^{-12}$ | 1.00 | 0 | 136 | 0.0415 | 1.00 | 0.00385 | 0.0625 |
| Berlin | $1.8 \cdot 10^{-12}$ | 1.00 | 0 | 164 | 0.0435 | 1.00 | 0.00385 | 0.0625 |

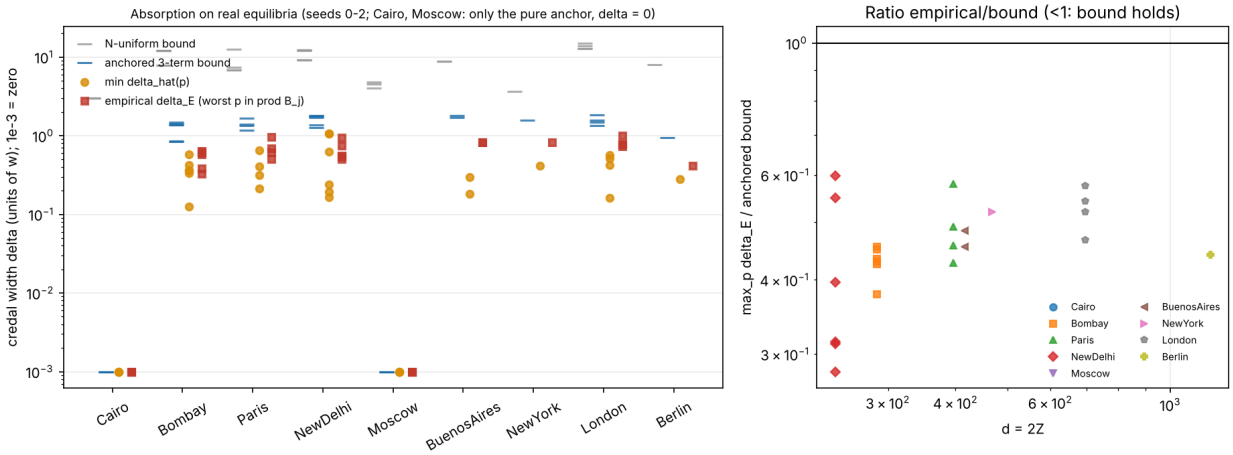



Figure F.6: Credal absorption on the city equilibria (three seeds). Left: $N$-uniform bound, anchored bound, smallest anchored radius and largest empirical E-threshold per anchor (Cairo and Moscow: pure anchors only, plotted at $10^{-3}$ for zero). Right: ratio of the largest empirical E-threshold to the anchored bound against $d = 2Z$; values below one mean the bound holds.

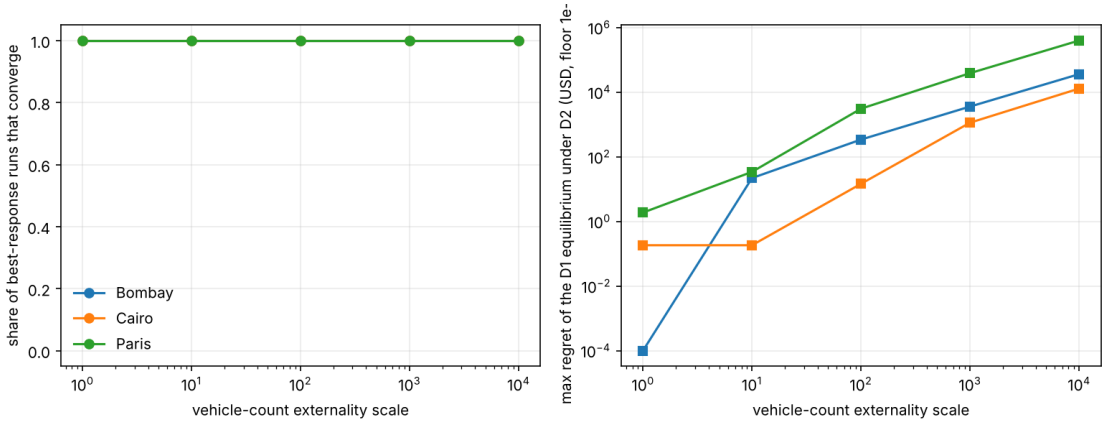



Figure F.7: City games of Cairo, Mumbai and Paris (three seeds) with a vehicle-count externality of increasing scale. Left: share of best-response runs that converge to a pure equilibrium. Right: largest regret of the D1 equilibrium when the game is of type D2 (USD).